



Hasim

Green Hydrogen Engineering Suite · version 2.0.0

User Manual

Engineering workstation for green hydrogen – from the electrochemical cell to the plant fence: drawings, streams, equipment, annual simulation, economics, component library, impact assessment, safety and design reports.

Hasim Green Hydrogen Engineering Suite File Edit View Simulate Drawings Tools Help

Case B - 100 MW AEL, islanded hybrid, ammonia export — Hasim 2.0.0 Owner: Prof. Hani Muhsen

Home Drawings Analysis Tools Help

New Open Save Solve Auto-solve Undo Redo Solve

Project Explorer

- Case B - 100 MW AEL, isla...
 - Overview
 - Design basis
 - Drawings
 - PFD-001 Process flow di...
 - SLD-001 Electrical single...
 - SNK-001 Energy balance...
 - SNK-002 Water & mass b...
 - CEL-002 Cell cross-secti...
 - STK-001 Stack, module...
 - WTR-001 Water-treatme...
 - CMP-001 Compression t...
 - LAY-001 Plot plan & safe...
 - Data
 - Stream table 21
 - Equipment list 26
 - Simulation
 - Annual simulation 8760 h
 - Economics
 - Impact assessment
 - Design checks 12/12
 - Design report
 - Engineering tools
 - Cell & Stack Lab

PFD-001 - Process flow diagram

Stream data Print SVG

PROPERTIES

EL-101 (x 20)
Electrolyser modules · Electrolysis

Design basis Show on drawing

Equipment list

Required equipment. It cannot be removed because it is the core of the plant.

DATASHEET

Technology	Alkaline
Modules × rating [MW]	20 × 5
Stacks per module	1
Cells per stack	230
Cell active area [cm²]	26 000
Current density (rate...)	0.400 CM 4.4
Cell voltage (rated, B...	1.86 CM 4.4
Stack current / voltage	10400 A / 428.3 V CM 6.1
Stack power [kW]	4455 CM 6.1
Operating T / p(H ₂) / ...	80 °C / 30 / 30 bar
Faradaic efficiency	0.998 CM 7.6
SEC stack / system (r...	49.6 / 53.7 CM 7.5/7.8
H ₂ production (rated...	1935 CM 4.2
Stack replacement in...	90 000 CM 6.6
Minimum load (desig...	20 % / 20 %

Messages 4 Design checks 12/12

- 20:42:13 WT-101 Positive scaling index (LSI) 1.86, indicative for seawater concentrate) - acid or antiscalant against CaCO₃ scaling (DM 8.6).
- 20:42:13 Solver System solved: 26 equipment items, 21 streams, 8760-h dispatch in 52 ms.
- 20:42:13 Licence Licensed to Prof. Hani Muhsen - Owner - perpetual
- 20:42:13 Hasim Hasim - Green Hydrogen Engineering Suite 2.0.0 (Web edition) - © 2026 Prof. Hani Muhsen. All rights reserved.

Solved - 52 ms 20 x 5 MW AEL, islanded H₂ 8339 t/a LCOH 7.66 EUR/kg CI 0.10 kg/kg Licensed to Prof. Hani Muhsen - Owner - perpetual © 2026 Prof. Hani Muhsen. All rights reserved. Owner & developer: Prof. Hani Muhsen.

Owner and developer: Prof. Hani Muhsen

Desktop · Portable · Web · Microsoft Store editions · September 2026

137 executable equations · 16 simulation levels · 9 engineering drawings · 34 library components · 318 validation checks

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About this manual

Chapters 1–3 get you started and explain the workstation. Chapters 4–10 follow the engineering workflow: design basis and library components, solver, drawings, data, annual simulation, economics and impact assessment, checks and report. Chapter 11 describes the engineering tools – among them the Component library and the Model builder –, chapters 12–13 the verification and the modelling assumptions. Step-by-step modelling exercises are provided in the companion document *Hasim – Model-Building Examples*.

1 Introduction

Hasim is an engineering workstation for green hydrogen projects. From a compact **design basis** – technology, number of modules, power supply, site, water source, storage, delivery and economics – it builds and solves a consistent engineering model of the complete plant and presents it the way engineers work: as drawings on sheets with title blocks, stream and equipment tables, datasheets, an annual 8760-hour simulation, economics, automatic design checks and a printable design report. Every change of an input re-solves the whole model within a fraction of a second.

The engineering content is taken from two references: the *Green Hydrogen Calculations Manual* (Edition 3, “CM”) and the *Green Hydrogen Plant Design Manual* (Edition 1, “DM”). Hasim implements 137 of their numbered equations (90 from CM, 47 from DM) as executable functions; results are traceable to these equation numbers throughout the program and in the reports.

Plants and value chains can also be built from the component level. The **component library** contains 34 generic components – PV plants, wind turbines, batteries, water plants, electrolyser stacks, power electronics, balance of plant, compressors, storage vessels, pipelines, tube trailers, refuelling stations and conversion plants – each with technical, economic, environmental and social data and the sources of its values; you add your own. The **component wizard** chooses the equipment step by step and shows the effect of every choice, and the **impact assessment** reports the technical, economic, environmental and social impacts of the plant or chain.

1.1 What Hasim produces

- **Nine engineering drawings:** process flow diagram with stream table (PFD-001), electrical single-line diagram (SLD-001), energy and mass Sankey diagrams (SNK-001/002), cell cross-section (CEL-001), stack and module arrangement (STK-001), water-treatment train (WTR-001), compression train (CMP-001) and plot plan with safety distances (LAY-001).
- **Stream table** with temperature, pressure, mass and volume flow and composition of every numbered stream, plus the energy streams.
- **Equipment list** with tags following the DM conventions and a datasheet for every item.
- **Annual simulation** of 8760 hours with a live, hour-by-hour power-flow view.
- **Economics:** LCOH, CAPEX build-up and a sensitivity (tornado) analysis that re-solves the whole system.
- **Impact assessment:** technical performance and checks; investment, NPV, IRR and local content; life-cycle carbon intensity, embodied emissions, water and land; construction, manufacturing and permanent jobs, water-stress risk and the safety exposure of neighbours.
- **Design checks** against the limits of both manuals, of the chosen components and of international codes, standards and regulations (IEC/ISO, EU, US), a **codes and standards screening** – RFNBO, hazardous areas, major-accident regulations, pressure equipment, separation distances, power quality – and a complete **design report**.
- **Component library** with 34 generic components and your own, and the **component wizard** for plant projects and Model-builder chains.
- **Engineering tools:** Cell & Stack Lab, Equation library, Experiments, Component library, Model builder, Script console and Validation.

1.2 Simulation levels

The equations are organised in sixteen levels that follow the value chain from the electrochemical cell to the plant, the market and safety:

Level	Name	Equations
1	Properties & units	10
2	Thermodynamics	6
3	Cell	8
4	Stack & power electronics	9
5	System efficiency & balances	10
6	Renewable supply	12
7	Water supply & treatment	12
8	Cooling & heat	5
9	Compression & purification	8
10	Storage	14
11	Transport	5
12	Power-to-X & end use	12
13	Economics	7
14	Environment & certification	3
15	Safety & materials	13
16	Design basis & reliability	3

2 Installation, editions and licensing

Hasim is delivered in four editions built from the same program. All run on the local computer: projects, inputs and results never leave the device. The Desktop, Portable and Web editions need no internet connection after download; the Microsoft Store edition uses it for installation and updates and works offline in between.

Edition	Package	How to start	Best for
Desktop	Hasim_Desktop_v2.0.0.zip	Unzip, double-click Hasim.vbs (Windows) or run hasim.sh (macOS/Linux). Optional Start-menu shortcuts with Create_Shortcuts.cmd .	Daily engineering work in a dedicated window
Portable	Hasim_Portable_v2.0.0.zip	Double-click Hasim_Portable.html – one self-contained file.	USB sticks, restricted computers, classrooms
Web / intranet	Hasim_Web_v2.0.0.zip	Copy the folder to a web server; open index.html . Installable as an app, works offline after the first visit.	Company intranets, teaching labs
Microsoft Store	Hasim in the Microsoft Store (Windows 10/11)	Buy it in the Microsoft Store (search “Hasim”) – one price, paid once – and start it from the Start menu. It starts as the full version; updates arrive automatically (\$2.4).	Individual engineers and researchers on Windows
Online evaluation	hasim website ► Try online	Opens in the browser; projects stay in the browser.	A first look without downloading

2.1 First start

On the first start Hasim shows the welcome screen with the licence agreement. After you accept it, the 14-day evaluation starts automatically (Figure 2.1). The example project “Case B – 100 MW AEL, islanded hybrid, ammonia export” is loaded and solved.

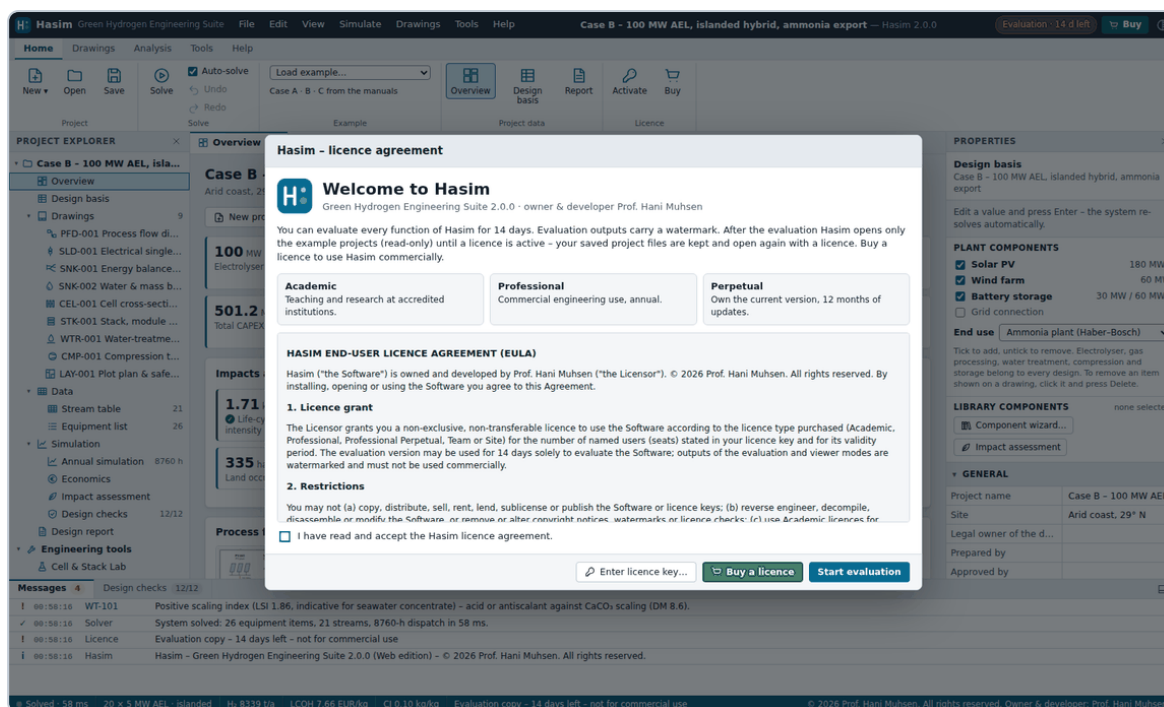


Figure 2.1 – Welcome screen with the licence agreement, shown on first start.

2.2 Evaluation, viewer and licensed modes

Function	Evaluation (14 days)	Viewer (after evaluation)	Licensed
Example projects, drawings, tables, charts	✓	✓ (read-only)	✓
Edit the design basis, solve	✓	–	✓
Save and open projects (.hasim)	✓	–	✓
Export SVG / CSV / HTML, print	✓ watermarked (also on printouts)	–	✓
Engineering tools	✓	Validation only	✓
Drawing title block	“Evaluation copy – not for commercial use”	“Viewer – not licensed”	“Licensed to ...”

The licence state is shown by the badge in the title bar and in the status bar. Evaluation outputs carry a diagonal watermark and must not be used commercially (Figure 2.2). Hasim reminds you during the last three days of the evaluation. Your own project files are kept after the evaluation and open again as soon as a licence is active.

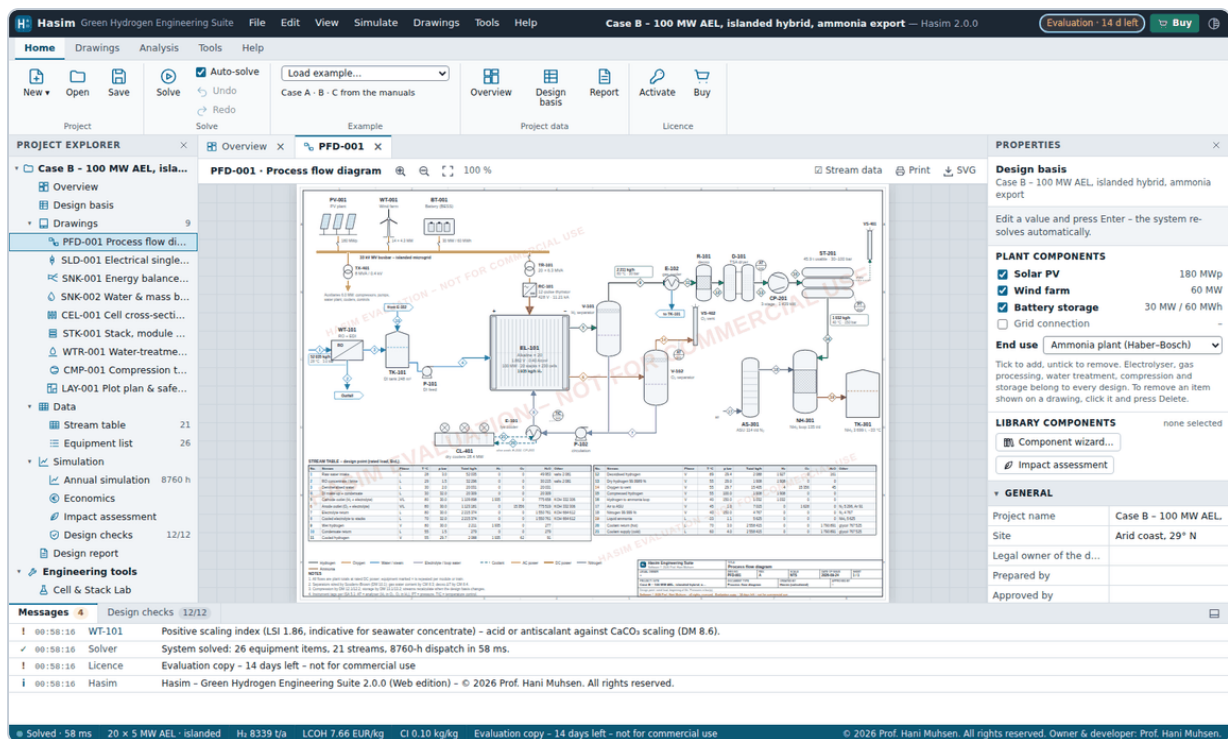


Figure 2.2 – Evaluation mode: the licence badge in the title bar and the watermark on the drawing.

2.3 Activating a licence key

Licence keys are used in every edition; Hasim bought in the Microsoft Store needs no key (\$2.4).

- 1 Buy a licence on the Hasim website. The licence key is shown on the order-confirmation page and sent by e-mail. It starts with **HASIM-**.
- 2 In Hasim, click the licence badge in the title bar or choose **Help ► Activate licence**.
- 3 Paste the complete key (or load a **.hasimlic** file) and press **Activate** (Figure 2.3).
- 4 The badge turns green and shows the licensee. Drawings and reports now carry “Licensed to ...” in the title block and no watermark.

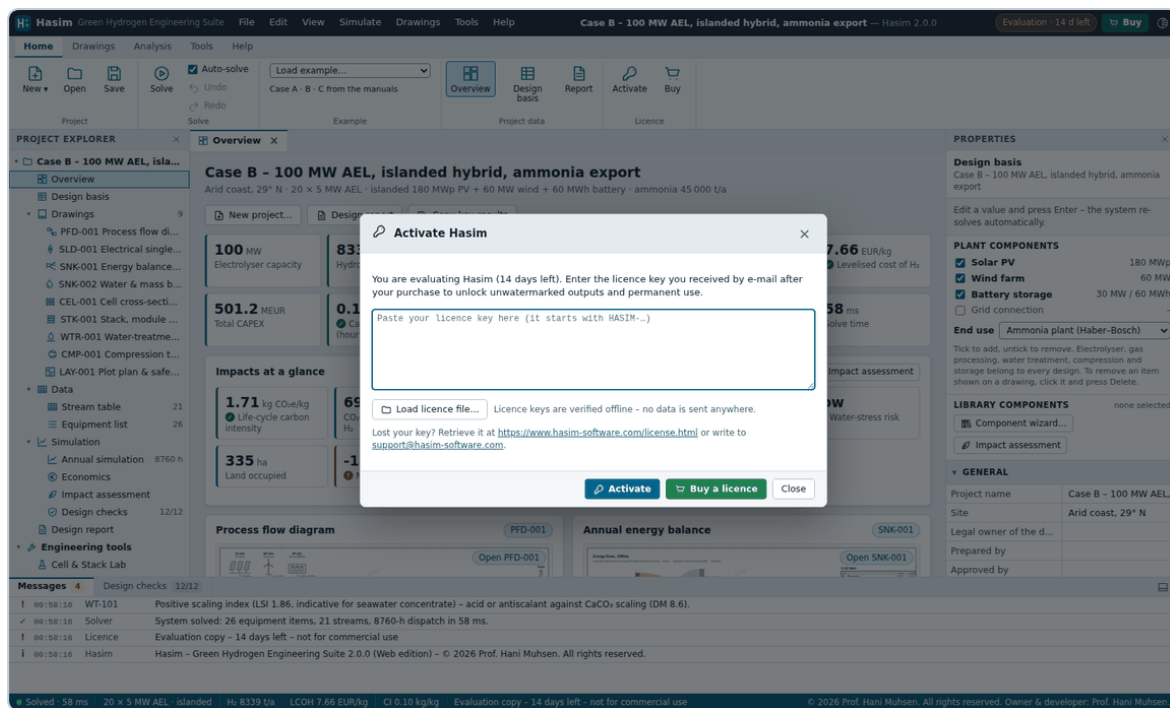


Figure 2.3 – The activation dialog. Keys are digitally signed and verified offline.

Good to know

Licence keys are verified offline with a digital signature (ECDSA P-256); no account or internet connection is needed. A key can be used on up to three computers of the same named user (see the EULA). To move to a new computer, activate the same key there. Lost keys can be retrieved on the website with the e-mail address used for the purchase. Subscriptions renew automatically; after each renewal you receive a new key by e-mail – activate it before the old one expires. A Professional Perpetual licence covers every version released during its update period; earlier versions remain available on the download page. Hasim re-checks the licence whenever its window regains focus. If the browser blocks local storage for Hasim, the activation dialog warns that the licence plan stays active only for the current session.

2.4 Buying Hasim in the Microsoft Store

In the Microsoft Store Hasim is a one-time purchase: one price, paid once – a lifetime licence for one user with all updates of the Store edition. Microsoft handles payment, receipts, taxes and refunds; there is no licence key to enter and no sign-in.

- 1 Try Hasim first: its web address in a browser runs as the free 14-day evaluation. **Buy** there explains the purchase and opens the Hasim page in the Microsoft Store (Figure 2.4).
- 2 Buy Hasim in the Microsoft Store with your Microsoft account and install it – a desktop app of about 100 MB.
- 3 Start Hasim from the Start menu: it starts as the full version, with the badge “Licensed · Microsoft Store” and no watermark. A click on the badge shows the licence (Figure 2.5).

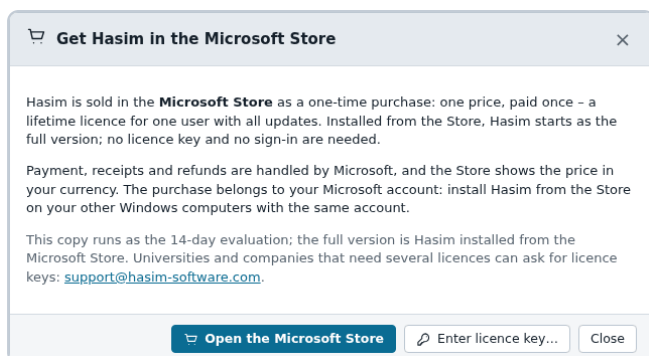


Figure 2.4 – Buy in the free evaluation: the one-time purchase in the Microsoft Store.

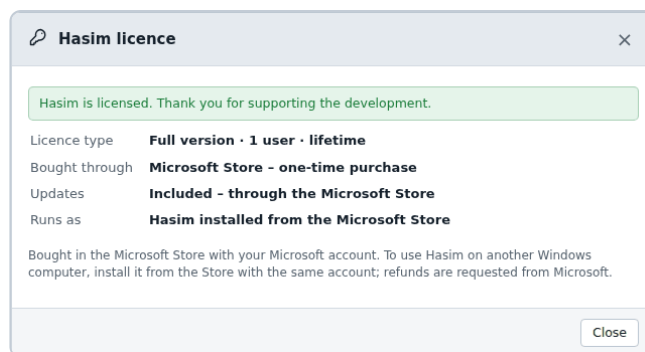


Figure 2.5 – Hasim installed from the Store: full version, lifetime licence, one-time purchase.

Good to know

Hasim from the Microsoft Store is a desktop app that contains the whole program: it works without an internet connection and starts as the full version – no key, no sign-in. On your other Windows computers, install it from the Store with the same Microsoft account. The web address in a browser always runs as the evaluation. **Help ► User Manual (PDF)** opens the manuals (a copy is saved in Documents\Hasim). Refunds are requested from Microsoft. Universities and companies that need several licences use licence keys (§2.3), which work in every edition.

2.5 System requirements

- Windows 10/11, macOS 12 or newer, or a current Linux desktop. The Microsoft Store edition needs 64-bit Windows 10 (version 1809 or later) or Windows 11 and about 400 MB of disk space.
- A current browser engine: Microsoft Edge, Google Chrome, Firefox or Safari (the desktop edition uses the Edge engine included in Windows).
- 4 GB RAM; a screen of 1366 × 768 or larger is recommended. Hasim also adapts to tablets and phones (Figure 3.6).

3 The engineering workstation

Hasim uses the layout of engineering workstations such as process simulators: menus and a ribbon at the top, the project explorer on the left, tabbed documents in the centre, the properties panel on the right, the message log at the bottom and a status bar (Figure 3.1).

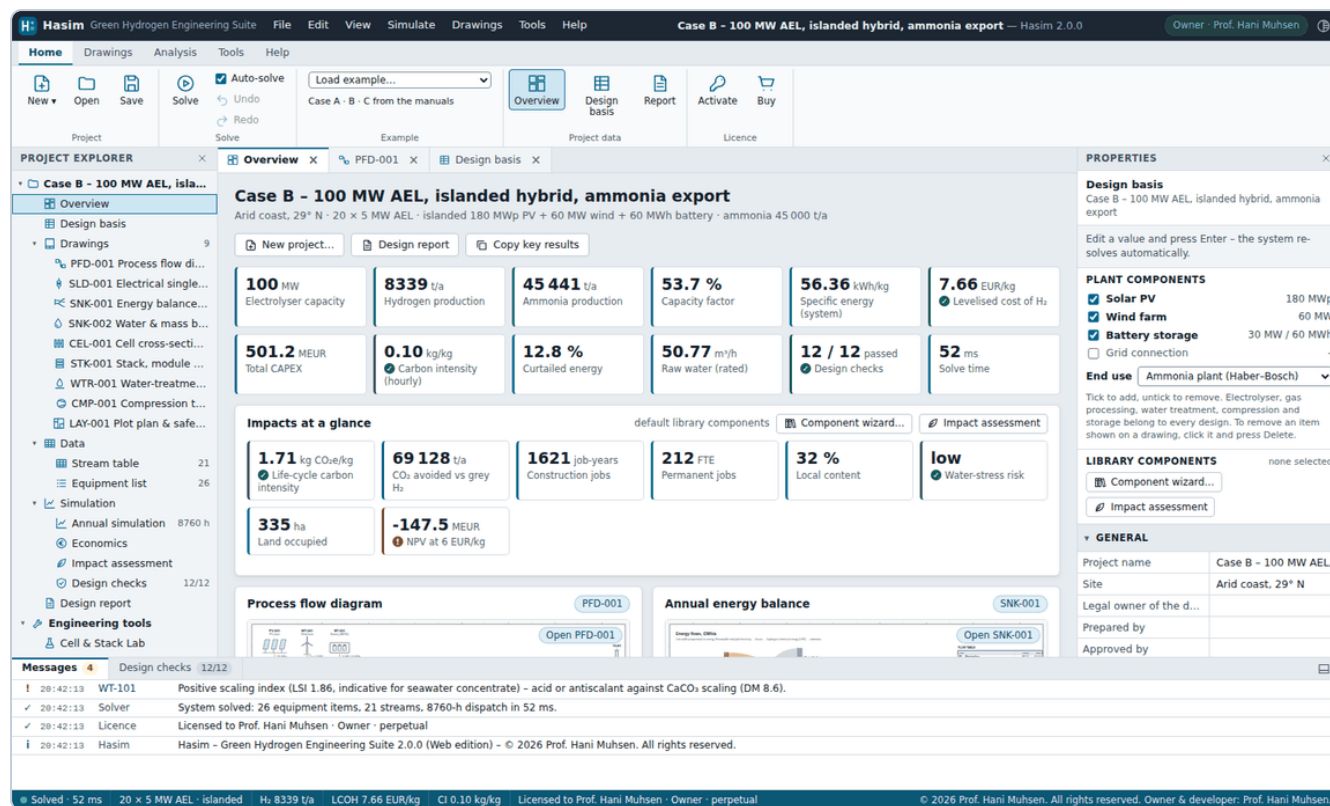


Figure 3.1 – The Hasim workstation with the project overview of Case B.

Area	Purpose
Title bar	Hasim logo, menus (File, Edit, View, Simulate, Drawings, Tools, Help), project name with “●” for unsaved changes, licence badge, Buy button and theme switch.
Ribbon	Tabs Home, Drawings, Analysis, Tools and Help with the most used commands, grouped as in engineering software.
Project explorer	Tree of all project documents – overview, design basis, the nine drawings, stream table, equipment list, annual simulation, economics, impact assessment, design checks, report – and the engineering tools.
Documents	Open documents as tabs. Drawings open in a zoom/pan viewer; tables and charts scroll.
Properties	The design basis as an editable property grid, or the datasheet of the selected equipment or stream.
Messages	Solver messages, warnings and input changes; a second tab lists the design checks.
Status bar	Solver state and time, plant summary, hydrogen production, LCOH, carbon intensity, licence and copyright.

3.1 Menus and ribbon

The **File** menu creates new projects with the **New project (wizard)** or from the examples, opens and saves **.hasim** project files, exports the stream table, equipment datasheets, hourly results and the design report, and prints the active document (Figure 3.2). **Edit** provides undo/redo of input changes; **View** toggles the panels and the theme; **Simulate** solves, controls auto-solve and opens the annual simulation, economics, impact assessment, checks and report; **Drawings** opens the drawings; **Tools** opens the engineering tools, the Component library and the component wizard of the plant; **Help** contains the user guide, licence functions and About.

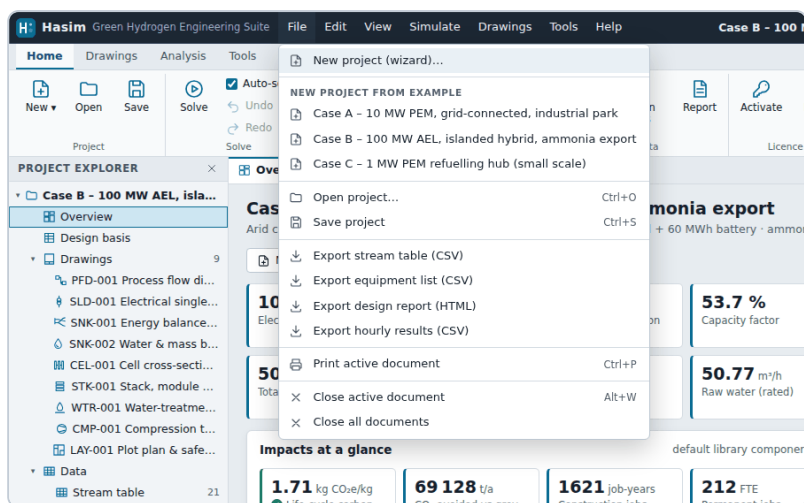


Figure 3.2 – File menu.

3.2 Project explorer and documents

Click an entry of the project explorer to open it as a tab. Tabs keep their state (zoom, scroll position) while you switch between them. Close a tab with its × button, the middle mouse button or **Ctrl+W**. The design-checks entry shows how many checks pass and turns amber when a check needs review.

3.3 Properties panel

Without a selection, the properties panel shows the **Plant components** (section 4.6), the **Library components** (section 4.7) and the design basis grouped into General, Electrolyser, Power supply, Site and weather, Grid connection, Water, Storage and delivery, Economics and Impact assessment. Click a group header to collapse it. Type a value and press **Enter**; values outside the allowed range are marked red and not applied. Inputs that differ from the example are marked with a blue bar. When you click equipment on a drawing or in the equipment list, the panel shows its datasheet with the manual reference of every value (Figure 3.3); a stream shows its conditions and composition.

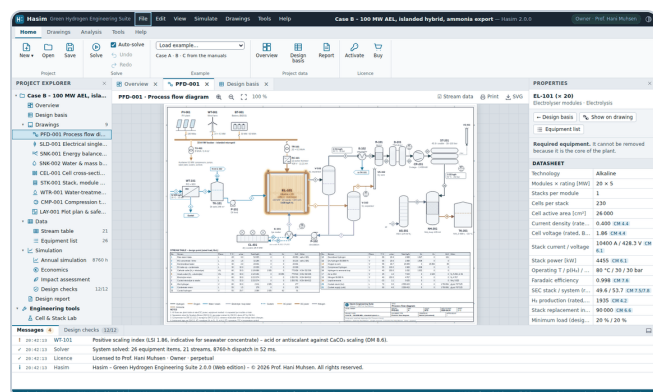


Figure 3.3 – EL-101 selected on the PFD; its datasheet appears in the properties panel.

PROPERTIES	
Stream 13	
Dry hydrogen 99.9989 %	
← Design basis	Show on PFD
Stream table	
CONDITIONS	
From	D-101
To	CP-201
Phase	vapour
Temperature	55 °C
Pressure	29 bar(a)
Mass flow	1908 kg/h
Volume flow	905 m ³ /h
Note	frost point – 70 °C
COMPOSITION (KG/H)	
H2	1908 (100.00 wt-%)

Figure 3.4 – Stream 13 (dry hydrogen) selected: conditions and composition.

3.4 Keyboard shortcuts

Keys	Action
F5	Solve the design basis
Ctrl+S / Ctrl+O	Save / open project
Ctrl+Z / Ctrl+Y	Undo / redo input changes
Ctrl+P	Print the active document
Ctrl+W	Close the active document
Ctrl+PageDown / PageUp	Next / previous document
Alt+1 / Alt+2 / Alt+3	Show or hide explorer / properties / messages
F1	User guide
Drawing viewer	Mouse wheel (hover or Ctrl) zooms, drag pans, double-click fits the sheet, click selects

3.5 Themes and small screens

Hasim follows the light or dark theme of the operating system; the theme button in the title bar switches manually. In the dark theme drawings are shown CAD-style on a dark sheet; exports always use the colours on screen. On narrow screens the explorer and the properties open as drawers from the title bar.

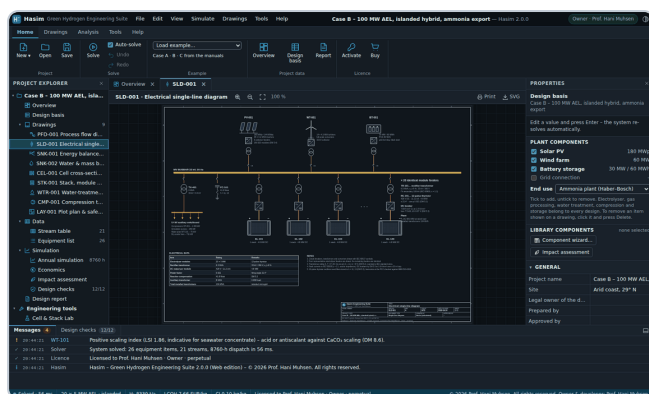


Figure 3.5 – Dark theme with the single-line diagram.

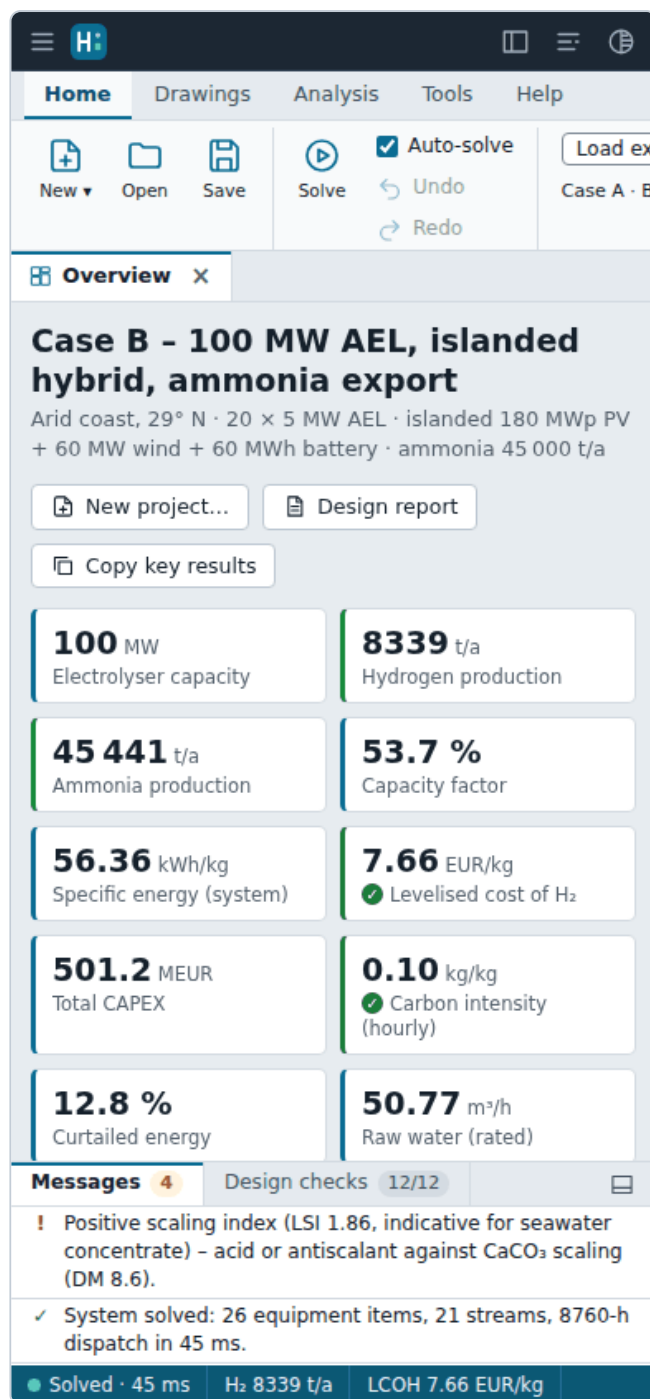


Figure 3.6 – Layout on a phone.

4 Projects and the design basis

4.1 Example projects

Three complete examples from the manuals are built in. Load them from **Home ► Example** or **File ► New project from example**.

	Case A	Case B	Case C
Plant	10 MW PEM, grid-connected, industrial park	100 MW AEL, islanded hybrid, ammonia export	1 MW PEM refuelling hub (small scale)
Electrolyser	2 × 5 MW PEM	20 × 5 MW AEL	1 × 1.25 MW PEM
Power supply	Grid + PPA (30 MWp PV, 15 MW wind)	Islanded: 180 MWp PV, 60 MW wind, 30 MW/60 MWh battery	Grid + PPA (2 MWp PV)
Hydrogen [t/a]	1 059	8 339	155
Capacity factor	68.3 %	53.7 %	80.1 %
System SEC [kWh/kg]	56.4	56.4	56.6
LCOH [EUR/kg]	10.13	7.66	15.19
CAPEX [MEUR]	35.2	501.2	8.8
Carbon intensity [kg CO ₂ e/kg]	3.13	0.10	17.81
Equipment items / streams	24 / 19	26 / 21	22 / 18
Design checks passed	15 of 16	12 of 12	15 of 16

4.2 Editing the design basis

Edit values in the properties panel or open the **Design basis** document, which shows all groups side by side (Figure 4.1). With **Auto-solve** on (Home ribbon), every accepted change re-solves the complete system; with auto-solve off, press **F5**. Changing the technology loads its minimum-load preset; switching the product to ammonia selects ammonia delivery. **Undo/Redo** (Ctrl+Z/Ctrl+Y) step through up to 100 changes. The message log records every change.

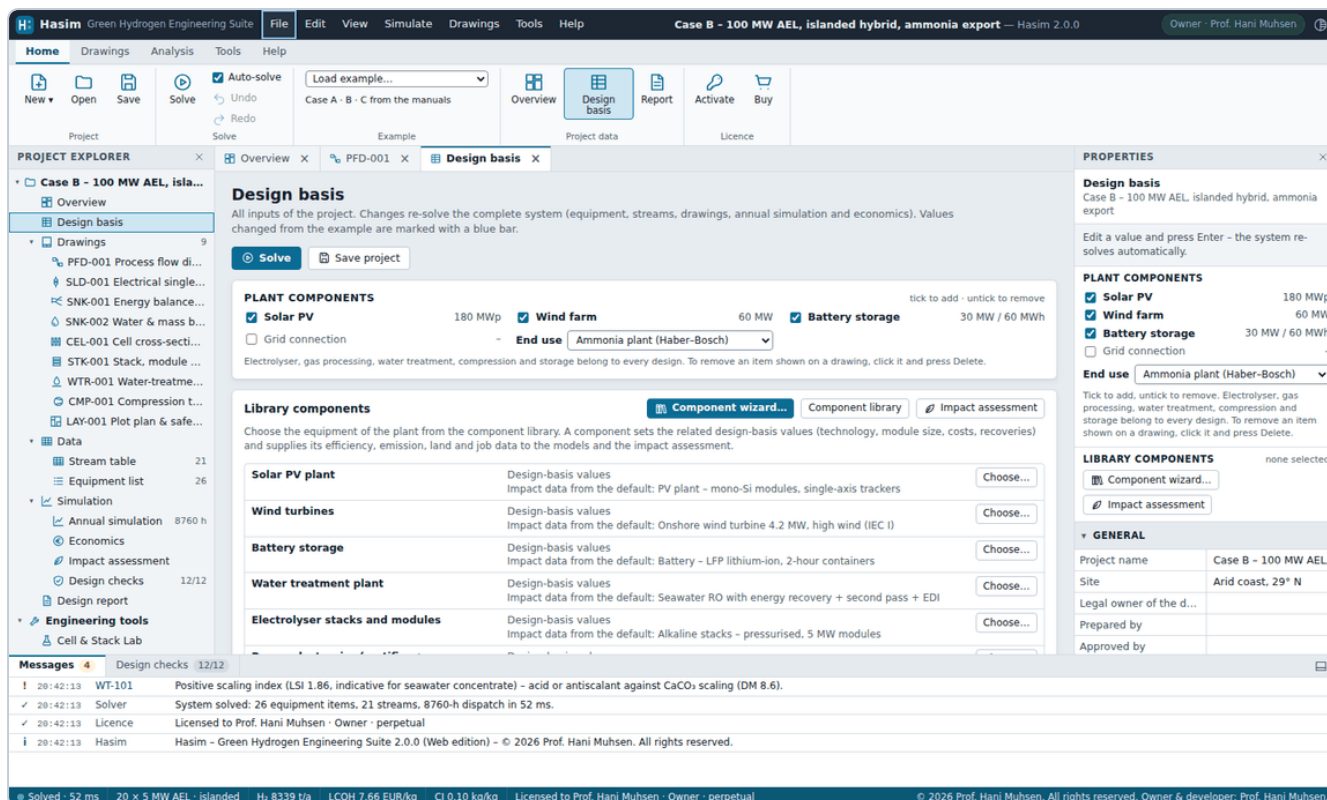


Figure 4.1 – The design-basis document.

4.3 Design-basis reference

The table lists every input with its unit, allowed range and meaning.

Input	Unit	Range / options	Meaning
General			
Project name			Project name shown in the title bar, drawing title blocks and the report.
Site			Site description printed on every drawing.
Legal owner of the documents			Legal owner of the documents – printed in the title block of every drawing (ISO 7200).
Prepared by			Person who prepared the drawings and the report (title block).
Approved by			Person who approved the drawings and the report (title block).
Product		Hydrogen / Ammonia	Hydrogen delivered as gas, or ammonia produced in a Haber–Bosch loop (adds ASU, synthesis loop and NH ₃ tank).
Demand	kg/d (H ₂) or t/a (NH ₃)	1 ... 10 000 000	Hydrogen demand in kg/d, or annual ammonia production in t/a; drives the sizing chain and the demand check.
Plant availability	–	0.50 ... 1	Fraction of the year the plant is available (DM 1.1). The annual simulation takes the modules out of service for maintenance one at a time, in six blocks per year, so that the time average equals this value.
Electrolyser			
Technology		PEM / Alkaline (AEL) / AEM / Solid oxide (SOEC)	Electrolysis technology; loads the cell, stack and operating-condition presets.

Input	Unit	Range / options	Meaning
Number of modules	–	1 ... 200	Number of identical electrolyser modules, each with its own transformer and rectifier.
Module rating	MW	0.10 ... 50	Rated DC power of one module.
Minimum load	–	0 ... 0.90	Minimum stable load of the dispatch. A lower value than the load at which hydrogen in oxygen reaches the 2 % trip (CM 7.7) is raised to that load; the H ₂ -in-O ₂ check reports the load actually used.
Design full-load hours	h/a	500 ... 8 760	Full-load hours of operation used by the sizing chain (DM 1.2), without downtime; the annual simulation gives CF × 8760 ÷ availability.
System CAPEX	EUR/kW	100 ... 10 000	Installed electrolyser system cost per kW.
Stack share of CAPEX	–	0 ... 0.80	Share of the electrolyser CAPEX spent on each stack replacement.
SOEC heat for steam raising		Electric boiler and heaters / External heat (e.g. waste heat)	SOEC only: heat for raising the steam and for operation below the thermoneutral voltage – electric boiler and heaters (counted in the SEC) or external heat such as waste heat (paid at the heat price).
Price of external heat (SOEC)	EUR/MWh	0 ... 500	Price of external heat for an SOEC.
Power supply			
Supply concept		Grid-connected + PPA / Islanded (off-grid)	Grid-connected with a renewable PPA, or an islanded PV–wind–battery microgrid (generation CAPEX then belongs to the project).
PV capacity	MWp	0 ... 100 000	PV DC capacity.
Wind capacity	MW	0 ... 100 000	Wind-farm capacity.
Battery power	MW	0 ... 10 000	Battery inverter power.
Battery energy	MWh	0 ... 100 000	Battery usable energy.
Renewable share of supply (sizing chain only)	–	0 ... 1	Renewable share of the purchased electricity assumed by the sizing chain; the annual simulation computes the grid share hour by hour.
Electricity price	EUR/MWh	0 ... 1 000	Price of purchased grid/PPA electricity.
Curtailement allowance (sizing chain only)	–	0 ... 0.90	Curtailement allowance of the sizing chain.
Surplus renewable energy (grid-connected)		Curtailed / not bought / Sold to the grid (pay-as-produced PPA)	Grid-connected plants: PPA electricity the electrolyser cannot use is curtailed (not bought), or sold to the grid at the export price (pay-as-produced PPA). Only a sold surplus can be matched with grid imports in the monthly correlation of the RFNBO rules (until the end of 2029).
Price of sold surplus	EUR/MWh	-100 ... 500	Price received for sold surplus electricity.
PV CAPEX (islanded)	EUR/kWp	0 ... 5 000	PV cost for islanded plants.
Wind CAPEX (islanded)	EUR/kW	0 ... 6 000	Wind cost for islanded plants.
Battery CAPEX	EUR/kWh	0 ... 3 000	Battery cost per kWh.
Site and weather			
Latitude	°	-66 ... 66	Latitude for the solar geometry of the synthetic year.

Input	Unit	Range / options	Meaning
Annual GHI	kWh/m ²	500 ... 2 800	Annual global horizontal irradiation – scales the solar year.
Mean wind speed at hub	m/s	2 ... 14	Mean wind speed at hub height (Weibull k = 2).
Mean ambient temperature	°C	-20 ... 40	Mean ambient temperature (PV temperature losses).
Cloudiness index	–	0 ... 1	Day-to-day variability of the solar resource, 0 clear ... 1 overcast.
Weather-year seed	–	1 ... 9 999	Random seed of the synthetic weather year – change it to test other years.
Design ambient (cooling)	°C	-10 ... 55	Design ambient temperature for the dry coolers.
Grid connection and electrical			
Grid import limit	MW	0 ... 10 000	Maximum grid import.
Grid use		Bridge to minimum load / Always full load / Fill to target load	Bridge: the grid keeps the electrolyser at minimum load when renewables are short; target: tops up to the target load; full: always rated load.
Target load (target mode)	–	0 ... 1	Target load for the “target” mode.
Grid carbon intensity	g/kWh	0 ... 1 200	Carbon intensity of grid electricity.
MV voltage	kV	6 ... 132	Medium-voltage level of the plant substation.
Grid fault level	MVA	0 ... 100 000	Short-circuit power of the grid (fault levels).
Auxiliary backup supply	MW	0 ... 1 000	Backup supply for the auxiliaries (e.g. of an islanded plant).
Auxiliary load (× rating)	–	0 ... 0.30	Auxiliary load as a fraction of the electrolyser rating.
Water			
Raw-water source		Municipal / fresh / Seawater	Raw-water source.
Raw-water TDS	g/L	0.01 ... 60	Total dissolved solids of the raw water.
RO recovery, pass 1	–	0.20 ... 0.95	Recovery of the first RO pass.
Recovery, pass 2 / EDI	–	0.50 ... 1	Recovery of the second pass / EDI polishing.
DI water consumption	L/kg	9 ... 20	Demineralised water per kg of hydrogen, including losses (DM 6.1).
Calcium	mg/L	0 ... 2 000	Calcium in the raw water (scaling).
Sulphate	mg/L	0 ... 6 000	Sulphate in the raw water (gypsum).
Alkalinity	mg/L	0 ... 2 000	Alkalinity (as CaCO ₃).
Raw-water pH	–	5 ... 10	Raw-water pH.
Water temperature	°C	1 ... 40	Raw-water temperature.
Raw-water price	EUR/m ³	0 ... 20	Price of raw water.
Storage and delivery			
Storage pressure	bar	10 ... 900	Maximum storage pressure (compressor discharge).
Minimum storage pressure	bar	1 ... 500	Minimum storage pressure (usable inventory).

Input	Unit	Range / options	Meaning
Compressor discharge temperature limit	°C	120 °C – API 618 6th ed., hydrogen, non-lubricated / 135 °C – API 618 6th ed., general / 150 °C – API 618 5th ed.	Highest discharge temperature of a compressor stage; sets the pressure ratio per stage and the number of stages (API 618: 120 °C for hydrogen-rich gas in non-lubricated machines).
Storage sizing		Days of full-load flow / From hourly inventory	Size the buffer as days of full-load flow, or from the 8760-hour inventory.
Buffer	d	0.05 ... 30	Buffer in days of full-load flow.
Offtake averaging window	h	0 ... 720	Offtake averaging window: 0 = constant offtake, 168 = weekly flexibility.
Offtake turndown	–	0 ... 1	Minimum offtake as a fraction of the mean offtake.
Storage CAPEX	EUR/kg	0 ... 5 000	Storage cost per kg of usable hydrogen.
Delivery		Pipeline / Ammonia plant / Refuelling station / At plant fence	Delivery route: pipeline, ammonia plant, refuelling station or plant fence.
Pipeline length	km	0 ... 2 000	Length of the product pipeline.
Delivery pressure	bar	1 ... 200	Delivery pressure at the customer.
Trailer loading	kg/d	0 ... 100 000	Hydrogen filled into tube trailers per day.
Economics			
WACC	–	0 ... 0.30	Weighted average cost of capital.
Economic life	a	5 ... 50	Economic life.
Other direct costs incl. battery and storage (× electrolyser)	–	0 ... 5	Other direct costs relative to the electrolyser CAPEX – balance of plant, civil works and EPC; the battery and the hydrogen storage are paid from this budget.
Fixed OPEX (share of CAPEX)	1/a	0 ... 0.20	Fixed O&M per year as a share of CAPEX.
Impact assessment			
Labour-productivity region		OECD (jobs × 1) / Middle East (jobs × 2) / Latin America (jobs × 2.3) / China (jobs × 1.3) / India (jobs × 1.4) / Developing Asia (jobs × 1.5) / Africa (jobs × 6.8) / Transition economies (jobs × 1.7)	Labour-productivity region; multiplies all employment factors of the impact assessment (section 9.3).
Local share of equipment supply	–	0 ... 1	Share of the equipment supply manufactured in the country (local content, local manufacturing jobs).
Local share of installation & civil works	–	0 ... 1	Share of installation and civil works carried out by local companies (local content, local construction jobs).
Local share of engineering & services	–	0 ... 1	Share of engineering, services, indirect and owner's costs spent in the country.
Baseline water stress (WRI Aqueduct)		Low (< 10 %) / Low-medium (10–20 %) / Medium-high (20–40 %) / High (40–80 %) / Extremely high (> 80 %)	Baseline water stress of the site (WRI Aqueduct 4.0); with the water source it gives the water-stress risk.
Distance storage – site boundary	m	0 ... 100 000	Distance from the storage yard to the site boundary, compared with the hazard distances of a storage leak.
Distance to nearest residents	m	0 ... 1 000 000	Distance from the storage yard to the nearest residents (exposure of neighbours).

Input	Unit	Range / options	Meaning
Hydrogen value (for NPV / IRR)	EUR/kg	0 ... 50	Value of the hydrogen for the NPV, IRR and payback of the impact assessment.

4.4 Saving and opening projects

File ► Save project (Ctrl+S) writes a `.hasim` file (JSON) containing the design basis and project information; **File ► Open project** reads it back and solves it. Hasim also keeps the current project in the browser storage of the edition you use, so it reappears at the next start. Design-basis JSON files of Hasim 1.x can be opened as well.

4.5 Creating a project from scratch

File ► New project (wizard) (also Home ► New ▼ and the **New project** button of the Overview) creates a plant from scratch in six steps: **Product** (hydrogen or ammonia, demand, end use), **Power supply** (grid-connected with PPA, or islanded; solar PV, wind, battery; grid use and prices), **Site & water** (climate preset or own values, municipal water or seawater), **Electrolyser** (technology, minimum load, CAPEX), **Storage & economics** and **Review**. The minimum load is preset to the lowest load, in steps of 5 %, at which hydrogen in oxygen stays below the 2 % trip; a lower value would be raised to the trip limit in the simulation. With **Size the plant for me**, Hasim runs the complete 8760-hour model repeatedly and sizes the electrolyser modules and the power supply so that the yearly production after gas processing meets the demand – whole modules first, then the solar and wind volumes close the remaining gap; alternatively enter the sizes yourself. The review shows the expected production, capacity factor and an indicative LCOH before the project is created.

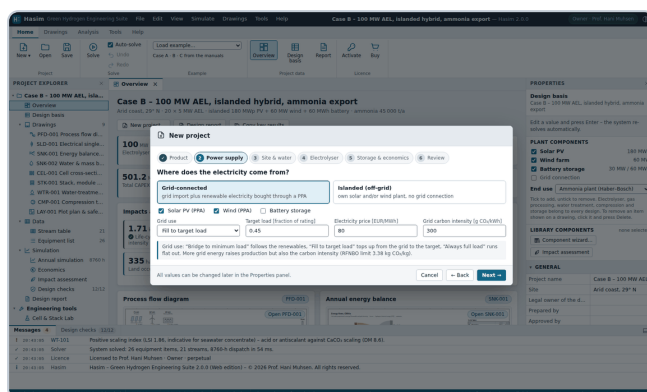


Figure 4.2 – New Project wizard, step 2: power supply.

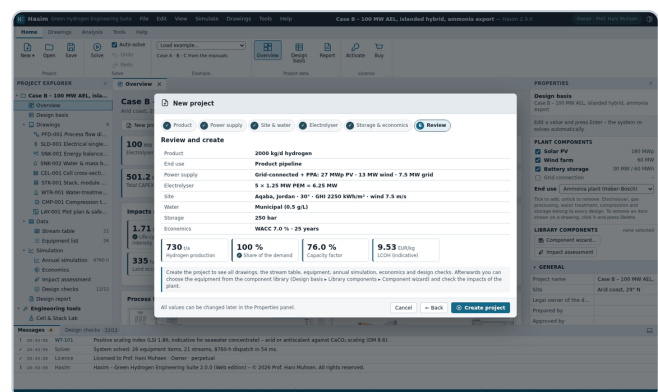


Figure 4.3 – Step 6: the plant sized to meet the demand.

A wizard project is an ordinary project. The blue change markers compare with the values chosen in the wizard, and **Edit ► Reset design basis to the wizard values** restores them. The Model-Building Examples document walks through a complete wizard project (“Start here”).

4.6 Adding and removing plant components

The **Plant components** list at the top of the Properties panel and of the Design basis document switches the optional equipment on and off: **Solar PV**, **Wind farm**, **Battery storage**, **Grid connection** and **Tube-trailer loading**, plus the **End use** (plant fence, product pipeline, refuelling station or ammonia plant). A removed component keeps its size, so ticking it again restores it; each change is one step of Undo. At least one power source must remain; removing the grid connection makes the plant islanded.

Equipment can also be removed where you see it: click it on a drawing or in the equipment list – its datasheet shows **Remove from plant** – or press **Delete** (Edit ► Remove selected equipment from the plant). Hasim asks for confirmation and re-solves the system. Equipment that every plant needs – electrolyser, gas processing, water treatment, compression and storage, and the electrical equipment that follows from the electrolyser rating – cannot be removed; its datasheet says why.

PROPERTIES

Design basis
Aqaba Green Hydrogen

Edit a value and press Enter – the system re-solves automatically.

PLANT COMPONENTS

☒ **Solar PV** 27 MWp

☒ **Wind farm** 13 MW

☐ Battery storage –

☒ **Grid connection** 7.5 MW import

☐ Tube-trailer loading –

End use Product pipeline

Tick to add, untick to remove. Electrolyser, gas processing, water treatment, compression and storage belong to every design. To remove an item shown on a drawing, click it and press Delete.

LIBRARY COMPONENTS none selected

Figure 4.4 – Plant components in the Properties panel.

am
100 %
Print
SVG

PROPERTIES

WT-001
Wind farm · Power supply

Design basis
Show on drawing

Equipment list

Remove from plant or press Delete

DATASHEET

Turbine	Onshore wind turbine 4.5 MW, large rotor for low wind (IEC III)
Turbines [MW]	3 × 4.3
Mean wind at hub [m/s]	7.5
Turbine class limit (a...)	7.5 IEC 61400-1
Weibull scale c [m/s]	8.46 CM 14.4
Net capacity factor	0.294 DM 4.2
Energy (Weibull) [G...]	33.49 DM 4.2
Simulated energy (ai...)	32.58 CM 14.3

Figure 4.5 – The wind farm WT-001 selected on the single-line diagram: its datasheet offers Remove from plant.

4.7 Library components of a plant

The equipment of a plant can be chosen from the **component library** (section 11.4). The **Library components** card of the Design basis document (Figure 4.6) – and its one-line version in the Properties panel – lists the equipment of the plant: solar PV, wind turbines, battery, water treatment, electrolyser stacks, power electronics, balance of plant, compressor and storage vessels and, depending on the end use, the pipeline, tube trailers, refuelling station or ammonia plant. Without a selection the plant keeps its design-basis values, and the impact assessment uses the **default component** of each type, shown in grey. The default follows the design basis: the stack and balance of plant of the chosen technology, the water plant for the salinity of the raw water, the turbine class for the site wind (IEC 61400-1), the vessel type for the storage pressure.

Library components		
Solar PV plant	PV plant – bifacial glass-glass modules, single-axis trackers POA gain 1.3 · 700 Wp · 580 EUR/kWp · 900 kg CO ₂ e/kWp	Hasim library Choose... x
Wind turbines	Onshore wind turbine 4.2 MW, high wind (IEC II) 4.2 MW · rated 13 m/s · 1250 EUR/kW · 448 kg CO ₂ e/kW	Hasim library Choose... x
Battery storage	Battery – LFP lithium-ion, 4-hour containers 4 h · RTE 0.870 · 300 EUR/kWh · 90.8 kg CO ₂ e/kWh	Hasim library Choose... x
Water treatment plant	Design-basis values Impact data from the default: Seawater RO with energy recovery + second pass + EDI	Choose...
Electrolyser stacks and modules	Alkaline stacks – large pressurised, 10 MW modules AEL · 10 MW modules · min. load 0.200 · 420 EUR/kW · 253.7 kg CO ₂ e/kW	Hasim library Choose... x
Power electronics (rectifier + transformer)	Design-basis values Impact data from the default: 12-pulse thyristor rectifier + transformer + harmonic filter	Choose...
Balance of plant (gas processing, circulation, cooling)	Design-basis values Impact data from the default: Balance of plant for alkaline – lye loop, separators, deoxo + dryer, dry coolers	Choose...
Hydrogen compressor	Design-basis values Impact data from the default: Reciprocating piston compressor, oil-free	Choose...
Hydrogen storage vessels	Design-basis values Impact data from the default: Steel pressure vessels (Type I), up to 300 bar	Choose...
Ammonia synthesis and air separation	Design-basis values Impact data from the default: Haber-Bosch loop + air separation (electrically driven)	Choose...

Figure 4.6 – Library components of Case B after choosing stacks, PV, wind turbines and battery; the other items use the defaults.

- 1 Press **Component wizard...** – on the card, in the Properties panel, in the Impact assessment or with **Tools ► Component wizard for the plant**. The wizard shows the equipment of the plant as steps; **Choose...** on the card opens it at one item.
- 2 Choose a component from the list on the left. **Hasim default – keep the design-basis values** removes a selection.
- 3 Set its size on the right – for example the number of modules, the PV or wind capacity or the battery power – and change parameters of the component if needed. An edited component is marked **modified**; the library itself stays unchanged. **Save as my component...** adds it to your library, **Reset to library values** undoes the edits.
- 4 **Effect on the model** at the bottom re-solves the complete plant – 8760-hour simulation included – with the components chosen so far: hydrogen, LCOH, investment, life-cycle carbon intensity, construction and permanent jobs, local content and checks passed (Figure 4.7).
- 5 **Apply components** applies all selections as one change; **Ctrl+Z** undoes it. The × on the card removes one selection – the design-basis values stay as they are.

Component wizard - Case B - 100 MW AEL, islanded hybrid, ammonia export

1 Solar PV plant 2 Wind turbines 3 Battery storage 4 Water treatment plant **5 Electrolyser stacks and modules** 6 Power electronics 7 Balance of plant

8 Hydrogen compressor 9 Hydrogen storage vessels 10 Ammonia synthesis and air separation

Plant equipment · choose a component

Electrolyser stacks and modules

Hasim default - keep the design-basis values
no library component; the design basis stays as it is

PEM stacks - 5 MW modules
PEM · 5 MW modules · min. load 0.100 · 675 EUR/kW · 137.5 kg CO₂e/kW
Hasim library · Proton-exchange membrane, Ir/Pt catalysts, 30 bar cathode

Alkaline stacks - pressurised, 5 MW modules
AEL · 5 MW modules · min. load 0.200 · 450 EUR/kW · 253.7 kg CO₂e/kW
Hasim library · Alkaline (30 wt-% KOH), nickel electrodes, 30 bar

Alkaline stacks - large pressurised, 10 MW modules
AEL · 10 MW modules · min. load 0.200 · 420 EUR/kW · 253.7 kg CO₂e/kW
Hasim library · Alkaline, large-area cells, 10 MW per module, 30 bar

AEM stacks - 1 MW modules
AEM · 1 MW modules · min. load 0.100 · 230 EUR/kW · 99.2 kg CO₂e/kW
Hasim library · Anion-exchange membrane, PGM-free catalysts, 10 bar

Solid-oxide stacks - 1 MW modules
SOEC · 1 MW modules · min. load 0.200 · 660 EUR/kW · 102.6

SIZE IN THIS PLANT
Number of modules [-]
= 20 × 5 MW = 100 MW

PEM stacks - 5 MW modules modified Save as my component...
Proton-exchange membrane, Ir/Pt catalysts, 30 bar cathode - edit a value to use it in this plant; the library stays unchanged.

TECHNICAL DATA USED BY THE MODEL

Technology

Module rating [MW]

Stack temperature [°C]

Cathode (hydrogen) pressure [bar]

Rated current density [A/cm²]

Area-specific resistance [Ω cm²]

Voltage degradation [μV/h]

Stack life [h]

Minimum load [-]

ECONOMIC DATA

Capital cost [EUR/kW]

Fixed O&M (share of CAPEX per year) [1/a]

Effect on the model - recalculated with the components chosen so far

8061 t/a Hydrogen

10.97 EUR/kg LCOH

637.7 MEUR Investment

0.02 kg CO₂e/kg GHG intensity (RFNBO)

1.90 kg CO₂e/kg Life-cycle CI

1941 job-years Construction jobs

211 FTE Permanent jobs

32.4 % Local content

13 / 13 Checks passed

Changes take effect when you press Apply - Ctrl+Z undoes them.

Cancel ← Back Next → **Apply components**

Figure 4.7 – Component wizard of Case B: PEM stacks with the minimum load edited to 0.35 (marked modified) and the effect on the whole plant.

A component sets the related design-basis values and supplies its data to the plant models:

Equipment	Design-basis values set	Model data used
Solar PV plant	PV CAPEX	Tracking gain, system losses, DC/AC ratio, temperature coefficient, module power, O&M
Wind turbines	Wind CAPEX	Power curve (cut-in, rated, cut-out wind speed), losses, turbine rating (number of turbines), O&M
Battery storage	Battery CAPEX; battery energy = power × storage duration	Round-trip efficiency, state-of-charge limits, O&M
Water treatment plant	Recoveries of both passes, water source; for a source of another kind also its typical salinity, composition and price	Maximum feed salinity (check)
Electrolyser stacks	Technology, minimum load, module rating and number of modules (same total rating), electrolyser CAPEX and stack share	Stack temperature and pressure, rated current density, ASR, voltage degradation, stack life
Power electronics	Electrolyser CAPEX and stack share	Rectifier efficiency and type (single-line diagram)
Balance of plant	Electrolyser CAPEX and stack share	Variable and fixed BoP consumption; technology (check)

Equipment	Design-basis values set	Model data used
Compressor	–	Isentropic efficiency, pressure ratio per stage, capacity per machine (number of machines), pressure rating (check)
Storage vessels	Storage CAPEX per kg	Pressure rating (check)
Pipeline, tube trailers, refuelling station, ammonia plant	–	Impact data; pipeline design pressure (check)

The electrolyser CAPEX of the design basis becomes the sum of stack, power electronics and balance of plant (EUR/kW), and the stack share the part of the stacks. When stacks of another technology are chosen, a balance of plant of the old technology is replaced by the matching one. The component checks of Chapter 10 verify the selection against the design, and the design report lists the components (section 8.4 of the report).

5 The system solver

Solving a design basis runs one deterministic sequence; typical solve times are 20–60 ms, including the 8760-hour simulation.

- 1 **Sizing chain** of the Design Manual: cell (CM 3–4), stack and module (CM 5–6), system efficiency (CM 7), renewables (DM 3–4), water (DM 6, 8), heat (DM 9), compression (DM 12), storage (DM 13), transport (DM 14), end use (DM 15–16), economics (DM 20), environment (CM 18) and safety (CM 19).
- 2 **Equipment**: every item receives a tag, quantity and datasheet – separators by Souders–Brown, coolers from duty and LMTD, deoxo and TSA dryers, compressors per stage, storage vessels, water plant, dry coolers, vents, end-use units and all electrical equipment.
- 3 **Streams**: the design-point mass and energy balance with numbered streams from raw water to product, including electrolyte loops, condensate, oxygen and coolant.
- 4 **Electrical one-line data**: rectifier type (12-pulse thyristor for AEL, IGBT active front-end otherwise), transformer ratings, DC bus, harmonics at the point of common coupling (IEEE 519-2022) and reactive power, short-circuit currents (IEC 60909), auxiliary transformer, PV/wind/battery/grid feeders.
- 5 **Annual simulation** of 8760 hours (Chapter 8) and the **annual energy and mass balances** that feed the Sankey diagrams; the energy balance closes to better than 10^{-6} GWh.
- 6 **Plot plan** with areas from the sizing chain, consequence distances of a storage leak (CM 19.1–19.8) and the separation distances of NFPA 2.
- 7 **Design checks**, messages and the **codes and standards screening** (Chapter 10).

Consistency

All documents of a project show the same solved state. When you change an input, every open document is marked as outdated and refreshed when you view it; the active document is refreshed immediately.

6 Engineering drawings

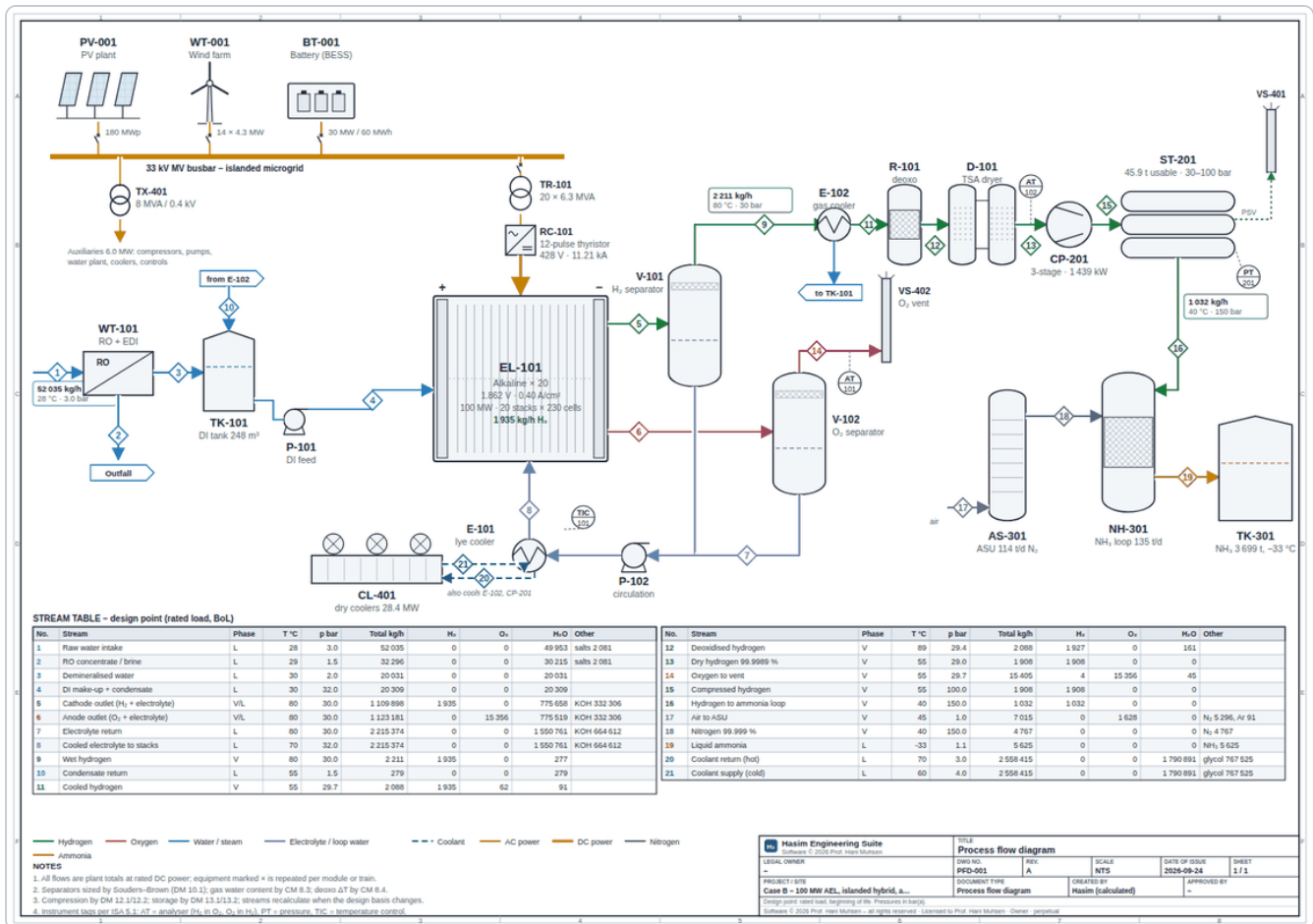
Drawings are generated on A3-proportion sheets with frame, zone markers and a title block with the data fields of ISO 7200: legal owner, identification number (drawing number), revision, date of issue, sheet, title, document type, creator and approver – the owner, creator and approver are taken from the design basis (group General) – plus scale, the basis of the drawing, the copyright line and the licence line. Electrical symbols follow IEC 60617. Line types and colours follow a consistent legend: hydrogen green, oxygen red, water blue, electrolyte violet, coolant teal, AC power amber, DC power brown.

6.1 Drawing viewer

Zoom with the mouse wheel while hovering (or with Ctrl), with the toolbar buttons or by double-clicking to fit. Drag to pan. Hover equipment or streams for a tooltip; click to show the datasheet in the properties panel. **SVG** exports the sheet as a vector drawing for CAD, reports and presentations; **Print** prints it. On the PFD, **Stream data** switches the stream call-outs on and off.

6.2 PFD-001 – Process flow diagram

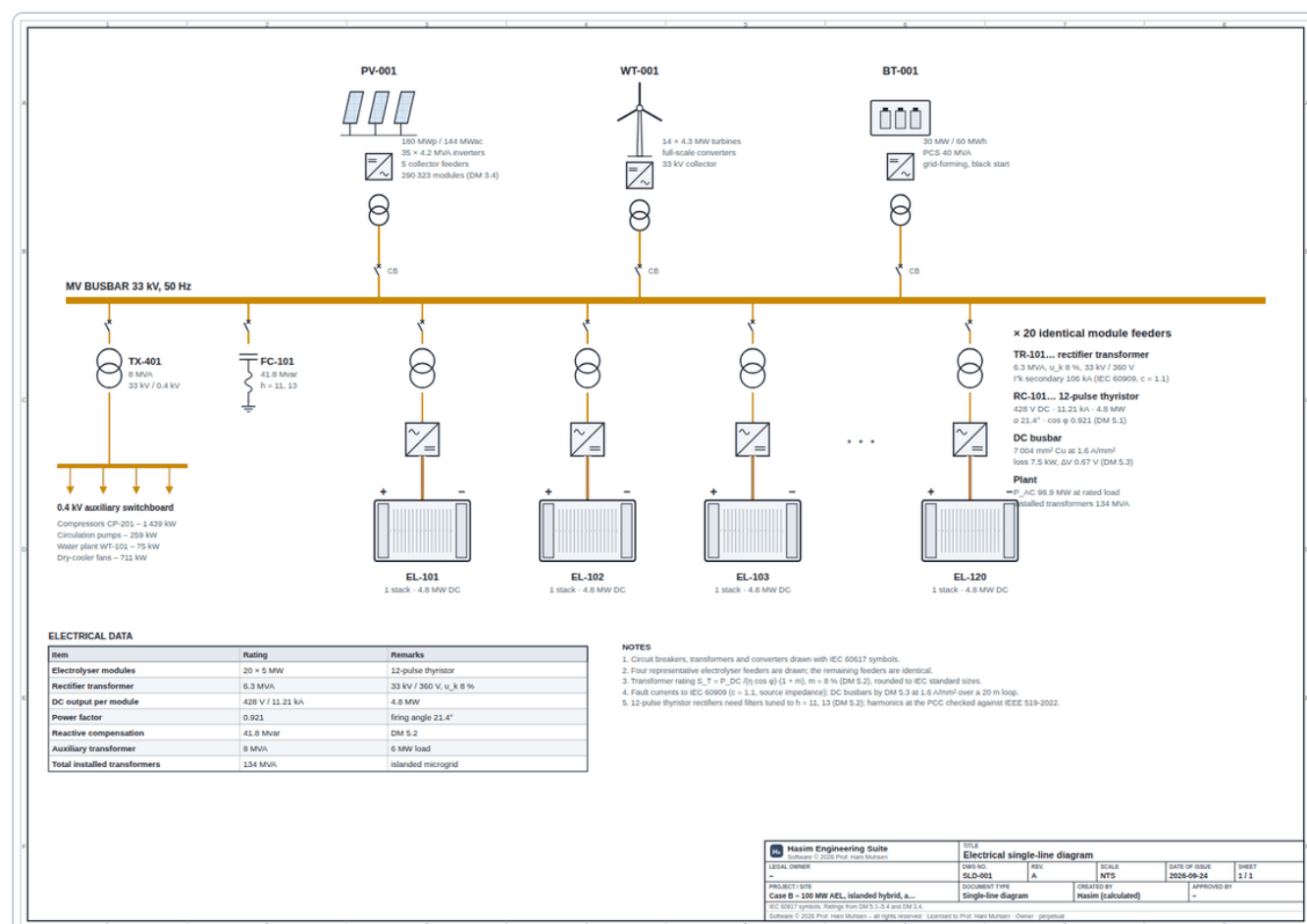
Power supply and MV bus, transformers and rectifiers, electrolyser modules with separators and electrolyte loop, gas processing (cooler, deoxo, TSA dryer), compression and storage, end use, water treatment and cooling, with numbered stream diamonds, instrument bubbles, stream call-outs and the stream table on the sheet.



PFD-001 for Case B.

6.3 SLD-001 – Electrical single-line diagram

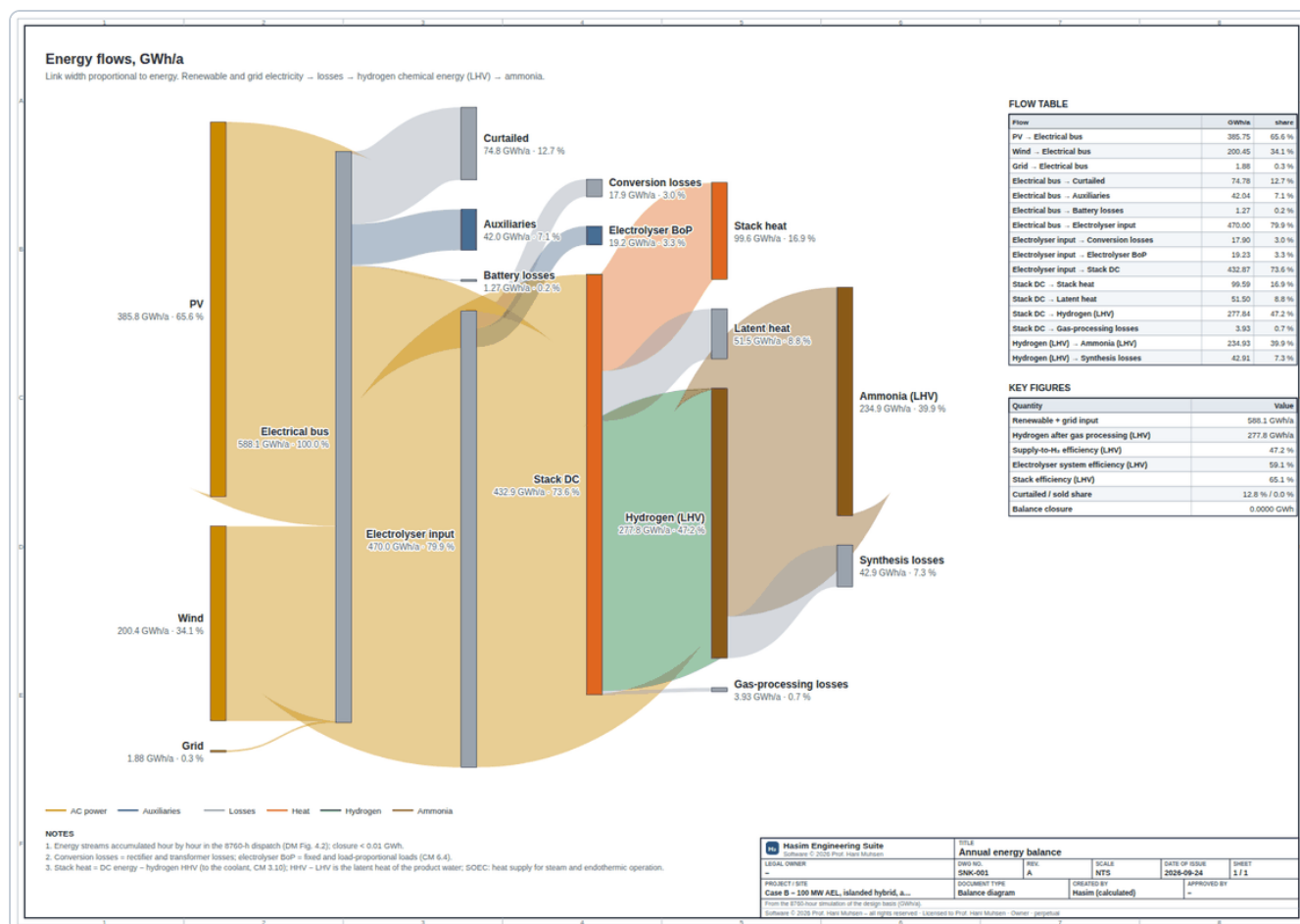
MV busbar with feeders for PV (inverter stations), wind turbines, battery PCS and grid; rectifier transformers with ratings and impedance; thyristor or IGBT rectifiers with DC data; harmonic filter/STATCOM for thyristor plants; auxiliary transformer and loads; ratings table.



SLD-001 for Case B.

6.4 SNK-001 – Annual energy balance (Sankey)

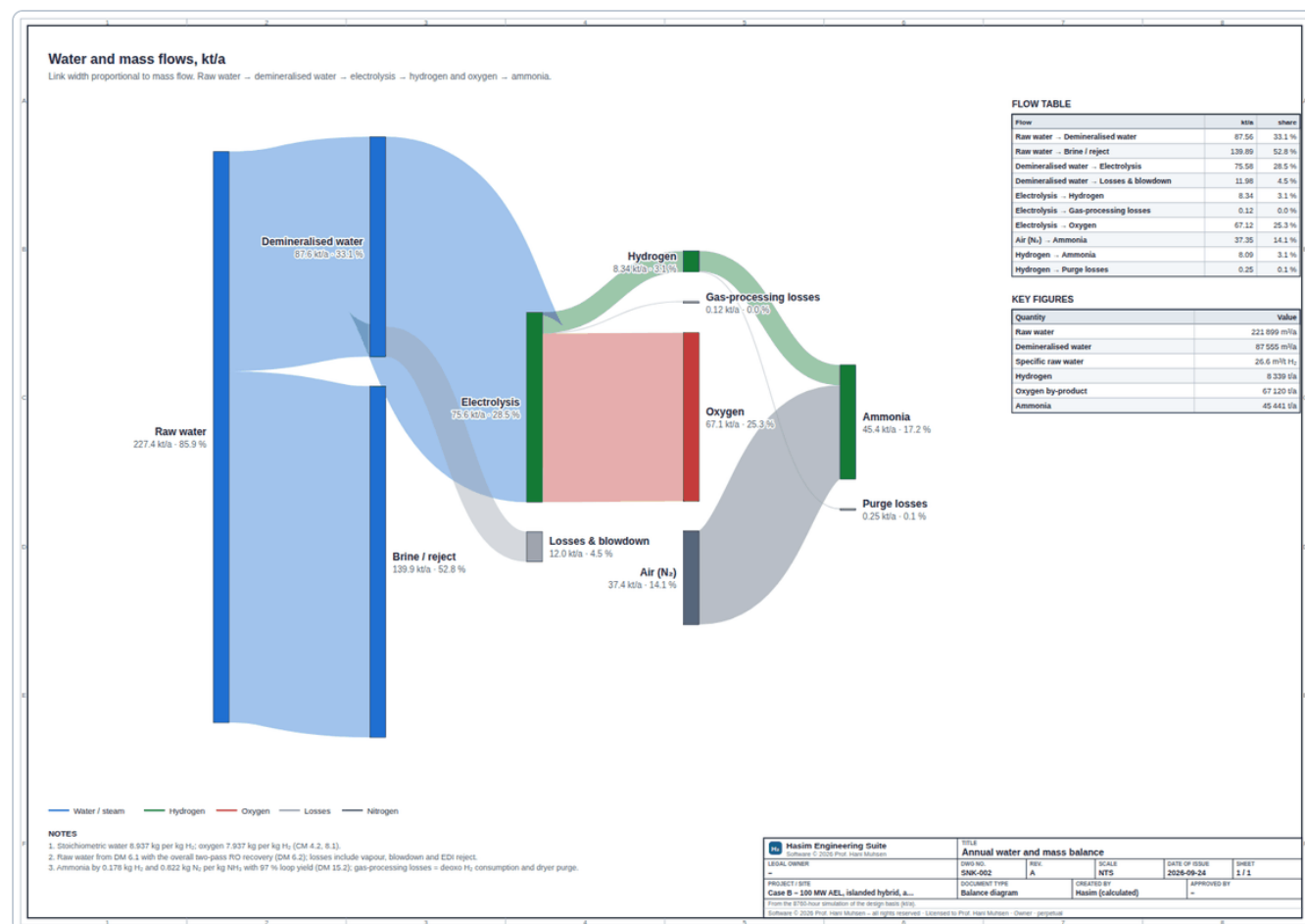
From PV, wind and grid electricity through curtailment, auxiliaries, battery and conversion losses to the stack DC power, stack heat and the chemical energy (LHV) of hydrogen – and ammonia for Power-to-X cases. Flow table and key figures on the sheet.



SNK-001 for Case B.

6.5 SNK-002 – Water and mass balance (Sankey)

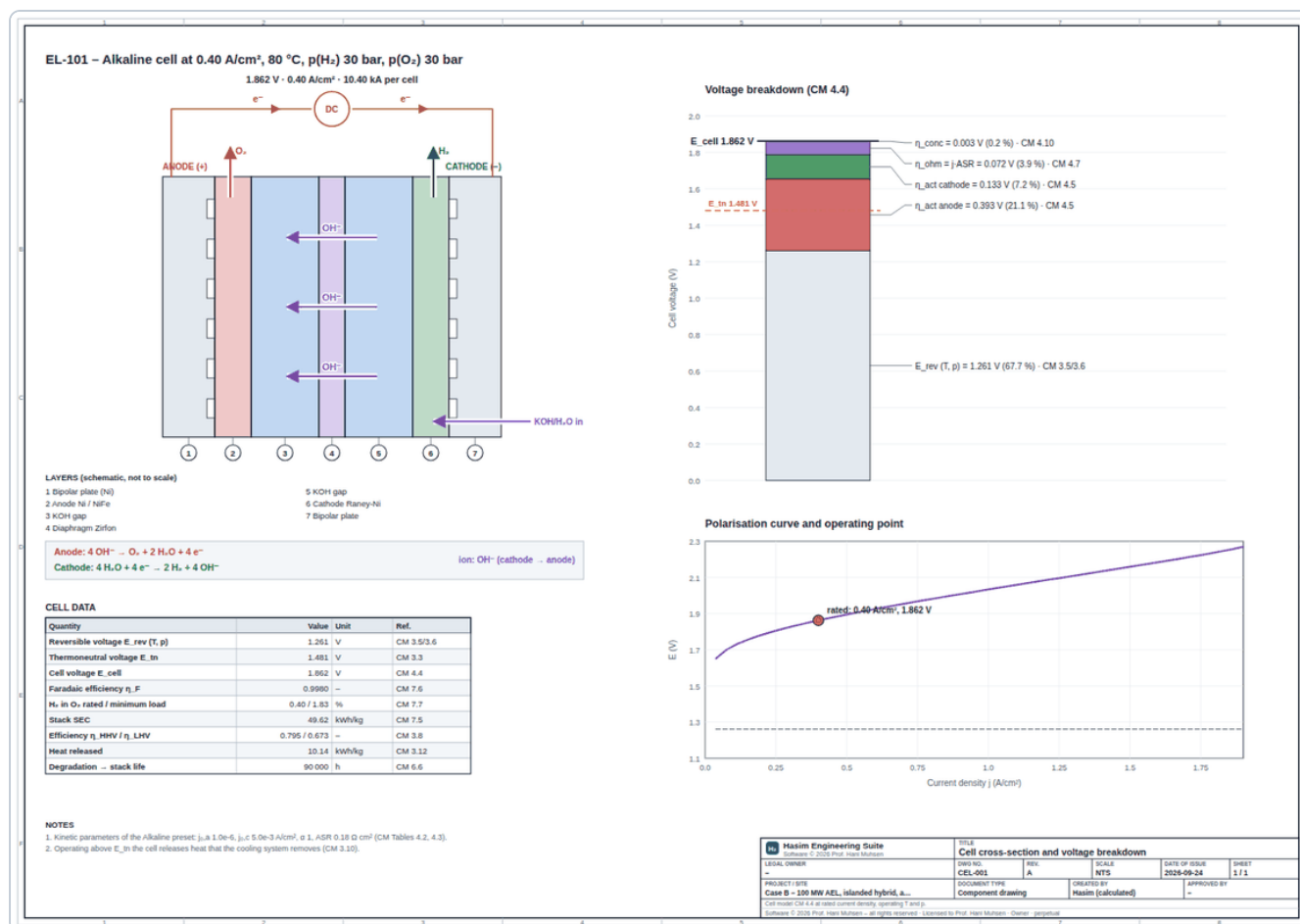
Raw water to demineralised water and brine, electrolysis to hydrogen and oxygen, losses and blowdown, and nitrogen/ammonia for ammonia plants.



SNK-002 for Case B.

6.6 CEL-001 – Cell cross-section and voltage breakdown

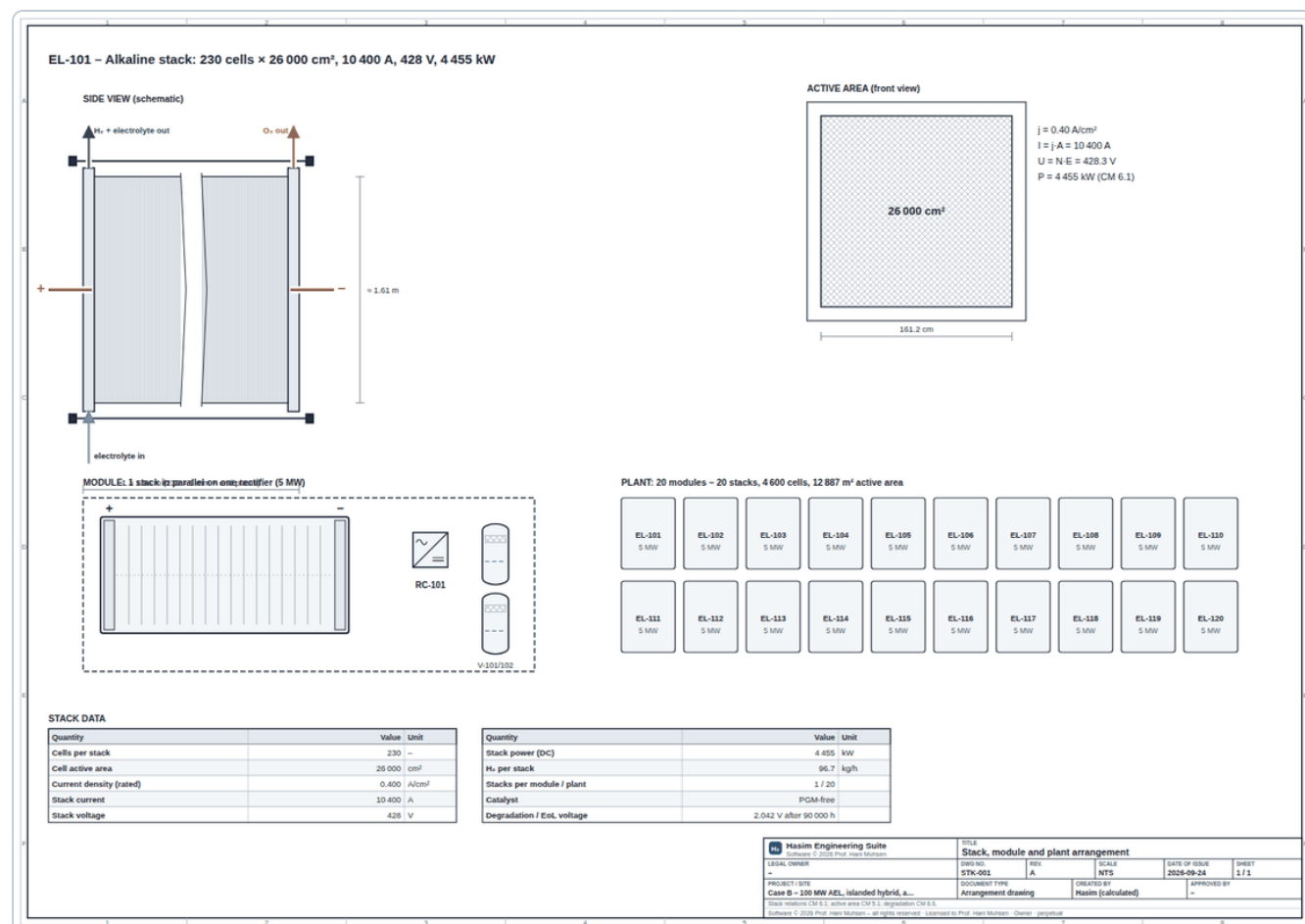
Cross-section of the technology's cell (electrodes, membrane or diaphragm, flows), the voltage breakdown at the operating point and the polarisation curve with the operating point.



CEL-001 for Case B.

6.7 STK-001 – Stack, module and plant arrangement

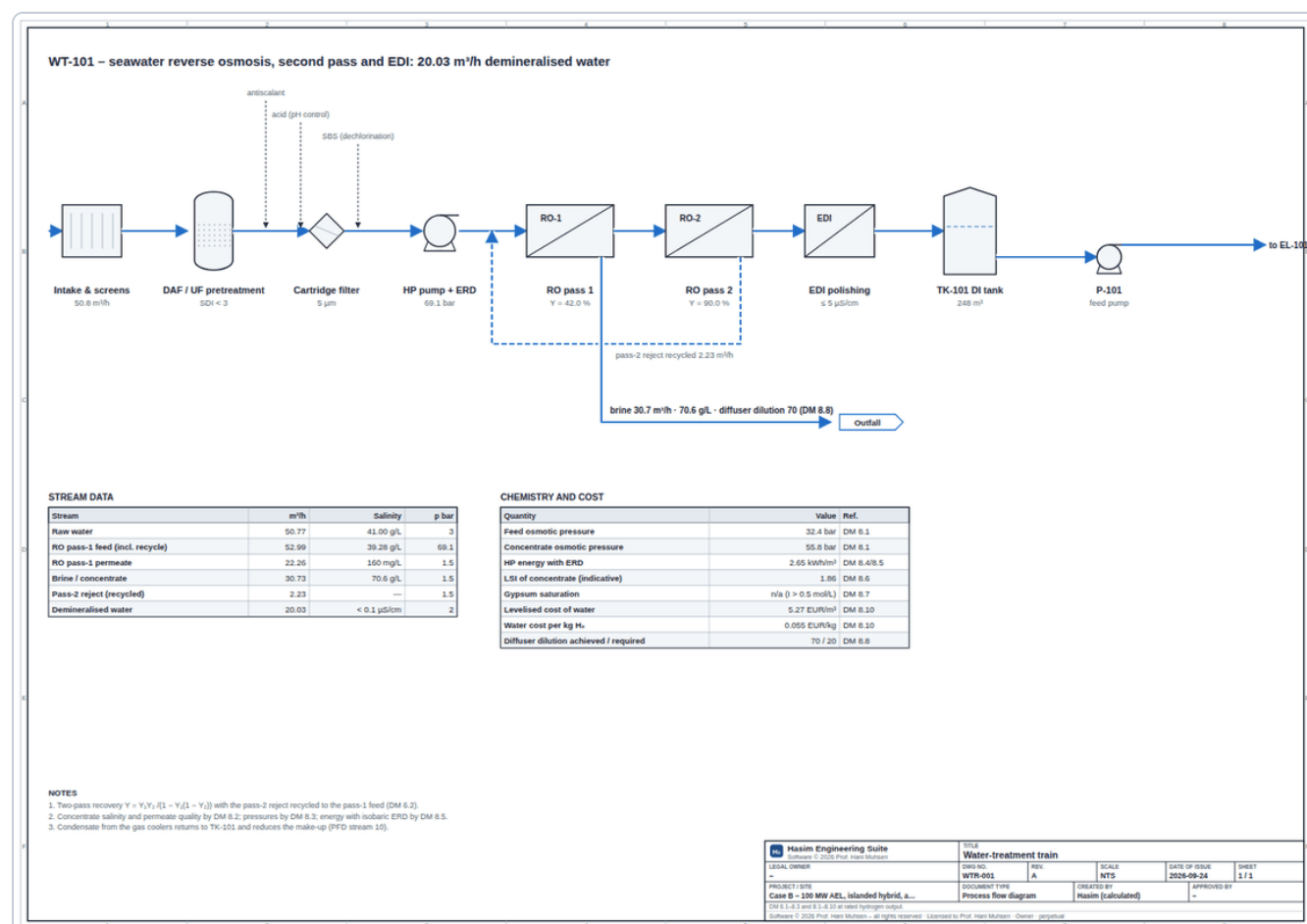
Cell stack with end plates and tie rods, module arrangement on the DC bus and the plant hierarchy from cell to plant with the relations of CM 6.1.



STK-001 for Case B.

6.8 WTR-001 – Water-treatment train

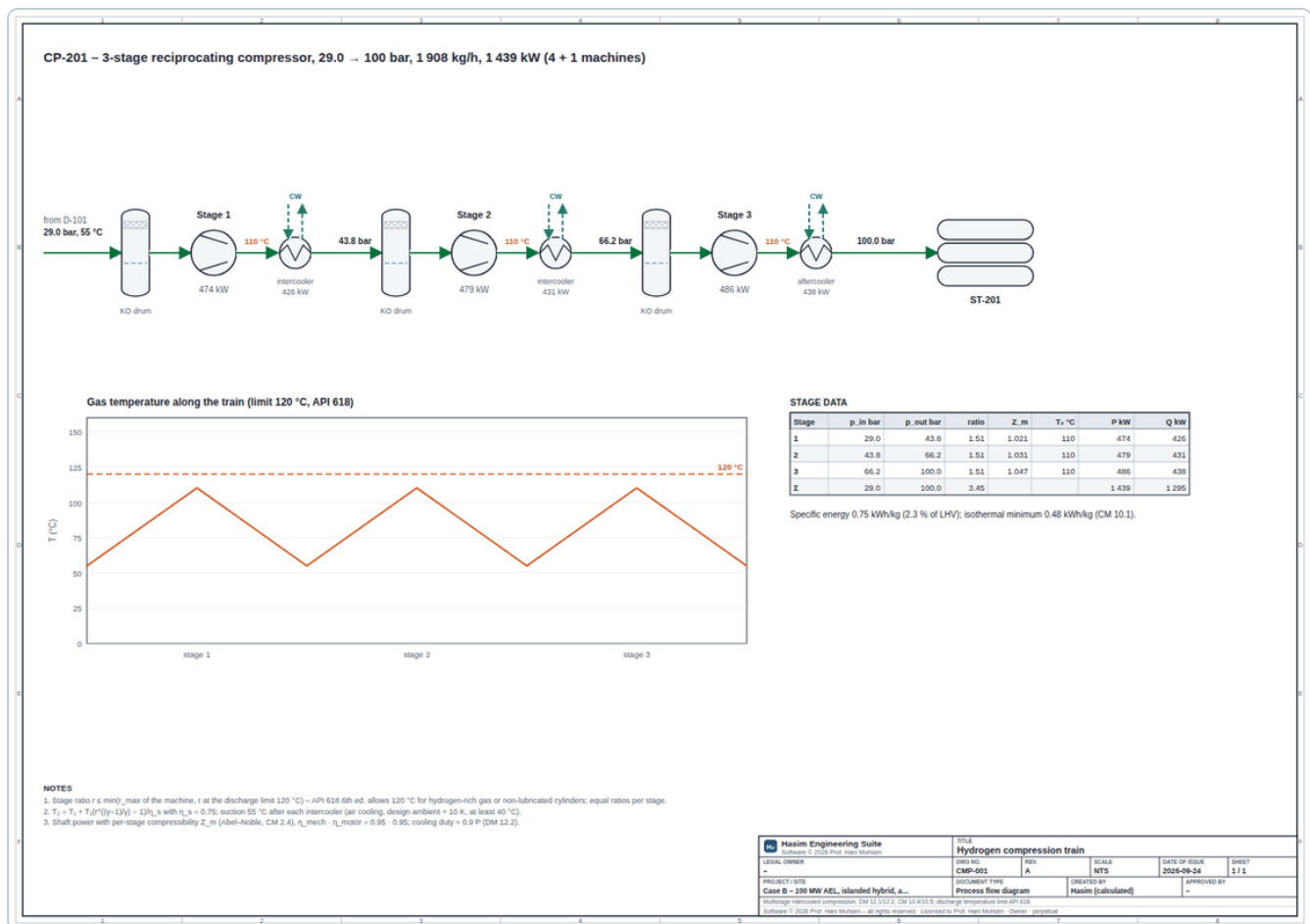
Intake, pre-treatment, one or two RO passes (with energy recovery for seawater), EDI polishing, DI tank, brine outfall with diffuser, flows and qualities; scaling indices and cost of water.



WTR-001 for Case B.

6.9 CMP-001 – Compression train

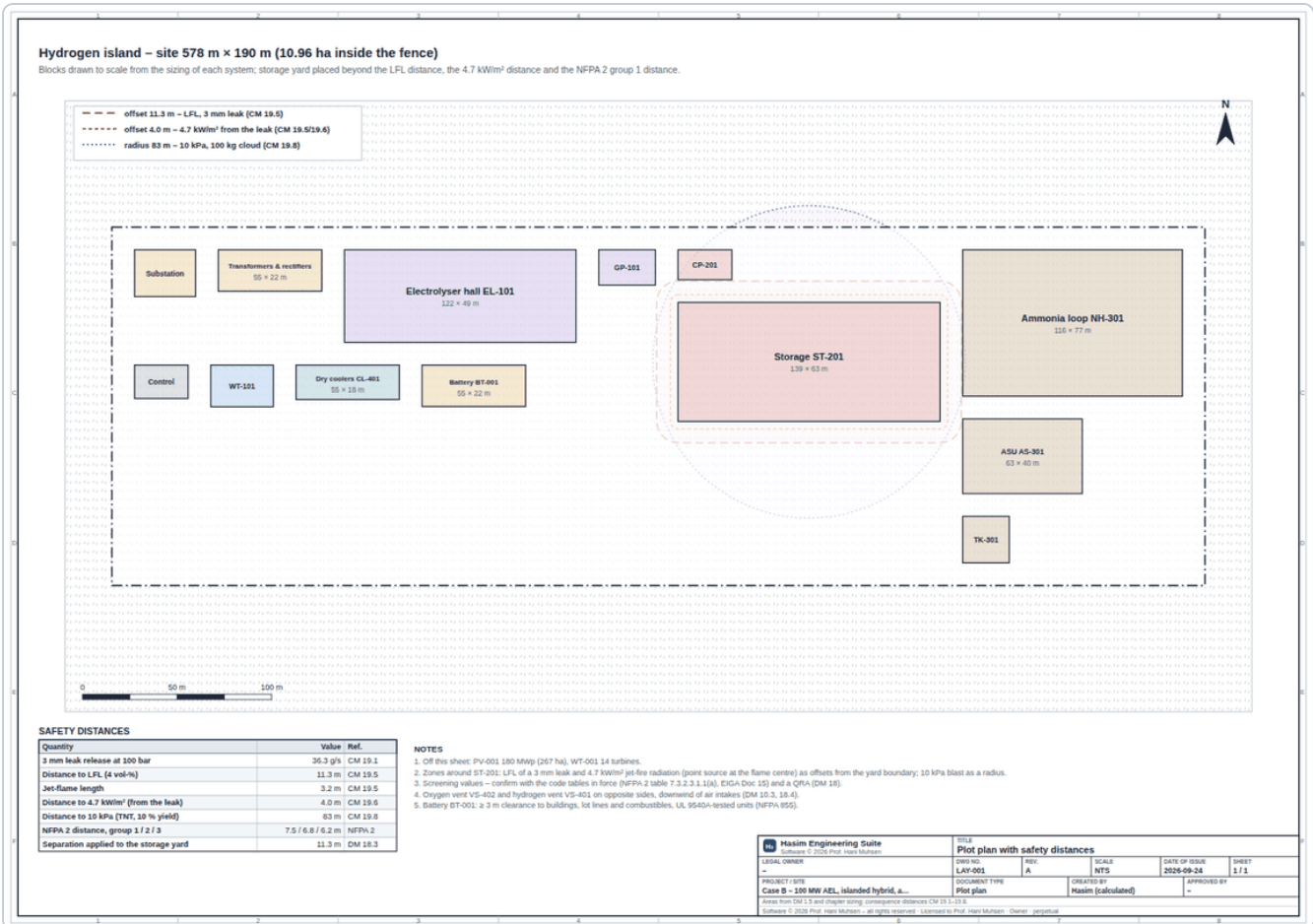
Multistage intercooled compression with stage pressures, temperatures and power, the p–h path and the specific compression energy.



CMP-001 for Case B.

6.10 LAY-001 – Plot plan with safety distances

Areas of substation, rectifiers, electrolyser hall, gas processing, water plant, coolers, control building, compressor and storage yard and end use, placed to scale with roads; zones for the LFL distance, the 4.7 kW/m^2 radiation distance and the 10 kPa overpressure radius of a storage leak; the storage yard is placed beyond the largest of these distances and of the NFPA 2 separation distance, and a battery keeps the NFPA 855 clearance.



LAY-001 for Case B.

7 Stream table and equipment list

7.1 Stream table

The stream table lists all numbered streams of the design point: from/to, phase, temperature, pressure, mass and volume flow, and the mass flow of each component (Figure 7.1). The coloured square shows the stream type. Click a row to inspect the stream, double-click to locate it on the PFD. **Export CSV** writes the complete table. The energy streams (AC input, DC power, stack heat, compressor power, heat rejection, hydrogen LHV) are listed below.

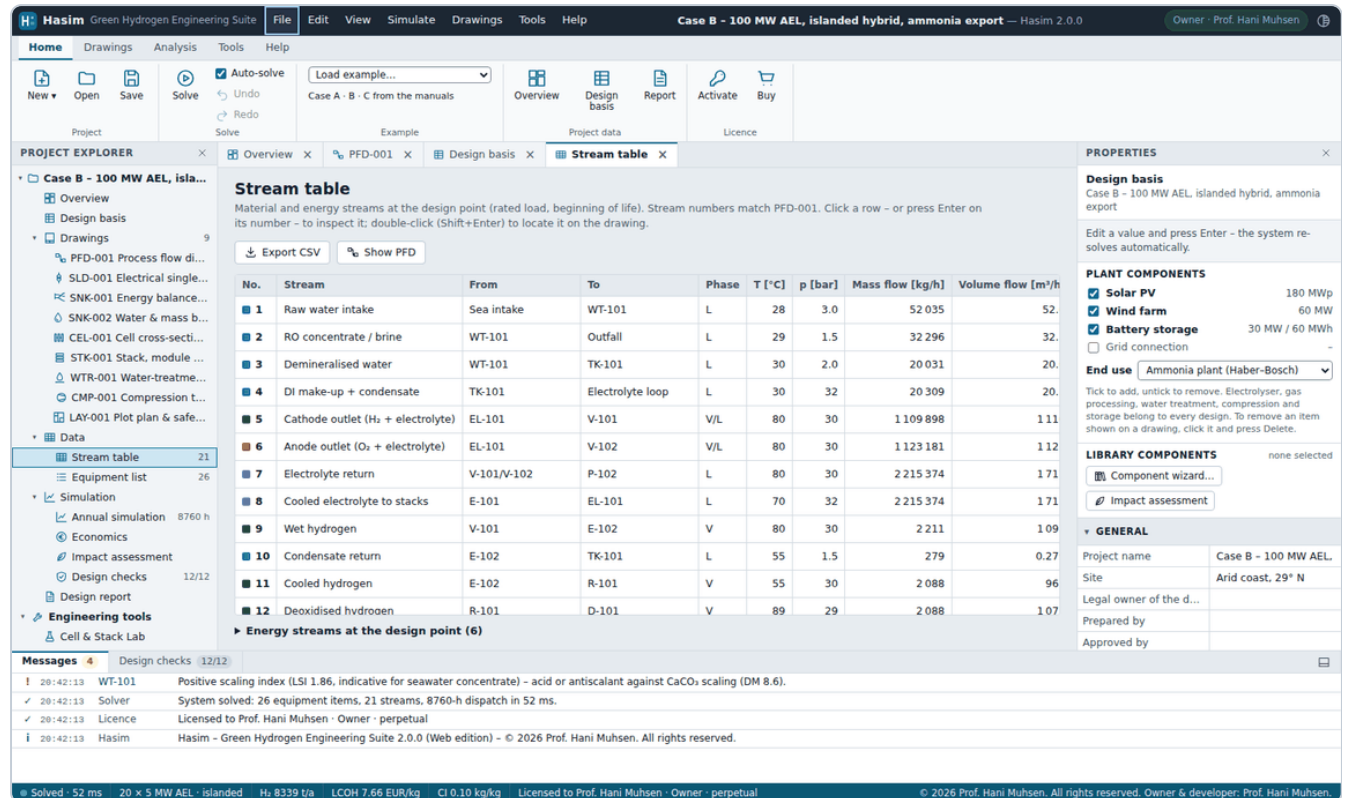


Figure 7.1 – Stream table of Case B.

No.	Stream	From → to	T [°C]	p [bar]	ṁ [kg/h]
1	Raw water intake	Sea intake → WT-101	28	3	52 035
2	RO concentrate / brine	WT-101 → Outfall	29	1.50	32 296
3	Demineralised water	WT-101 → TK-101	30	2	20 031
5	Cathode outlet (H ₂ + electrolyte)	EL-101 → V-101	80	30	1 109 898
9	Wet hydrogen	V-101 → E-102	80	30	2 212
13	Dry hydrogen 99.9989 %	D-101 → CP-201	55	29	1 908
14	Oxygen to vent	V-102 → VS-402	55	29.70	15 405
15	Compressed hydrogen	CP-201 → ST-201	55	100	1 908
16	Hydrogen to ammonia loop	ST-201 → NH-301	40	150	1 032
18	Nitrogen 99.999 %	AS-301 → NH-301	40	150	4 767
19	Liquid ammonia	NH-301 → TK-301	-33	1.10	5 625

7.2 Equipment list and datasheets

The equipment list groups the equipment by function – electrolysis, gas processing, compression, storage, water, utilities, safety, end use, power supply and electrical – with tag, description, quantity, key data and status. A warning chip indicates items with design warnings, for example scaling of the RO concentrate. Click an item to open its datasheet in the properties panel (Figure 3.3); **Export CSV** writes all datasheets.

Hasim Green Hydrogen Engineering Suite | File | Edit | View | Simulate | Drawings | Tools | Help | Case B - 100 MW AEL, islanded hybrid, ammonia export — Hasim 2.0.0 | Owner · Prof. Hani Muhsen

PROJECT EXPLORER

- Case B - 100 MW AEL, isla...
 - Overview
 - Design basis
 - Drawings
 - PFD-001 Process flow di...
 - SLD-001 Electrical single...
 - SNK-001 Energy balance...
 - SNK-002 Water & mass b...
 - CEL-001 Cell cross-secti...
 - STK-001 Stack, module ...
 - WTR-001 Water-treatme...
 - CMP-001 Compression t...
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 - Annual simulation 8760 h
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 - Cell & Stack Lab

Equipment list

26 items with DM tag conventions. Click an item – or press Enter on its tag – for its datasheet in the Properties panel; double-click (Shift+Enter) to locate it on a drawing.

Export CSV (datasheets)

Tag	Description	Qty	Key data	Status
ELECTROLYSIS				
EL-101	Electrolyser modules	20	Technology: Alkaline · Modules × rating: 20 × 5 MW · Stacks per module: 1	OK
GAS PROCESSING				
V-101	H ₂ / electrolyte separator	20	Per module: 20 × · Gas flow per module: 96.73 kg/h · Gas density: 2.03 kg/m ³	OK
V-102	O ₂ / electrolyte separator	20	Per module: 20 × · Gas flow per module: 767.8 kg/h · Gas density: 32.69 kg/m ³	OK
P-102	Electrolyte circulation pump	20	Flow (plant): 1717 m ³ /h · Flow per module: 85.87 m ³ /h · Head / efficiency: 30 m / 70 %	OK
E-101	Electrolyte cooler	20	Duty (BoL / end of stack life): 18.46 / 24.72 MW · Hot side: 80 → 70 °C · Cold side: 60 → 70 °C	OK
E-102	Hydrogen gas cooler / condenser	4	Trains: 4 × · Gas in → out: 80 → 55 °C · Condensate: 185.8 kg/h	OK
R-101	Deoxo reactor (Pd)	4	Trains: 4 × · O ₂ in feed: 2000 ppm(v) · Adiabatic ΔT: 33.6 K	OK
D-101	TSA dryer (twin bed)	4	Trains: 4 × · Water removed: 160.6 kg/h · Product frost point: -70 °C at 29 bar	OK
COMPRESSION				
CP-201	Hydrogen compressor (reciprocating, oil-free)	4	Machines: 4 + 1 · Suction / discharge: 29 → 100 bar · Suction temperature (intercooled, air): 55 °C	OK
STORAGE				

PROPERTIES

Design basis
Case B - 100 MW AEL, islanded hybrid, ammonia export

Edit a value and press Enter – the system resolves automatically.

PLANT COMPONENTS

- ☒ Solar PV 180 MWp
- ☒ Wind farm 60 MW
- ☒ Battery storage 30 MW / 60 MWh
- ☐ Grid connection -

End use Ammonia plant (Haber-Bosch)

Tick to add, untick to remove. Electrolyser, gas processing, water treatment, compression and storage belong to every design. To remove an item shown on a drawing, click it and press Delete.

LIBRARY COMPONENTS none selected

Component wizard...
Impact assessment

GENERAL

Project name	Case B - 100 MW AEL
Site	Arid coast, 29° N
Legal owner of the d...	
Prepared by	
Approved by	

Messages 4 | Design checks 12/12

- 20:42:13 WT-101 Positive scaling index (LSI 1.86, indicative for seawater concentrate) – acid or antiscalant against CaCO₃ scaling (DM 8.6).
- 20:42:13 Solver System solved: 26 equipment items. 21 streams. 8760-h dispatch in 52 ms.
- 20:42:13 Licence Licensed to Prof. Hani Muhsen · Owner · perpetual
- 20:42:13 Hasim Hasim – Green Hydrogen Engineering Suite 2.0.0 (Web edition) – © 2026 Prof. Hani Muhsen. All rights reserved.

● Solved · 52 ms · 20 × 5 MW AEL · islanded · H₂ 8339 t/a · LCOH 7.66 EUR/kg · CI 0.10 kg/kg · Licensed to Prof. Hani Muhsen · Owner · perpetual · © 2026 Prof. Hani Muhsen. All rights reserved. Owner & developer: Prof. Hani Muhsen.

Figure 7.2 – Equipment list of Case B.

8 Annual simulation

The annual simulation dispatches the plant hour by hour over a synthetic typical year: solar geometry with a Markov clearness model scaled to the annual GHI, and an autoregressive Weibull wind series scaled to the mean wind speed. The dispatch follows DM Fig. 4.2: renewable power first supplies the auxiliaries, then the electrolyser between minimum and rated load, then charges the battery; the remainder is curtailed. The battery bridges shortfalls to keep the electrolyser at minimum load; the grid tops up according to the grid mode. The storage inventory balances production against the offtake profile.

Result	Case A	Case B	Case C
Hydrogen [t/a]	1 059	8 339	155
Capacity factor	68.3 %	53.7 %	80.1 %
Full-load hours [h]	5 980	4 707	7 018
PV / wind / grid energy [GWh]	32.8 / 36.9 / 11.04	385.8 / 200.4 / 1.88	3.6 / 0 / 6.12
Curtailed share	22.8 %	12.8 %	4.3 %
Starts per year	1	221	7
Battery cycles [1/a]	0	163	0
Unserved auxiliary energy [MWh]	0	0.40	0

8.1 Models of the dispatch

The hourly dispatch contains the following models. They make the annual production the hydrogen that actually leaves the plant after gas processing, maintenance and ageing – the value compared with the demand by the design checks.

Model	What Hasim does	Reference
Solar resource	Hourly GHI → plane-of-array irradiance with the Erbs diffuse fraction, isotropic sky and ground reflection (albedo 0.2) for a fixed tilt or a single-axis tracker (tracking gain ≥ 1.18); the annual sum is scaled to gain $\times$ GHI. Cell temperature from the NOCT model and power with the temperature coefficient of the modules; DC/AC clipping at the inverters.	DM 3.2; Erbs et al. (1982)
Wind resource	Autoregressive Weibull series ($k = 2$) scaled to the mean wind at hub height; the power curve uses the wind speed corrected to the air density of the hour $(p/1.225)^{1/3}$.	DM 3.3, IEC 61400-12-1
Availability	Every module is out of service for maintenance in six blocks per year, the modules staggered in time, so that the time-average equals the plant availability; one module means whole-plant outages, many modules a small derating.	DM 2.3
Minimum load	The minimum load of the design basis, raised if necessary to the load at which hydrogen in oxygen reaches the 2 % trip (crossover by Fick permeation, CM 7.6/7.7).	CM 7.7, ISO 22734-1
Stack ageing	Stacks are replaced at their stated life or when the cell voltage has risen by 10 %, whichever comes first; the specific energy uses the life-average voltage rise (half of the rise at replacement). The replacements within the economic life enter the LCOH and the impact assessment.	CM 6.6, DM 9.1
Gas processing	Oxygen in the raw hydrogen is converted in the deoxo with twice its amount of hydrogen, and the TSA dryer loses 1 % as regeneration purge; the product is the stack output less these losses.	CM 8.4, DM 10
SOEC heat	Below the thermoneutral voltage the stack needs $k_{\text{SEC}}(E_{\text{tn}} - E)$ of heat per kg, and raising the steam about 6.6 kWh per kg of hydrogen – from electric heaters (in the SEC) or from external heat at the heat price.	CM 3.12

Model	What Hasim does	Reference
Grid and surplus	Grid use: bridge to the minimum load, fill to a target load or always full load. PPA electricity the electrolyser cannot use is curtailed or sold at the export price; only sold surplus supports the monthly RFNBO correlation.	DM 4.2, 4.6
Compression	Number of stages from the discharge-temperature limit (API 618: 120 °C for hydrogen), suction at 40 °C or 10 K above the ambient after the intercoolers.	CM 10.4, DM 12.1

8.2 Live power flow

The power-flow view shows any hour of the year: arrow widths scale with power, the battery shows its state of charge and the storage its inventory. Drag the slider, use the quick buttons (summer noon, winter night, windy day) or press **Play** to animate the year at 1 hour, 3 hours or 1 day per step (Figure 8.1).

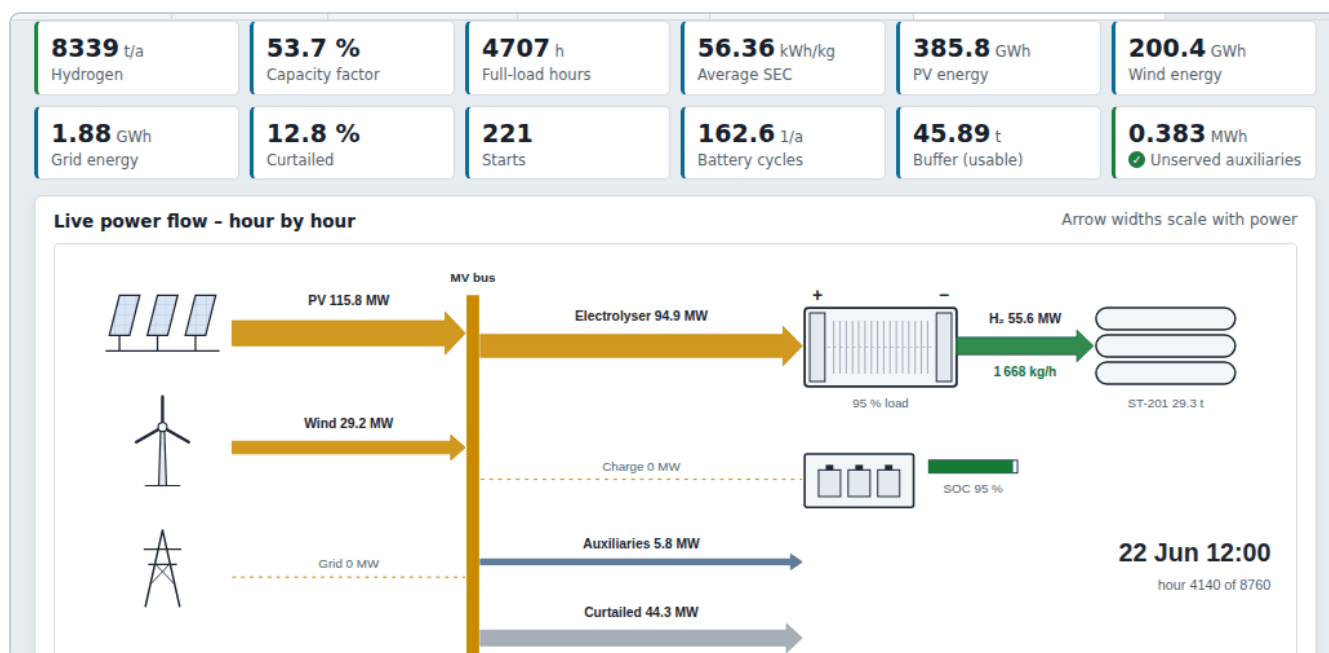


Figure 8.1 – Live power flow of Case B at noon on 22 June.

8.3 Charts

The document continues with the one-week power balance (choose the week with the slider), battery state of charge and hydrogen production, monthly energy, the electrolyser load-duration curve, the storage inventory and a heat map of the electrolyser load by hour of day and day of year (Figure 8.2). Every chart offers **Table**, **CSV** and **SVG**; **Export hourly CSV** writes all 8760 hours.

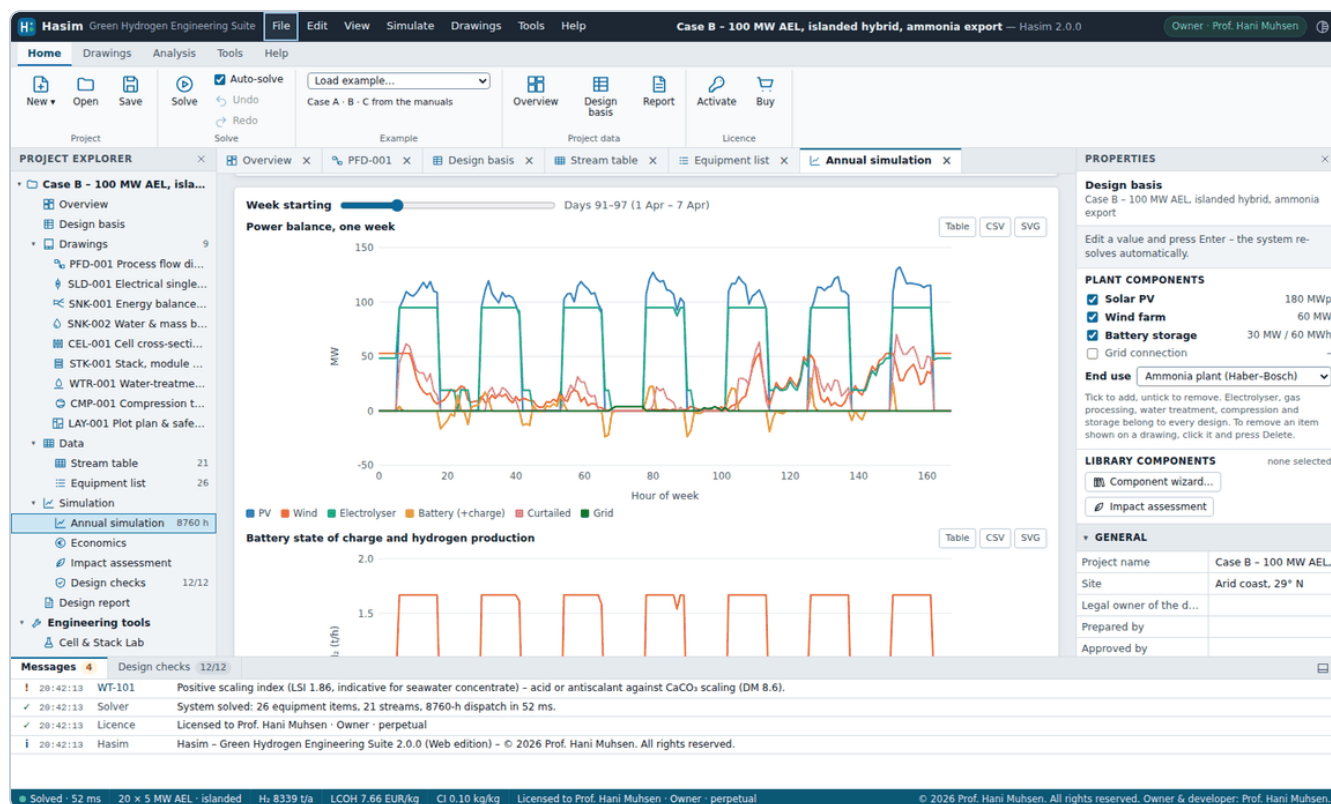


Figure 8.2 – Monthly energy, load-duration curve, storage inventory and load heat map.

9 Economics, sensitivity and impact assessment

The levelised cost of hydrogen follows DM 20.2 / CM 17.3: annualised CAPEX (capital recovery factor from WACC and life) plus the present value of the stack replacements of the annual simulation (CM 6.6) and of other replacements within the economic life (power electronics and battery, at the life of their components), fixed O&M, electricity – including a sold PPA surplus at the difference between the PPA and the export price –, external heat of an SOEC, water and by-product credits, divided by the simulated annual production after gas processing. The economics document shows the LCOH from the 8760-hour simulation and from the design chain, the CAPEX build-up and the LCOH breakdown. The stacks of Case A are replaced 5 times in 25 years (PEM, limited by the voltage degradation), those of Case B once (alkaline, stack life 90 000 h).

LCOH component [EUR/kg]	Case A	Case B	Case C
capital	2.86	5.63	5.78
stacks	0.83	0.13	0.99
other replacements	0.05	0.13	0.00
O&M	1.46	1.75	2.84
electricity	4.90	0.01	5.55
water	0.03	0.00	0.03
by-products	0.00	0.00	0.00
Total	10.13	7.66	15.19

9.1 Sensitivity analysis

Run sensitivity varies the key parameters one at a time – electrolyser CAPEX, WACC, other direct costs, fixed OPEX, stack share and, depending on the supply concept, electricity price and PPA share or PV/wind/battery CAPEX and the solar and wind resource – and re-solves the complete system for each case. The tornado chart ranks the parameters by their LCOH swing (Figure 9.1).

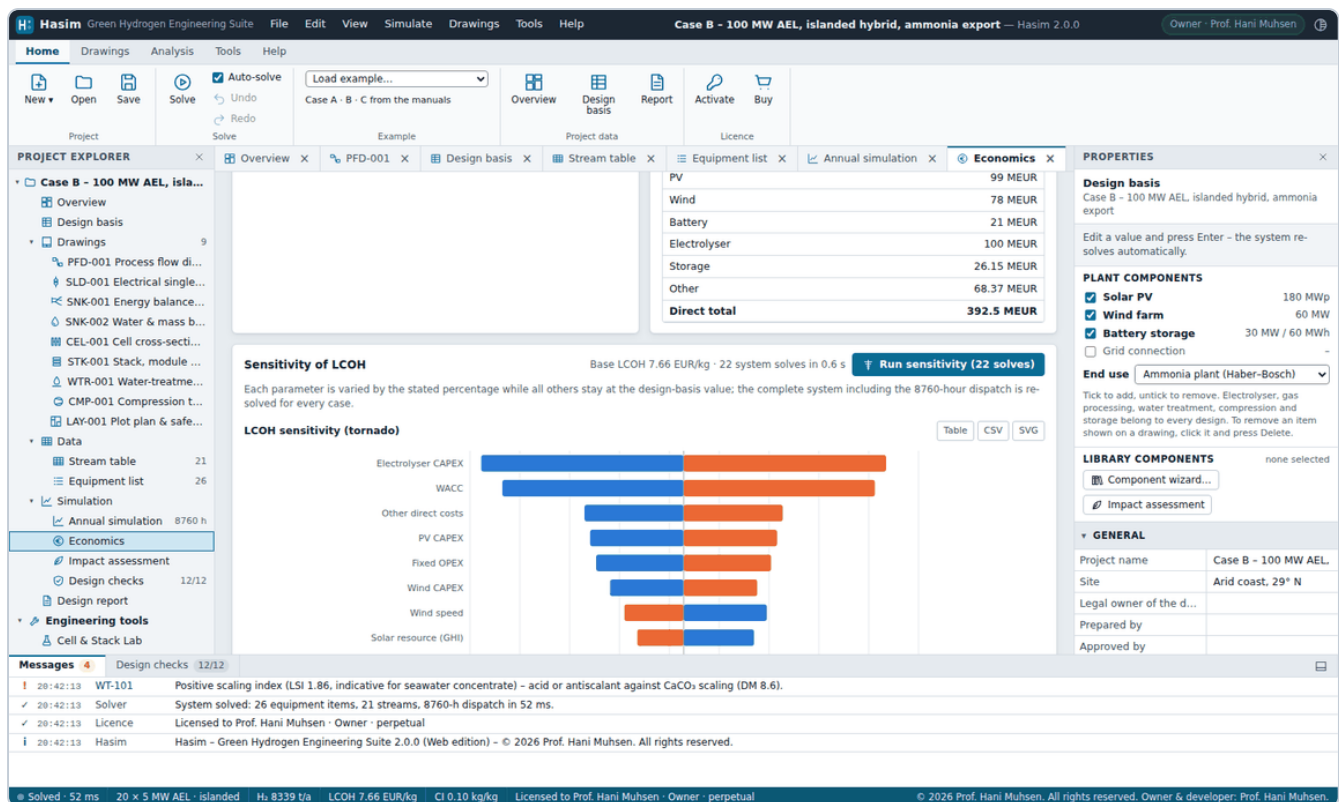


Figure 9.1 – LCOH sensitivity of Case B.

9.2 Impact assessment

Simulate ► Impact assessment (also **Analysis ► Impacts**, the project explorer and the Overview card *Impacts at a glance*, Figure 9.3) evaluates the solved plant together with its components (Figure 9.2). The buttons at the top show all sections or one of them:

- **Performance** – production, capacity factor, full-load hours, specific energy, plant electricity per kg and plant efficiency, renewable share, curtailment, starts, and every design and component check.
- **Economic** – investment and specific investment, a conversion plant that is levelised separately, LCOH, running cost, net present value (NPV), internal rate of return (IRR) and payback at the hydrogen value of the settings, local content and local spending; the investment by component and the component inventory (Figure 9.4).
- **Environmental** – life-cycle carbon intensity with its parts, the carbon intensity of the RFNBO design check, CO₂ avoided against grey hydrogen, embodied emissions, water withdrawal, water intensity and brine, land occupied and total project area, platinum-group metals; embodied emissions and land by component (Figure 9.5).
- **Social** – construction and manufacturing jobs (job-years) and their local part, permanent operation and maintenance jobs (full-time equivalents, FTE), jobs per MW, local content, the water-stress risk with the fresh water expressed as the domestic use of a number of people, land, the largest hazard distance of a storage leak, the Seveso III tiers of the hydrogen and ammonia inventories and the exposure of the site boundary and of the nearest residents (Figure 9.6).
- **Method** – the settings, notes on the inventory, the method and the sources used.

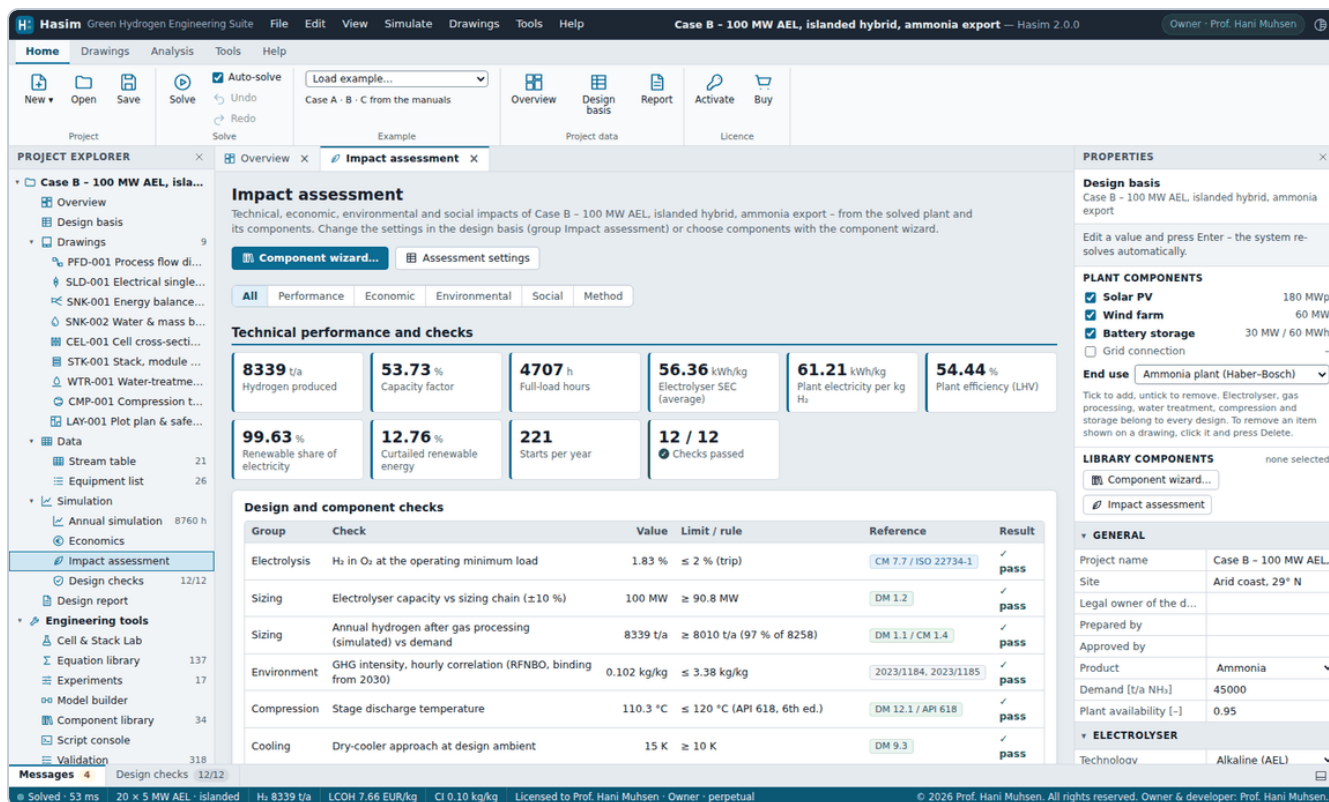


Figure 9.2 – The impact assessment of Case B: technical performance and the design checks.



Figure 9.3 – Impacts at a glance on the Overview of Case B.

Indicator	Case A	Case B	Case C
Labour-productivity region	OECD (× 1)	Middle East (× 2)	Middle East (× 2)
Water source · baseline water stress	fresh · low-medium	seawater · extremely high	fresh · high
Investment [MEUR]	35.2	501.2 + 74.2 ammonia plant	8.8
LCOH [EUR/kg]	10.13	7.66	15.19
NPV at 6 EUR/kg [MEUR] · IRR	-50.9 · -	-147.5 · 4.5 %	-14 · -
Local content of the investment	27.5 %	31.5 %	34.4 %
Life-cycle carbon intensity [kg CO ₂ e/kg] (electricity + equipment)	5.08 (3.13 + 1.95)	1.71 (0.10 + 1.61)	19.21 (17.81 + 1.40)
Carbon intensity of the RFNBO check [kg CO ₂ e/kg]	3.13	0.10	17.81
Embodied emissions of the equipment [kt CO ₂ e]	51.7	335.3	4.3
Water withdrawal [m ³ /a] · water-stress risk	13 234 · low	221 899 · low	2 269 · high
Land occupied · total project area [ha]	44.6 · 562	334.9 · 2 405	3.8 · 4
Construction · manufacturing jobs [job-years]	116 · 134	1 621 · 1 770	26 · 26
Permanent O&M jobs [FTE]	11.6	212.3	2.9
Seveso III (hydrogen · ammonia)	lower tier	upper tier · upper tier	below threshold

The table compares the three examples with the default components. At a hydrogen value of 6 EUR/kg only Case B earns back part of its investment (IRR 4.5 %, below its WACC of 8 %); at a hydrogen value equal to the LCOH the NPV is exactly zero, so the levelised cost and the cash-flow view are consistent. Case B, with its own PV and wind plants in the Middle East region, creates about 1 600 construction job-years and 212 permanent jobs. Per MW of electrolyser, construction needs 11.6 (A), 16.2 (B) and 20.6 (C) job-years – small plants carry more fixed equipment per MW. Case C draws fresh water in a region of high water stress: the risk is flagged *high*.

Life-cycle carbon intensity and the RFNBO check

The **life-cycle carbon intensity** adds the embodied emissions of all equipment – including PV and wind plants bought through a PPA – to the emissions of the electricity and of transport. It is a screening life-cycle value for comparing designs and components. The **design check** of the carbon intensity (CM 18.1, DM 4.6) follows the RFNBO methodology of Delegated Regulation (EU) 2023/1185: grid electricity at the emission factor of the grid, renewable electricity at zero, plus transport and conversion – the emissions of manufacturing the equipment are not counted. It decides whether the hydrogen qualifies as renewable. For Case A the two values are 5.08 and 3.13 kg CO₂e/kg.

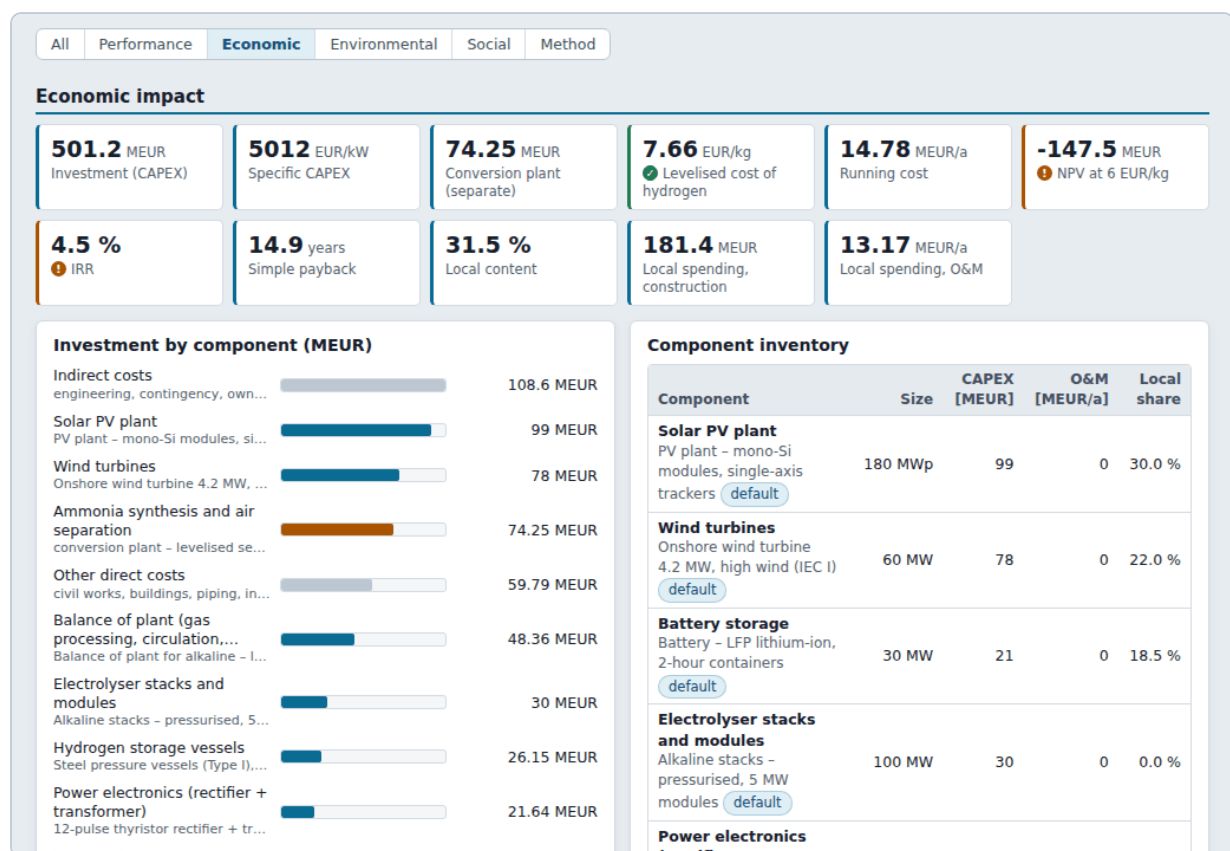


Figure 9.4 – Economic impact of Case B: investment by component and the component inventory.

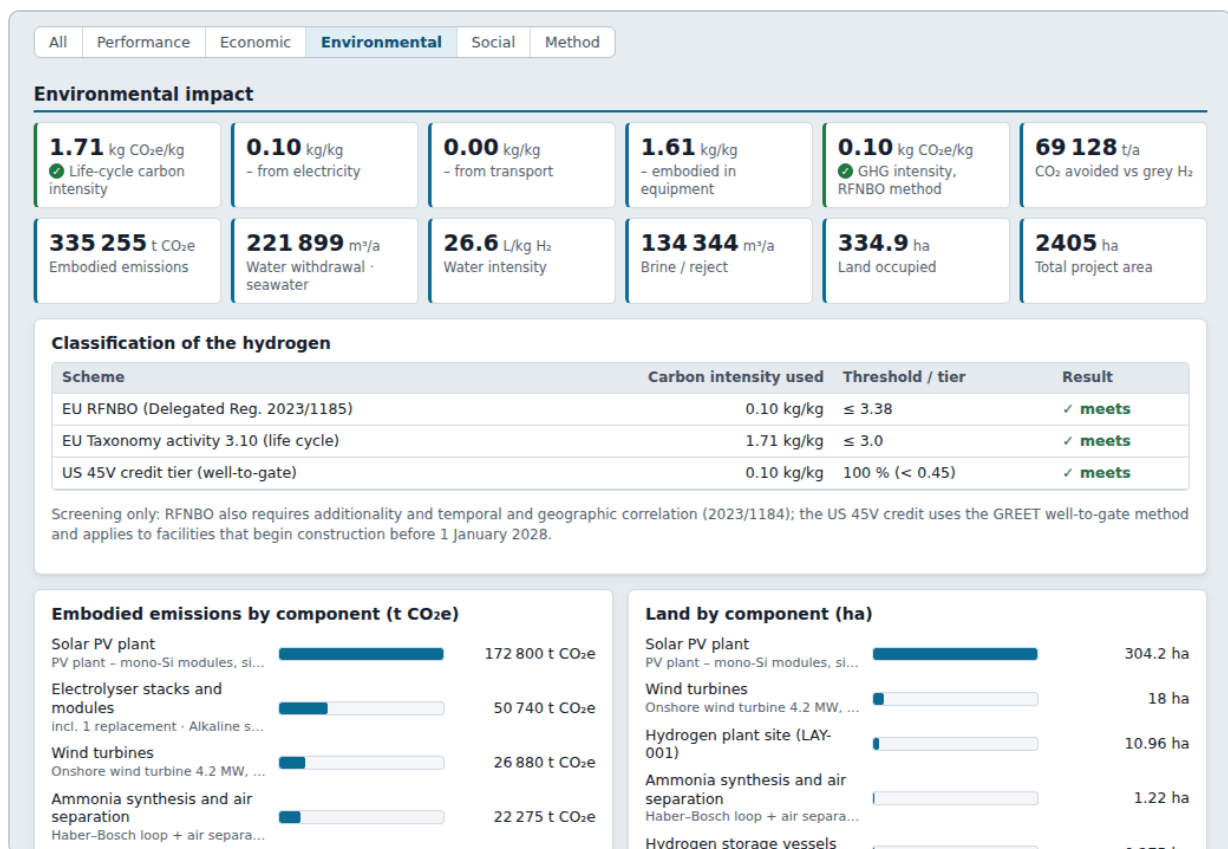


Figure 9.5 – Environmental impact of Case B: carbon intensity, embodied emissions, water and land.

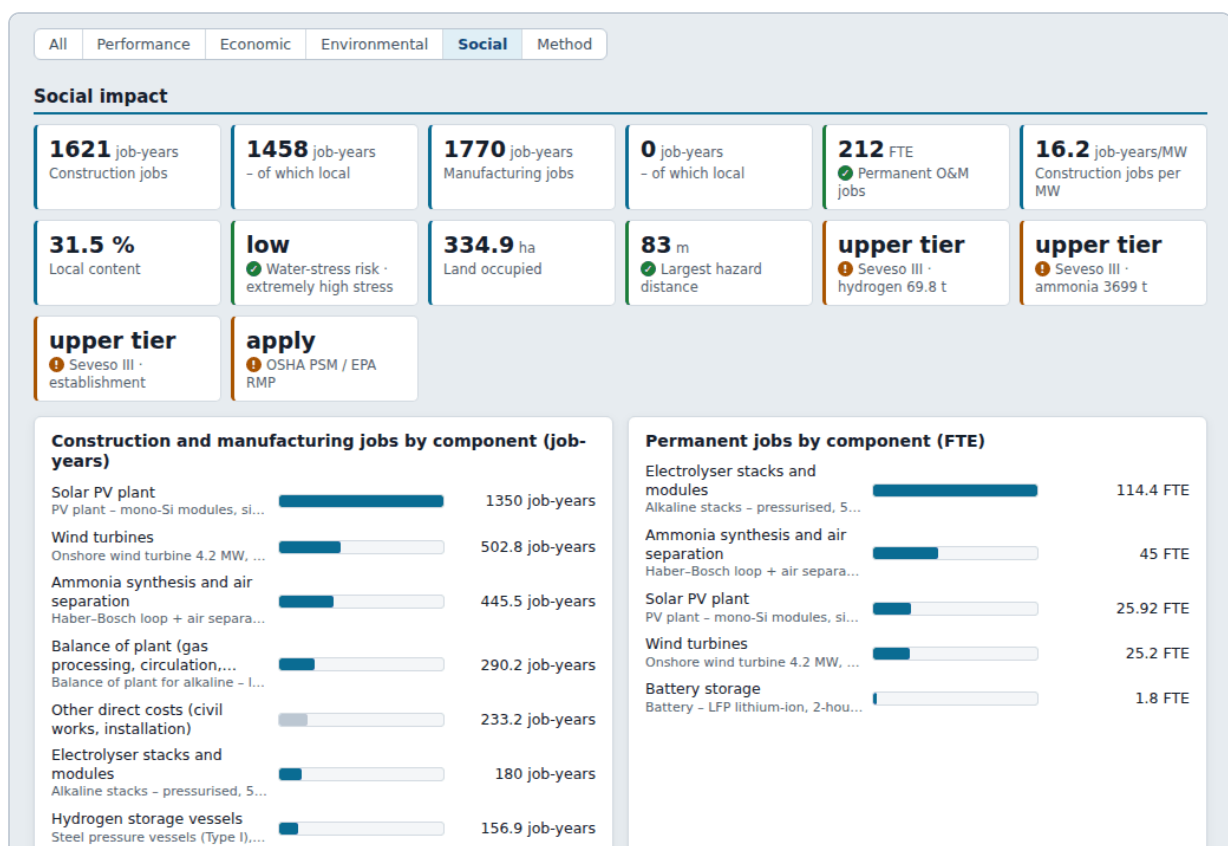


Figure 9.6 – Social impact of Case B: jobs by component, local content, water-stress risk and safety.

Assessment settings opens the design basis at the group **Impact assessment**: labour-productivity region, local shares of equipment supply, installation and services, baseline water stress, the distances from the storage yard to the site boundary and to the nearest residents, and the hydrogen value for NPV and IRR. The New Project wizard presets the region and the water stress from the climate of the site. The **design report** contains the assessment as section 8 (economic, environmental, social, equipment and components).

9.3 Method of the impact assessment

The assessment combines the solved model with an **inventory**: every equipment item with its size, investment, fixed O&M and component. In a plant project, PV, wind, battery, electrolyser and storage carry the costs of the design basis – the electrolyser is split into stacks, power electronics and balance of plant with the cost shares of their components. The water plant, compressor, pipeline, trailers and refuelling station are priced with the unit costs of their components inside the other direct costs of the design basis; if they exceed them, they are scaled down and a note shows the value of “Other direct costs (× electrolyser)” that includes them in full. An ammonia plant is an item outside the hydrogen plant: it is levelised separately as the cost of ammonia (CM 17.6) but counted in local content, jobs, emissions and land. In a Model-builder chain every block adds its equipment and costs.

Indicator	Calculation
Jobs	$J = f \times Q \times m$. The employment factor f of the component refers to its size Q (per MWp, MW, MWh, m ³ /d, kg ...), to its investment in MEUR, to the hydrogen output in MW (LHV) or to the number of drivers; m is the multiplier of the labour-productivity region. Construction and installation and manufacturing in job-years, operation and maintenance in full-time equivalents (FTE). Other direct costs add 1.5 job-years per MEUR for construction and, on their 30 % equipment share, 1.5 job-years per MEUR for manufacturing.
Local content	Share of the investment (incl. a conversion plant) spent in the country: $\Sigma \text{CAPEX} \times (\text{equipment share} \times \text{local equipment supply} + \text{installation share} \times \text{local installation} + \text{services share} \times \text{local services})$; other direct costs split 30/60/10 %, indirect and owner's costs at the local share of services. Local construction jobs = construction jobs × local installation share; local manufacturing jobs = manufacturing jobs × local equipment share; local O&M spending = fixed O&M × local installation share.
Embodied emissions	$E = e \times Q \times (1 + n)$ with n replacements over the economic life (stacks: replacements of the annual simulation). Other direct costs are counted with 0.3 kg CO ₂ e per EUR (spend-based).
Carbon intensity	$CI_{\text{life}} = CI_{\text{electricity}} + CI_{\text{transport}} + E \div (\text{hydrogen per year} \times \text{economic life})$. CO ₂ avoided = $(10 - CI_{\text{life}}) \times \text{hydrogen per year}$, against steam methane reforming at about 10 kg CO ₂ e/kg (CM 18.2).
Water	Raw-water withdrawal of the annual simulation, water intensity in litres per kg of hydrogen, brine = raw water – demineralised water. Fresh water is also expressed as the domestic use of people at 100 litres per person and day (UN/WHO: 50–100 L).
Water-stress risk	Seawater → low. Brackish water → medium in high or extremely high baseline water stress, otherwise low. Fresh water → high in high or extremely high stress, medium in medium-high stress, otherwise low.
Land	Land factor × size of every component plus, for a plant, the site of the plot plan (LAY-001), which contains the electrolyser area. The total project area adds the spacing area of the wind farm, which remains usable for other purposes.
Safety exposure	Distances of a 3 mm leak from the storage at its maximum pressure: lower flammability limit (CM 19.5), thermal radiation of 4.7 kW/m ² (CM 19.6) and 10 kPa overpressure of a cloud of up to 100 kg (CM 19.8), compared with the distances to the site boundary and to the residents. Seveso III (2012/18/EU): hydrogen against 5 t / 50 t and ammonia in the product tank (30 days of production) against 50 t / 200 t, combined with the summation rule of Annex I note 4 (lower / upper tier); OSHA PSM and EPA RMP apply above 10 000 lb (4.5 t).
NPV, IRR, payback	$\text{NPV} = -(\text{CAPEX} + \text{present value of stack replacements}) + (\text{revenue} - \text{running cost}) \times \text{annuity factor (WACC, life)}$; revenue = hydrogen × hydrogen value; running cost = fixed O&M + electricity + water – by-product credits. IRR: discount rate at which NPV = 0; payback = investment ÷ annual net cash flow.

Setting (design basis ► Impact assessment)	Default	Options
Labour-productivity region	OECD (× 1)	OECD × 1, Middle East × 2, Latin America × 2.3, China × 1.3, India × 1.4, Developing Asia × 1.5, Africa × 6.8, Transition economies × 1.7
Local share of equipment supply	0 %	0 ... 100 %
Local share of installation, civil works	90 %	0 ... 100 %
Local share of engineering, services	50 %	0 ... 100 %
Baseline water stress	Medium-high (20–40 %)	Low (< 10 %), Low-medium (10–20 %), Medium-high (20–40 %), High (40–80 %), Extremely high (> 80 %)
Distance storage – site boundary	100 m	0 ... 100 km
Distance to the nearest residents	1000 m	0 ... 1 000 km
Hydrogen value for NPV and IRR	6 EUR/kg	0 ... 50 EUR/kg

The regional multipliers express the lower labour productivity outside the OECD (more workers for the same equipment) after Rutovitz & Usher (2010), interpolated to 2026 [4]; the employment factors themselves are OECD values [3]. The water-stress classes are those of WRI Aqueduct 4.0 [16]. Choose the region and the stress class of your site – the wizard does it from the climate preset.

9.4 Default values and sources

The library values are **screening values** for concept studies – typical published figures of 2022–2025 or documented Hasim assumptions. Every component lists its sources (Component library ► *Sources of the default values*). Replace the values with vendor quotations, environmental product declarations (EPDs) and local labour studies for your project: duplicate a component to your library and edit it (section 11.4). The table lists all components with the units of their category line; the shares split the investment into equipment / installation and civil works / engineering and services for the local content. Numbers refer to the list of sources below.

Component	CAPEX	O&M [1/a] · life [a]	Embodied	Land	Jobs C&I	Jobs mfg.	Jobs O&M	Shares [%]	Sources
Solar PV plant – size in MWp · CAPEX EUR/kWp · embodied kg CO ₂ e/kWp · land ha/MWp · jobs per MWp									
PV plant – mono-Si modules, fixed tilt	500	0.015 · 30	960	1.15	1.29	2.46	0.072	60 / 25 / 15	2, 3, 7, 9, 10, 20
PV plant – mono-Si modules, single-axis trackers	550	0.015 · 30	960	1.69	1.29	2.46	0.072	60 / 25 / 15	2, 3, 7, 9, 10, 20
PV plant – bifacial glass-glass modules, single-axis trackers	580	0.015 · 30	900	1.69	1.29	2.46	0.072	60 / 25 / 15	3, 7, 9, 20
Wind turbines – size in MW · CAPEX EUR/kW · embodied kg CO ₂ e/kW · land ha/MW (direct) · jobs per MW									
Onshore wind turbine 5 MW class, medium wind (IEC II)	1300	0.025 · 25	448	0.3	2.65	1.54	0.21	72 / 20 / 8	2, 3, 8, 11, 19, 20
Onshore wind turbine 4.5 MW, large rotor for low wind (IEC III)	1400	0.025 · 25	448	0.3	2.65	1.54	0.21	72 / 20 / 8	3, 8, 11, 19, 20
Onshore wind turbine 4.2 MW, high wind (IEC I)	1250	0.025 · 25	448	0.3	2.65	1.54	0.21	72 / 20 / 8	3, 8, 11, 19, 20

Component	CAPEX	O&M [1/a] · life [a]	Embodied	Land	Jobs C&I	Jobs mfg.	Jobs O&M	Shares [%]	Sources
Battery storage – size in MW · CAPEX EUR/kWh · embodied kg CO ₂ e/kWh · land m ² /MWh · jobs per MW									
Battery – LFP lithium-ion, 2-hour containers	350	0.02 · 15	90.8	20	0.53	0.31	0.03	75 / 15 / 10	2, 3, 12, 20
Battery – LFP lithium-ion, 4-hour containers	300	0.02 · 15	90.8	15	0.53	0.31	0.03	75 / 15 / 10	3, 12, 20
Water treatment plant – size in m ³ /d · CAPEX EUR per m ³ /d · embodied kg CO ₂ e per m ³ /d · land m ² per m ³ /d · jobs per m ³ /d									
Seawater RO with energy recovery + second pass + EDI	3500	0.05 · 25	1050	0.8	1.5/MEUR	1.5/MEUR	0	60 / 30 / 10	2, 20
Brackish-water RO + EDI polishing	2000	0.05 · 25	600	0.5	1.5/MEUR	1.5/MEUR	0	60 / 30 / 10	20
Municipal water: RO + EDI polishing	1500	0.05 · 25	450	0.4	1.5/MEUR	1.5/MEUR	0	60 / 30 / 10	2, 20
Electrolyser stacks and modules – size in MW · CAPEX EUR/kW · embodied kg CO ₂ e/kW · land m ² /MW (plant incl. BoP) · jobs per MW									
PEM stacks – 5 MW modules	675	0.025 · 25	137.5	250	1.5/MEUR	1.5/MEUR	0.9/MW H ₂	100 / 0 / 0	1, 6, 13, 14, 15, 20
Alkaline stacks – pressurised, 5 MW modules	450	0.025 · 25	253.7	350	1.5/MEUR	1.5/MEUR	0.9/MW H ₂	100 / 0 / 0	1, 6, 13, 15, 20
Alkaline stacks – large pressurised, 10 MW modules	420	0.025 · 25	253.7	330	1.5/MEUR	1.5/MEUR	0.9/MW H ₂	100 / 0 / 0	1, 6, 13, 20
AEM stacks – 1 MW modules	230	0.025 · 25	99.2	300	1.5/MEUR	1.5/MEUR	0.9/MW H ₂	100 / 0 / 0	1, 6, 13, 15, 20
Solid-oxide stacks – 1 MW modules	660	0.025 · 25	102.6	400	1.5/MEUR	1.5/MEUR	0.9/MW H ₂	100 / 0 / 0	1, 6, 13, 15, 20
Power electronics (rectifier + transformer) – size in MW · CAPEX EUR/kW · embodied kg CO ₂ e/kW · land m ² /MW · jobs per MW									
IGBT active front-end rectifier + transformer	200	0.01 · 20	60	0	1.5/MEUR	1.5/MEUR	0	100 / 0 / 0	1, 20
12-pulse thyristor rectifier + transformer + harmonic filter	170	0.01 · 20	51	0	1.5/MEUR	1.5/MEUR	0	100 / 0 / 0	2, 20
Balance of plant (gas processing, circulation, cooling) – size in MW · CAPEX EUR/kW · embodied kg CO ₂ e/kW · land m ² /MW · jobs per MW									
Balance of plant for PEM – water loop, separators, deoxo + dryer, dry coolers	625	0.025 · 25	187.5	0	1.5/MEUR	1.5/MEUR	0	80 / 20 / 0	1, 20
Balance of plant for alkaline – lye loop, separators, deoxo + dryer, dry coolers	380	0.025 · 25	114	0	1.5/MEUR	1.5/MEUR	0	80 / 20 / 0	1, 20
Balance of plant for AEM	720	0.025 · 25	216	0	1.5/MEUR	1.5/MEUR	0	80 / 20 / 0	1, 20
Balance of plant for SOEC – steam generator, recuperators, hot box	1340	0.025 · 25	402	0	1.5/MEUR	1.5/MEUR	0	80 / 20 / 0	1, 20
Hydrogen compressor – size in kW · CAPEX EUR/kW · embodied kg CO ₂ e/kW · land m ² /kW · jobs per kW									
Reciprocating piston compressor, oil-free	3000	0.04 · 20	900	0	1.5/MEUR	1.5/MEUR	0	80 / 15 / 5	2, 20
Diaphragm compressor for small flows and high pressure	4500	0.05 · 20	1350	0	1.5/MEUR	1.5/MEUR	0	80 / 15 / 5	20
Hydrogen storage vessels – size in kg · CAPEX EUR/kg usable · embodied kg CO ₂ e/kg usable · land m ² /kg usable · jobs per kg									
Steel pressure vessels (Type I), up to 300 bar	600	0.02 · 30	180	0.06	1.5/MEUR	1.5/MEUR	0	70 / 25 / 5	2, 20
Composite vessels (Type III/IV), 350–500 bar	900	0.02 · 25	270	0.025	1.5/MEUR	1.5/MEUR	0	70 / 25 / 5	2, 20

Component	CAPEX	O&M	Embodied	Land	Jobs C&I	Jobs mfg.	Jobs O&M	Shares [%]	Sources
		[1/a] · life [a]							
Hydrogen pipeline – size in km · CAPEX MEUR/km · embodied t CO ₂ e/km · land ha/km (right of way) · jobs per km									
Steel pipeline DN300 (L360 / X52), buried	1.2	0.01 · 40	360	2	1.5/MEUR	1.5/MEUR	0.05	40 / 50 / 10	2, 20
Steel pipeline DN100, short connection	0.65	0.01 · 40	195	1.5	1.5/MEUR	1.5/MEUR	0.05	40 / 50 / 10	2, 20
Tube trailers – size in trailers · CAPEX MEUR per trailer · embodied t CO ₂ e per trailer · land m ² per trailer · jobs per trailer									
Tube trailer – steel tubes, 200 bar	0.35	0.03 · 20	105	0	0	1.5/MEUR	1 × drivers	100 / 0 / 0	1, 20
Tube trailer – composite tubes, 500 bar	0.9	0.03 · 20	270	0	0	1.5/MEUR	1 × drivers	100 / 0 / 0	1, 20
Refuelling station – size in t/d · CAPEX MEUR per t/d · embodied t CO ₂ e per t/d · land m ² per t/d · jobs per t/d									
Refuelling station 70 MPa – booster, 900 bar cascade, pre-cooling	3	0.05 · 20	900	1200	1.5/MEUR	1.5/MEUR	1.5	70 / 25 / 5	2, 20
Ammonia synthesis and air separation – size in t/a · CAPEX EUR per t/a · embodied kg CO ₂ e per t/a · land m ² per t/a · jobs per t/a									
Haber–Bosch loop + air separation (electrically driven)	1650	0.03 · 25	495	0.27	1.5/MEUR	1.5/MEUR	0.0005	60 / 30 / 10	1, 2, 20
Methanol synthesis – size in t/a · CAPEX EUR per t/a · embodied kg CO ₂ e per t/a · land m ² per t/a · jobs per t/a									
Methanol synthesis from captured CO ₂	900	0.03 · 25	270	0.2	1.5/MEUR	1.5/MEUR	0.0005	60 / 30 / 10	1, 20
Fuel-cell power system – size in kW · CAPEX EUR/kW · embodied kg CO ₂ e/kW · land m ² /kW · jobs per kW									
PEM fuel-cell power system	2000	0.03 · 10	600	0.05	1.5/MEUR	1.5/MEUR	0	85 / 10 / 5	1, 20

Jobs: construction and installation (C&I) and manufacturing in job-years, operation and maintenance in FTE – per size unit of the category line, per MEUR of investment (/MEUR), per MW of hydrogen output (/MW H₂) or per driver of the trailer trips (× drivers), for OECD labour productivity. Shares: equipment / installation and civil works / engineering and services of the investment.

Sources

- Green Hydrogen Calculations Manual, Ed. 3 (CM) – cell, stack, system and cost equations used by Hasim.
- Green Hydrogen Plant Design Manual, Ed. 1 (DM) – plant design values used by Hasim (e.g. water-plant and storage costs, example cases).
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- Lin X. et al. (2024). Environmental impact analysis of lithium iron phosphate batteries for energy storage in China. *Frontiers in Energy Research* 12, 1361720.

13. Wei et al. (2024). Comparative life cycle analysis of electrolyzer technologies for hydrogen production: manufacturing and operations. *Joule* 8(12).
14. U.S. Department of Energy (status 2022). Technical targets for proton exchange membrane electrolysis – PGM loading 0.8 g/kW, stack 450 of 1 000 USD/kW system cost.
15. Corbeau A.-S., Merz A.-K. (2023). Demystifying electrolyzer production costs. Center on Global Energy Policy, Columbia University – stacks are 19–60 % of electrolyser system cost.
16. Kuzma S. et al. (2023). Aqueduct 4.0: updated decision-relevant global water risk indicators. World Resources Institute – baseline water stress classes.
17. Directive 2012/18/EU (Seveso III), Annex I Part 2 – qualifying quantities: hydrogen 5 t / 50 t, ammonia (anhydrous) 50 t / 200 t (lower / upper tier).
18. UN General Assembly resolution 64/292 and WHO guidance: 50–100 litres of water per person per day meet most basic needs.
19. IEC 61400-1: wind-turbine classes I / II / III for annual mean wind speeds at hub height of 10 / 8.5 / 7.5 m/s.
20. Hasim assumption – a documented screening value (User Manual, section 9.4); replace it with project or vendor data.

Documented Hasim assumptions [20]

- Process equipment without a published factor (water plant, power electronics, balance of plant, compressors, vessels, pipelines, trailers, refuelling stations, conversion plants) and the other direct costs: embodied emissions of 0.3 kg CO₂e per EUR of investment (spend-based), 1.5 construction and 1.5 manufacturing job-years per MEUR.
- Costs of components that the manuals do not give (bifacial PV, the low-wind and high-wind turbines, 4-hour batteries, brackish-water RO, diaphragm compressors, large alkaline modules) are typical 2025 market values; the balance of plant is the technology cost of CM minus stacks and power electronics.
- Seveso III and OSHA PSM / EPA RMP: the hydrogen inventory is the storage at maximum pressure and at the lowest site temperature, plus tube trailers, the refuelling cascade and the process hold-up; the ammonia tank holds 30 days of production. Grey hydrogen for the avoided CO₂: 10 kg CO₂e/kg. Domestic water use: 100 L per person and day.

10 Design checks and the design report

After each solve Hasim checks the design against the rules of both manuals and against the codes, standards and regulations named in each check (section 10.1). Failed checks appear in the design-checks document, in the bottom panel and as warnings in the message log. The limit column states the acceptance criterion – the production check, for example, accepts 97 % of the demand.

Group	Check	Value	Limit	Reference	Case B
Electrolysis	H ₂ in O ₂ at the operating minimum load	1.83 %	≤ 2 % (trip)	CM 7.7 / ISO 22734-1	pass
Sizing	Electrolyser capacity vs sizing chain (±10 %)	100 MW	≥ 90.8 MW	DM 1.2	pass
Sizing	Annual hydrogen after gas processing (simulated) vs demand	8 339 t/a	≥ 8010 t/a (97 % of 8258)	DM 1.1 / CM 1.4	pass
Environment	GHG intensity, hourly correlation (RFNBO, binding from 2030)	0.10 kg/kg	≤ 3.38 kg/kg	2023/1184, 2023/1185	pass
Compression	Stage discharge temperature	110.3 °C	≤ 120 °C (API 618, 6th ed.)	DM 12.1 / API 618	pass
Cooling	Dry-cooler approach at design ambient	15 K	≥ 10 K	DM 9.3	pass
Water	Brine diffuser dilution (≤ +1.5 g/L at the edge of the mixing zone)	69.62 –	≥ 20	DM 8.8	pass
Power	Unserved auxiliary energy	0.38 MWh/a	= 0 MWh	DM 4.3	pass
Power	Energy balance closure	0 GWh	< 0.01 GWh	solver	pass
Power	Site wind within the turbine class (IEC I)	8.60 m/s	≤ 10 m/s	IEC 61400-1	pass
Safety	Storage separation to the process area	12 m	≥ 11.3 m (LFL, 4.7 kW/m ² , NFPA 2 group 1)	CM 19 / NFPA 2	pass
Safety	Storage distance to the site boundary	39.81 m	≥ 7.5 m (NFPA 2 group 1, lot line)	NFPA 2 (2026)	pass

The site wind is always checked against the class of the wind turbine – the chosen one or the default for the site (IEC 61400-1). When library components are chosen (section 4.7), **component checks** are added in the group *Components*: the feed salinity within the design of the water plant, balance of plant and stacks of the design-basis technology, the storage pressure within the rating of the vessels and of the compressor, the delivery pressure within the design pressure of the pipeline, and the battery cycles over its life within its cycle life (Figure 10.2).

Design checks

Automatic verification of the solved design against the limits of the Calculations Manual (CM), the Plant Design Manual (DM) and the codes, standards and regulations named in each check, followed by a screening against codes and regulations.

12 Checks | 12 Passed | 0 To review | 1 Equipment warnings

Group	Check	Value	Unit	Limit / rule	Refer
Electrolysis	H ₂ in O ₂ at the operating minimum load	1.83	%	≤ 2 % (trip)	CM 7.1
Sizing	Electrolyser capacity vs sizing chain (±10 %)	100	MW	≥ 90.8 MW	DM 1.1
Sizing	Annual hydrogen after gas processing (simulated) vs demand	8339	t/a	≥ 8010 t/a (97 % of 8258)	DM 1.1
Environment	GHG intensity, hourly correlation (RFNBO, binding from 2030)	0.102	kg/kg	≤ 3.38 kg/kg	2023/7
Compression	Stage discharge temperature	110.3	°C	≤ 120 °C (API 618, 6th ed.)	DM 12
Cooling	Dry-cooler approach at design ambient	15	K	≥ 10 K	DM 9.1
Water	Brine diffuser dilution (≤ +1.5 g/L at the edge of the mixing zone)	69.62	-	≥ 20	DM 8.1
Power	Unserved auxiliary energy	0.383	MWh/a	= 0 MWh	DM 4.1
Power	Energy balance closure	1.65e-11	GWh	< 0.01 GWh	solver
Power	Site wind within the turbine class (IEC I)	8.6	m/s	≤ 10 m/s	IEC 61
Safety	Storage separation to the process area	12	m	≥ 11.3 m (LFL, 4.7 kW/m², NFPA 2 group 1)	CM 19
Safety	Storage distance to the site boundary	39.81	m	≥ 7.5 m (NFPA 2 group 1, lot line)	NFPA 2

Equipment warnings

WT-101 Positive scaling index (LSI 1.86, indicative for seawater concentrate) - acid or antiscaling against CaCO₃ scaling (DM 8.6).

System solved: 26 equipment items, 21 streams, 8760-h dispatch in 52 ms.

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Figure 10.1 – Design checks and equipment warnings.

Design checks

Automatic verification of the solved design against the limits of the Calculations Manual (CM), the Plant Design Manual (DM) and the codes, standards and regulations named in each check, followed by a screening against codes and regulations.

14 Checks | 14 Passed | 0 To review | 1 Equipment warnings

Group	Check	Value	Unit	Limit / rule	Refer
Electrolysis	H ₂ in O ₂ at the operating minimum load	1.83	%	≤ 2 % (trip)	CM 7.1
Sizing	Electrolyser capacity vs sizing chain (±10 %)	100	MW	≥ 90.8 MW	DM 1.1
Sizing	Annual hydrogen after gas processing (simulated) vs demand	8295	t/a	≥ 8010 t/a (97 % of 8258)	DM 1.1
Environment	GHG intensity, hourly correlation (RFNBO, binding from 2030)	0.078	kg/kg	≤ 3.38 kg/kg	2023/7
Compression	Stage discharge temperature	110.3	°C	≤ 120 °C (API 618, 6th ed.)	DM 12
Cooling	Dry-cooler approach at design ambient	15	K	≥ 10 K	DM 9.1
Water	Brine diffuser dilution (≤ +1.5 g/L at the edge of the mixing zone)	69.62	-	≥ 20	DM 8.1
Power	Unserved auxiliary energy	0.507	MWh/a	= 0 MWh	DM 4.1
Power	Energy balance closure	1.83e-11	GWh	< 0.01 GWh	solver
Power	Site wind within the turbine class (IEC I)	8.6	m/s	≤ 10 m/s	IEC 61
Safety	Storage separation to the process area	12	m	≥ 11.3 m (LFL, 4.7 kW/m², NFPA 2 group 1)	CM 19
Safety	Storage distance to the site boundary	39.81	m	≥ 7.5 m (NFPA 2 group 1, lot line)	NFPA 2
Components	Stack component matches the design-basis technology	AEL		AEL	comp
Components	Battery cycles over its life	2236		≤ 6000 cycles	comp

Equipment warnings

Tag	Equipment	Warning
WT-101	Seawater RO + EDI water plant	Positive scaling index (LSI 1.86, indicative for seawater concentrate) - acid or antiscaling against CaCO ₃ scaling (DM 8.6).

Codes, standards and regulations - screening

Figure 10.2 – Design checks of Case B with library components: the component checks are added in the group Components.

10.1 Codes, standards and regulations

The checks and the **screening** below them in the design-checks document (Figure 10.3) apply the following international, European and US rules at concept level. The screening tables list the RFNBO criteria, the hazardous areas, the major-accident regulations, the pressure equipment, the separation distances and the power-quality and electrical items of the solved plant; the design report contains them as section 7.1.

Framework	Standard or regulation	Applied by Hasim
ISO	ISO 22734-1 – hydrogen generators using water electrolysis	H ₂ in O ₂ at the operating minimum load: alarm 1 %, trip 2 % (CM 7.7); the dispatch raises the minimum load to the trip limit
ISO / SAE	ISO 14687:2025 grade D, ISO 19880-1, SAE J2601	Refuelling stations: fuel quality after deoxo and dryer (total impurities ≤ 300 μmol/mol, H ₂ O and O ₂ ≤ 5 μmol/mol), cascade pressure above the vehicle MOP, fuel temperature window T40, hot-fill end pressure
IEC	IEC 60079-10-1:2020 with ATEX 1999/92/EC	Hazardous areas: source of release, grade, zone, extent (LFL distance of a secondary release), gas group IIC / IIA, temperature class T1, equipment protection level
IEC / IEEE	IEC 60909-0, IEEE 519-2022, IEC 61400-1, IEC 60617	Short-circuit currents with c = 1.1 and the source impedance of the grid; current TDD and voltage THD at the point of common coupling; turbine class for the site wind; drawing symbols
ISO	ISO 7200	Title block of the drawings: legal owner, identification, revision, date of issue, document type, creator, approver
EU	Delegated Regulations (EU) 2023/1184 and 2023/1185 (RFNBO)	GHG intensity ≤ 3.38 kg CO ₂ e/kg with hourly correlation (binding from 2030) and, with a sold PPA surplus, monthly correlation (until the end of 2029); additionality, geographic correlation and grid-electricity rules listed for review
EU	EU Taxonomy, activity 3.10	Life-cycle GHG intensity ≤ 3.0 kg CO ₂ e/kg in the impact assessment (section 9.2)
EU	Seveso III Directive 2012/18/EU, PED 2014/68/EU	Lower and upper tier of the establishment with the summation rule; category of the storage vessels (group 1 gas)
US	ASME B31.12 (option A) with EIGA Doc 121, ASME BPVC VIII-1	Pipeline wall with the hydrogen performance factor, corrosion allowance 1 mm and at least 6.4 mm, hoop stress ≤ 30 % SMYS; vessel shell by UG-27, thick wall by Appendix 1-2
US	API 618 (6th edition)	Discharge temperature of every compressor stage ≤ 120 °C for hydrogen-rich gas (selectable 135 °C or 150 °C)
US	NFPA 2 (2026), NFPA 855 / UL 9540A	Separation distances of bulk gaseous hydrogen for the exposure groups 1–3 against the plot plan; clearance of the battery
US	OSHA PSM (29 CFR 1910.119), EPA RMP (40 CFR 68), IRA § 45V	Threshold quantity 10 000 lb for hydrogen and ammonia; 45V credit tier from the well-to-gate GHG intensity (impact assessment)

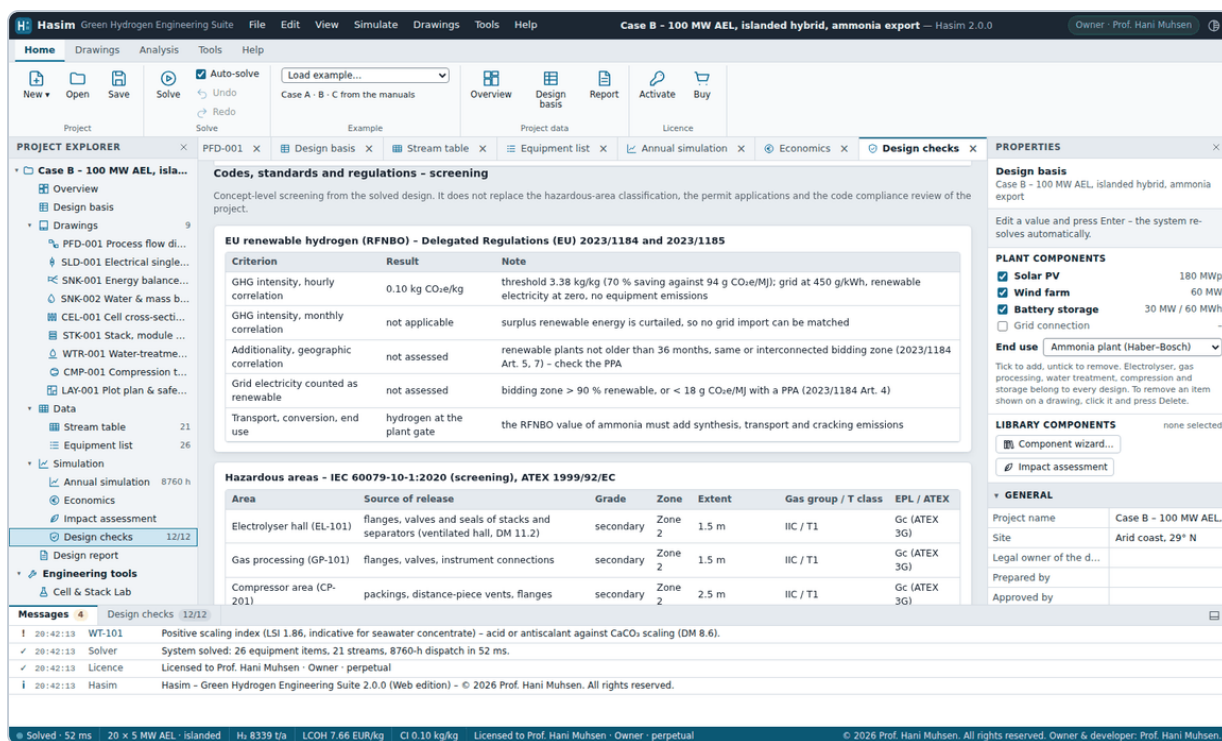


Figure 10.3 – Codes, standards and regulations: the screening of Case B.

Screening, not certification

The checks and tables are concept-level screening values from the solved design. They do not replace the hazardous-area classification and the dilution assessment of the project, the QRA, the design codes of the equipment suppliers, the permit applications or the certification of the hydrogen by an accredited scheme. Verify every code against the edition in force at the site.

10.2 Design report

The design report assembles the cover information with licence line, a summary, the complete design basis, all sizing sections with their equation references, the equipment list, the stream table, the monthly performance, the design checks, the impact assessment with the equipment and its components, all nine drawings and a disclaimer. Print it to paper or PDF with **Print / save as PDF**, or export a self-contained HTML file.

The screenshot displays the Hasim Green Hydrogen Engineering Suite interface. The main window shows the 'Design report' for 'Case B - 100 MW AEL, islanded hybrid, ammonia export'. The report includes a summary table and a list of plant components.

Design report
Case B - 100 MW AEL, islanded hybrid, ammonia export
 Arid coast, 29° N

Document: Hasim design report Hasim_Case_B_100_MW_AEL_islanded_hybrid_ammonia_export.RPT-001, rev. A
 Date of issue: 2026-09-24
 Prepared with: Hasim 2.0.0 - Green Hydrogen Engineering Suite
 Licence: Licensed to Prof. Hani Muhsen - Owner - perpetual
 Copyright: Software © 2026 Prof. Hani Muhsen. Report content belongs to the licensee.

1. Summary

Quantity	Value
Electrolyser	20 × 5 MW AEL
Power supply	Islanded hybrid
Hydrogen production (8760 h)	8339 t/a
Capacity factor (full load hours)	93.7% / 8207 h

PLANT COMPONENTS

- ☒ Solar PV: 180 MWp
- ☒ Wind farm: 60 MW
- ☒ Battery storage: 30 MW / 60 MWh
- ☐ Grid connection: -

End use: Ammonia plant (Haber-Bosch)

LIBRARY COMPONENTS (none selected)

GENERAL

Project name	Case B - 100 MW AEL
Site	Arid coast, 29° N
Legal owner of the d...	
Prepared by	
Approved by	

Messages

- 20:42:13 WT-101 Positive scaling index (LSI 1.86, indicative for seawater concentrate) - acid or antiscalant against CaCO₃ scaling (DM 8.6).
- 20:42:13 Solver System solved: 26 equipment items, 21 streams, 8760-h dispatch in 52 ms.
- 20:42:13 Licence Licensed to Prof. Hani Muhsen - Owner - perpetual
- 20:42:13 Hasim Hasim - Green Hydrogen Engineering Suite 2.0.0 (Web edition) - © 2026 Prof. Hani Muhsen. All rights reserved.

Status Bar: Solved - 52 ms | 20 × 5 MW AEL - islanded | H₂ 8339 t/a | LCOH 7.66 EUR/kg | CI 0.10 kg/kg | Licensed to Prof. Hani Muhsen - Owner - perpetual | © 2026 Prof. Hani Muhsen. All rights reserved. Owner & developer: Prof. Hani Muhsen.

Figure 10.4 – The design report.

11 Engineering tools

11.1 Cell & Stack Lab

Explores the electrochemistry of each technology: reversible voltage $E_{rev}(T, p)$ (CM 3.5–3.6), activation, ohmic and concentration losses (CM 4.4–4.10), Faradaic efficiency and gas crossover (CM 7.6–7.7), stack and system specific energy (CM 7.5, 7.8). Edit any parameter; the polarisation curve, loss breakdown, SEC, part-load efficiency, H_2 -in- O_2 and a table of operating points update immediately.

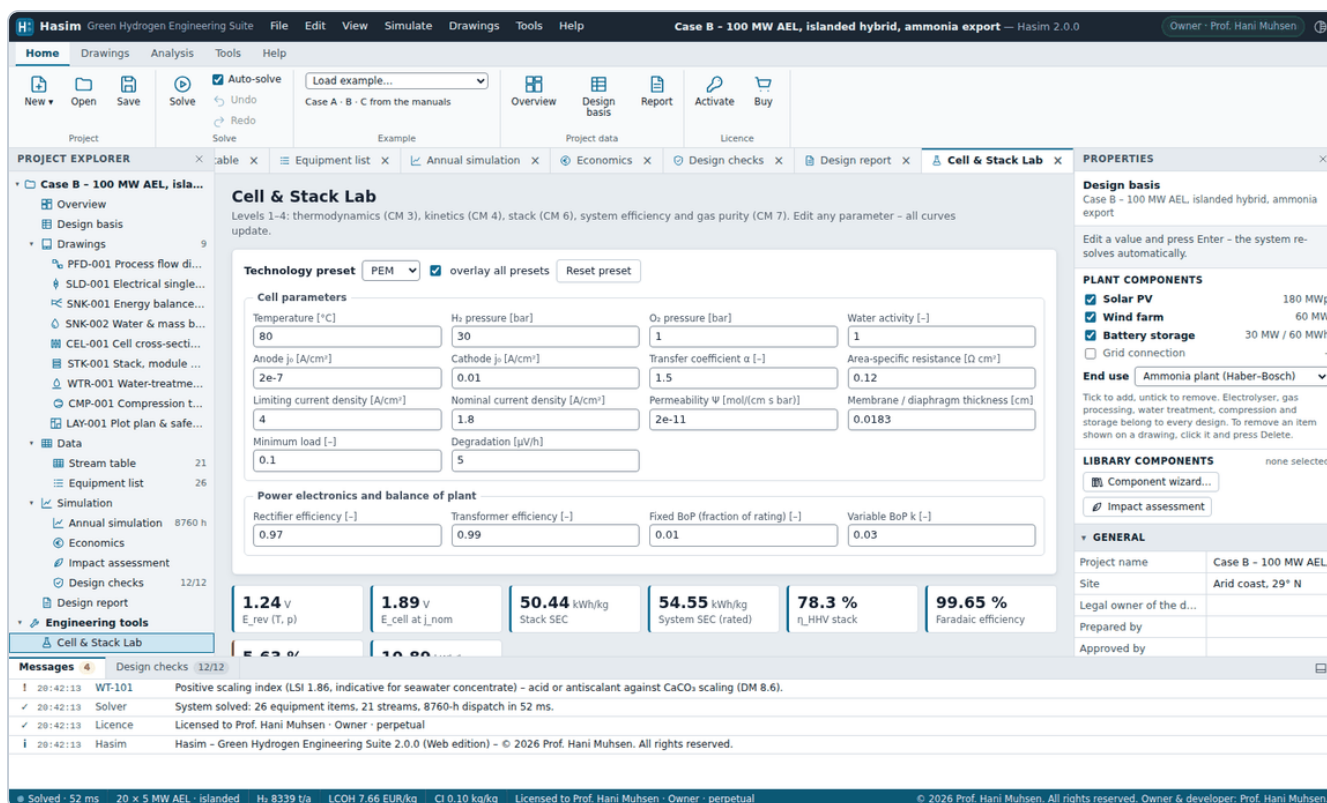


Figure 11.1 – Cell & Stack Lab.

j [A/cm ²]	E_{rev} [V]	η_a [V]	η_c [V]	η_{ohm} [V]	E_{cell} [V]	SEC stack [kWh/kg]
0.25	1.235	0.285	0.065	0.030	1.616	44.1
0.50	1.235	0.299	0.079	0.060	1.675	45.1
1	1.235	0.313	0.093	0.120	1.766	47.3
1.50	1.235	0.321	0.102	0.180	1.845	49.3
2	1.235	0.327	0.107	0.240	1.920	51.2
2.50	1.235	0.332	0.112	0.300	1.994	53.1
3	1.235	0.335	0.116	0.360	2.067	55.1

Table 11.1 – Polarisation of the PEM preset (80 °C) computed by the cell model of CM 4.4.

11.2 Equation library

All 137 equations as live calculators, filterable by level and manual and searchable by title, symbol or number. Each equation shows its formula, inputs, outputs and the equivalent script call; the parametric sweep varies one input and plots any outputs.

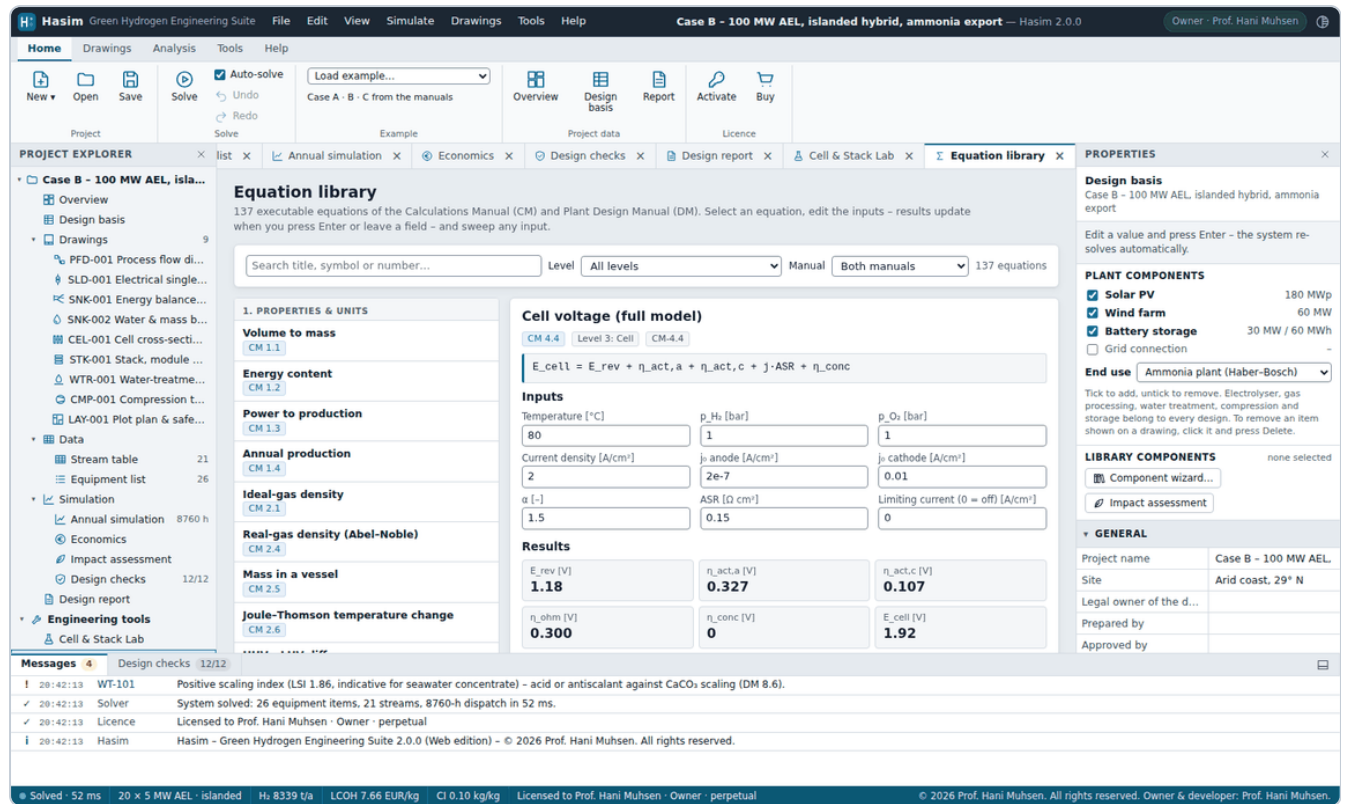


Figure 11.2 – Equation library.

11.3 Experiments

17 ready-made numerical experiments combine many equations; each has parameters, charts and result tables.

Experiment	Level	References
Polarisation curves of the four technologies	3	CM 3.5, 3.6, 4.4–4.10, 7.5
Temperature and pressure effects on the PEM cell	2	CM 3.5, 3.6, 4.5, 4.8
Part-load efficiency, Faradaic losses and gas purity	5	CM 6.4, 7.6, 7.7, 7.8, DM 10.2
Degradation over the stack life	4	CM 6.6, DM 9.1
Hybrid PV–wind sizing sweep (DM Table 4.1 replica)	6	DM Fig. 4.2, 4.3; CM 14.5; DM 20.2
Battery sizing for an islanded plant	6	DM 4.4, Fig. 4.2
LCOH sensitivity (tornado)	13	CM 17.1, 17.3
LCOH map: capacity factor × electricity price	13	CM 17.3
Compression energy vs pressure and stages	9	CM 10.1, 10.2, 10.4, 10.6
Real vs ideal gas: storage density	10	CM 2.1, 2.4, 2.5, 9.1
Weather-year Monte Carlo: P50 / P90 production	6	DM 3.1, 4.3
Transport: pipeline vs tube trailers	11	CM 13.1, 13.3, 20.1
GHG intensity vs grid share (RFNBO / Taxonomy / 45V)	14	CM 18.1, 18.3; Del. Reg. (EU) 2023/1185

Experiment	Level	References
Consequences of leaks: hole size and pressure	15	CM 19.1, 19.5, 19.6, 19.8
Cost outlook with learning curves	13	CM 17.8, 17.3
Desalination: RO energy vs recovery	7	DM 8.1, 8.2, 8.4, 8.5
Storage need vs offtake flexibility	10	CM 14.7, DM 13.1, 13.2

11.4 Component library

Tools ► Component library (Tools ribbon ► Components) holds 34 generic components in 15 categories, grouped like a plant – power supply (solar PV, wind turbines, battery storage), water treatment, electrolyser (stacks and modules, power electronics, balance of plant), compression and storage, delivery (pipeline, tube trailers, refuelling station) and conversion and end use (ammonia, methanol, fuel cells) – together with your own components. Choose a category on the left: the table compares all its components side by side (Figure 11.3); click a column to select one. The data of a component are grouped as follows:

Component library

34 generic components with technical, economic, environmental and social data, plus your own. Compare components, duplicate one to edit its values, and use them in the Model builder and in plant projects.

5 components - click a column (or its name) to select it

Parameter	PEM stacks - 5 MW modules Hasim library	Alkaline stacks - pressurised, 5 MW modules Hasim library	Alkaline stacks - large pressurised, 10 MW modules Hasim library	AEM stacks - 1 MW modules Hasim library	Solid-oxide stacks - 1 MW modules Hasim library
Technology	PEM	AEL	AEL	AEM	SOEC
Module rating [MW]	5	5	10	1	1
Stack temperature [°C]	80	80	80	60	800
Cathode (hydrogen) pressure [bar]	30	30	30	10	1
Rated current density [A/cm²]	1.8	0.400	0.400	1	0.800
Area-specific resistance [Ω cm²]	0.120	0.180	0.180	0.250	0.250
Voltage degradation [µV/h]	5	2	2	20	10
Stack life [h]	80 000	90 000	90 000	20 000	30 000
Minimum load [-]	0.100	0.200	0.200	0.100	0.200
Platinum-group metals [Ir + Pt] [g/kW]	0.800	0	0	0	0
Capital cost [EUR/kW]	675	450	420	230	660
Fixed O&M [1/a]	0.025	0.025	0.025	0.025	0.025
Economic life [a]	25	25	25	25	25
Embodied emissions [kg CO ₂ e/kW]	137.5	253.7	253.7	99.2	102.6
Land [m²/MW (plant incl. BoP)]	250	350	330	300	400
Construction jobs [job-years]	1.5 per MEUR	1.5 per MEUR	1.5 per MEUR	1.5 per MEUR	1.5 per MEUR
Manufacturing jobs [job-years]	1.5 per MEUR	1.5 per MEUR	1.5 per MEUR	1.5 per MEUR	1.5 per MEUR
O&M jobs [FTE]	0.9 per MW H ₂	0.9 per MW H ₂	0.9 per MW H ₂	0.9 per MW H ₂	0.9 per MW H ₂
CAPEX shares: equipment / installation / services	100 / 0 / 0 %	100 / 0 / 0 %	100 / 0 / 0 %	100 / 0 / 0 %	100 / 0 / 0 %

Figure 11.3 – The component library with the five electrolyser stacks side by side (Properties panel hidden with Alt+2).

Group	Content	Used by
Technical data used by the model	e.g. module rating, current density, ASR, degradation, stack life and minimum load of stacks; tracking gain, losses and DC/AC ratio of PV; power curve of wind turbines; round-trip efficiency of batteries; recoveries of water plants; isentropic efficiency and pressure ratio of compressors	Plant models and block models
Limits used by the component checks	Turbine class limit, maximum feed salinity, pressure ratings, capacity per machine, cycle life, technology of the balance of plant	Component checks
Datasheet	Hub height and rotor diameter, vessel and compressor type, power factor	Information
Economic data	Capital cost in the unit of the category, fixed O&M as a share of CAPEX, economic life	Costs of blocks and plants, NPV
Environmental data	Embodied GHG emissions of manufacture, transport and installation, land occupied, platinum-group metals, total area of wind farms	Environmental impact
Social data	Employment factors for construction and installation, manufacturing and operation and maintenance with their basis; CAPEX shares of equipment, installation and services	Jobs and local content

Your own components

- 1 Select a component and press **Duplicate to edit**; give the copy a name. It appears in the same category marked *my library* (Figure 11.4).
- 2 Change its values, name and technology and press **Save changes**. **Delete** removes it from your library – projects and models that use it keep their own copy.
- 3 **Apply to the plant project** uses the component in the current plant; **Use in the Model builder** puts it into the selected block of the Model builder (or the first block with this kind of equipment).
- 4 **Export my library** writes your components to a JSON file, **Import...** reads one – for example vendor data shared in your team. Imported components are checked: unknown fields are ignored and every value is limited to its allowed range.

PEM stack - vendor quotation 2027
Duplicate
Save changes
Delete
Apply to the plant project
Use in the Model builder

IDENTIFICATION
Name: PEM stack - vendor quotation 2027
Technology: Proton-exchange membrane, Ir/Pt catalysts, 30 bar cathode
Source: User - based on "PEM stacks - 5 MW modules"

TECHNICAL DATA USED BY THE MODEL
Technology: PEM
Module rating [MW]: 5
Stack temperature [°C]: 80
Cathode (hydrogen) pressure [bar]: 30
Rated current density [A/cm²]: 1.8
Area-specific resistance [Ω cm²]: 0.12
Voltage degradation [μV/h]: 5
Stack life [h]: 80000
Minimum load [-]: 0.1

ECONOMIC DATA
Capital cost [EUR/kW]: 560
Fixed O&M (share of CAPEX per year) [1/a]: 0.025
Economic life [years]: 25

ENVIRONMENTAL DATA
Platinum-group metals (Ir + Pt) [g/kW]: 0.8
Embodied GHG emissions (manufacture, transport, installation) [kg CO₂e/kW]: 137.5
Land occupied [m²/MW (plant incl. BoP)]: 250

Figure 11.4 – A component of your library, duplicated from the PEM stack and edited with a vendor price.

Components of the Hasim library are read-only. A value changed in the component wizard changes only the copy used by that project or model. Your library is kept in the browser storage of the edition you use – export it to keep a backup or to take it to another computer.

11.5 Model builder

Builds value chains from blocks – Power supply (PV / wind / grid), Water treatment (RO + polishing), Electrolyser plant, Compressor, Compressed storage buffer, Pipeline, Tube-trailer delivery, Ammonia synthesis (Haber-Bosch), Methanol synthesis, Fuel cell (power-to-power), Refuelling station 700 bar, Gas-grid blending, Economics & carbon (LCOH, CI). The stream (hydrogen, pressure, energy, cost, carbon) passes from block to block; the electrolyser block runs the 8760-hour model with the power supply before it. The builder opens with the chain of Case B (islanded ammonia export); **Load example model...** offers further chains.

- 1 **New empty model**, then **start with a standard chain** or add blocks from the palette. The palette is grouped by stage (power and water, hydrogen production, compression and storage, delivery and end use, evaluation) and every block is inserted at its place in the chain.
- 2 Click a block to edit its parameters and components in the editor below the palette. **← Move / Move →** change its position.
- 3 Delete a block with its **x**, with **Delete block** or with the Delete key; **Undo** (Ctrl+Z while the builder is active) restores the chain.
- 4 The order checks under the chain explain problems – a missing electrolyser, water treatment after the electrolyser, blocks that receive no hydrogen, Economics not at the end – and **Fix order** sorts the blocks into flow order.
- 5 With **Live results** the results update half a second after every change; **Run model** recalculates and scrolls to them. **Save model / Open model** store the chain with its components and settings as a JSON file.

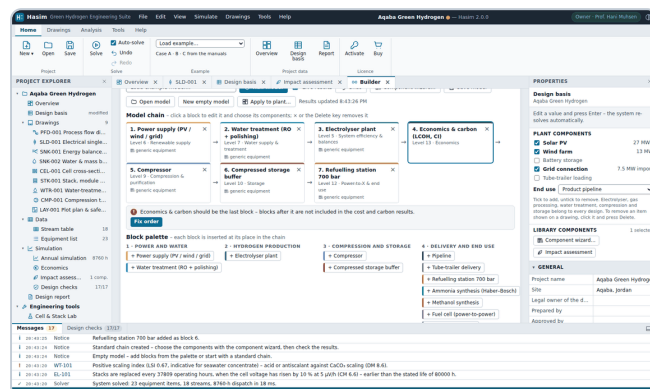


Figure 11.5 – Model builder: chain, order checks and palette.

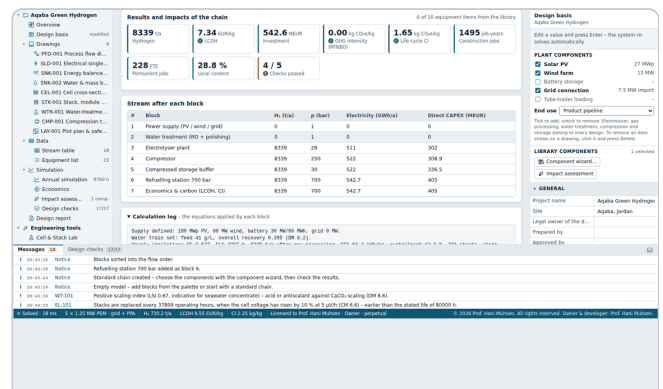


Figure 11.6 – Results of a chain.

Components of the blocks

Blocks with equipment – power supply, water treatment, electrolyser, compressor, storage, pipeline, trailers, refuelling station, ammonia, methanol and fuel cell – list their equipment under **Components of this block** (Figure 11.7). Without a component a block uses its own parameters (*Generic – block values*) and the assessment uses the Hasim default. **Component wizard...** in the block editor or the toolbar walks through the equipment of all blocks of the chain (Figure 11.8): choose a component, set its size in the model – for example the PV capacity, the mean wind speed of the site or the electrolyser rating – edit its parameters and watch the effect on the whole chain. **Apply components** stores the selection in the blocks as one step of Undo. Block parameters defined by a component are

shown grey and locked; their tooltip names the component. Change them in the wizard, or return to the generic block with ×.

3. Electrolyser plant ← Move Move → ✕ Delete block

Runs the 8760-h dispatch model (DM Fig. 4.2) with the cell-level polarisation model of the chosen technology.

Components of this block Component wizard...

Component	Value	Action
Electrolyser stacks and modules	Alkaline stacks – large pressurised, 10 MW modules Hasim library	Choose... ✕
Power electronics (rectifier + transformer)	Generic – block values	Choose...
Balance of plant (gas processing, circulation, cooling)	Generic – block values	Choose...

⚠ Grey fields are set by the components.

Technology (PEM/AEL/AEM/SOEC)	Rated power [MW]	Minimum load [-]	System CAPEX [EUR/kW]
AEL	100	0.2	970
Auxiliaries [MW]	Availability [-]	Stack temperature (blank = preset) [°C]	Cathode pressure (blank = preset) [bar]
6	0.95	80	30

Figure 11.7 – The electrolyser block with a library stack: technology, minimum load, temperature, pressure and CAPEX are set by the components (grey).

Component wizard - Case B - islanded ammonia export

1. Power supply (PV / wind / grid) · choose a component

Wind turbines

Generic - use the block values
no library component; the block parameters are used and the assessment uses the Hasim default

Onshore wind turbine 5 MW class, medium wind (IEC II)
5 MW · rated 12 m/s · 1300 EUR/kW · 448 kg CO₂e/kW
Hasim library · Three-bladed upwind turbine, ~150 m rotor, 120 m hub

Onshore wind turbine 4.5 MW, large rotor for low wind (IEC III)
4.5 MW · rated 10.5 m/s · 1400 EUR/kW · 448 kg CO₂e/kW
Hasim library · Low-specific-rating turbine, ~163 m rotor, 125 m hub

Onshore wind turbine 4.2 MW, high wind (IEC I)
4.2 MW · rated 13 m/s · 1250 EUR/kW · 448 kg CO₂e/kW
Hasim library · High-specific-rating turbine for strong winds, ~117 m rotor, 100 m hub

SIZE IN THIS MODEL

Wind capacity [MW]

Mean wind speed at hub (site) [m/s]

= 14 turbines of 4.2 MW

Onshore wind turbine 4.2 MW, high wind (IEC I) Hasim library Save as my component...

High-specific-rating turbine for strong winds, ~117 m rotor, 100 m hub - edit a value to use it in this model; the library stays unchanged.

TECHNICAL DATA USED BY THE MODEL

Turbine rating [MW]

Cut-in wind speed [m/s]

Rated wind speed [m/s]

Cut-out wind speed [m/s]

Wake, availability and electrical losses [-]

LIMITS USED BY THE COMPONENT CHECKS

Turbine class limit: mean wind at hub (IEC 61400-1) [m/s]

DATASHEET (INFORMATION ONLY)

Hub height [m]

Rotor diameter [m]

ECONOMIC DATA

Effect on the model - recalculated with the components chosen so far

8201 t/a Hydrogen	8.32 EUR/kg LCOH	595.8 MEUR Investment	0.00 kg CO ₂ e/kg GHG intensity (RFNBO)	1.58 kg CO ₂ e/kg Life-cycle CI	1613 job-years Construction jobs	204 FTE Permanent jobs	32.0 % Local content
-----------------------------	----------------------------	---------------------------------	--	--	--	----------------------------------	--------------------------------

5 / 6 Checks passed

Changes take effect when you press Apply - Ctrl+Z undoes them.

Cancel ← Back Next → Apply components

Figure 11.8 – The component wizard of the Model builder at the wind turbines of the power-supply block.

The results start with **Results and impacts of the chain** – hydrogen, LCOH, investment, life-cycle carbon intensity, construction and permanent jobs, local content and checks – followed by the stream after each block, the calculation log, the monthly production and the complete impact assessment of section 9.2 (Figure 11.9). The chain checks cover the gas purity at the minimum load (CM 7.7, DM 10.2), unserved auxiliary energy (DM 4.3), compressor discharge temperature (DM 12.1), storage and pipeline capacity, the carbon intensity (CM 18.1) and the

component checks – module rating, balance-of-plant technology, turbine class, battery cycles, feed salinity, pressure ratings, number of compressors and trailer pressure. **Assessment settings** above the results hold the settings of section 9.3 and an optional production target. **Apply to plant...** transfers the library components of the model to the plant project – the plant keeps its sizes – and opens its impact assessment.

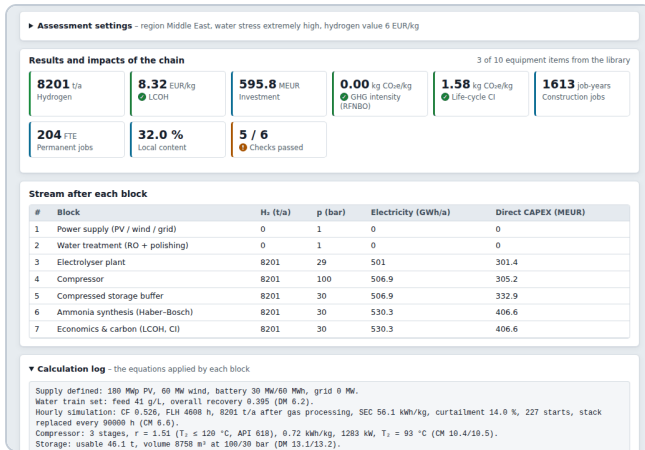


Figure 11.9 – Results and impacts of the chain with three library components.

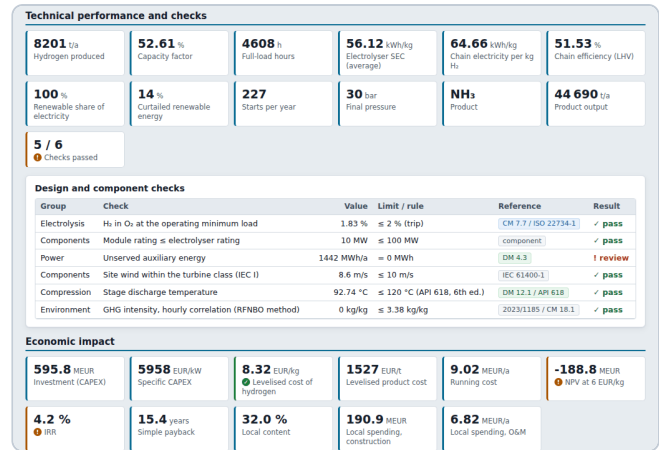


Figure 11.10 – Technical performance and the chain checks: the auxiliaries of the islanded chain are not supplied at night.

11.6 Script console and API

Automates studies in JavaScript. The console provides `H2` (engine), `H2F` (drawings), `print()`, `table()` and `plot()`. Scripts run in strict mode and have no access to the application window, its storage or the licence; results are exported with the Table/CSV/SVG buttons of the charts and tables they create. The example *Impacts of a design with library components* applies components to Case B and tabulates its impacts.

Call	Returns
<code>H2.solveSystem(basis)</code>	Complete system: equipment, streams, electrical data, annual simulation, balances, layout, checks, codes and standards screening (<code>compliance</code> , <code>hazAreas</code> , <code>seveso</code>) and KPIs
<code>H2.CASES.A B C</code>	Example design bases
<code>H2.run(id, inputs)</code> , <code>H2.sweep(id, key, a, b, n, fixed)</code>	One equation, or a sweep of one input
<code>H2.cell(tech, j)</code> , <code>H2.system(tech, P_kW, opts).atLoad(L)</code>	Cell model and system operating points
<code>H2.hourly(cfg)</code> , <code>H2.designPlant(basis)</code> , <code>H2.runFlowsheet(blocks)</code>	Annual simulation, design chain, block model
<code>H2F.pfd(sys)</code> , <code>H2F.sld(sys)</code> , <code>H2F.DRAWINGS</code>	Drawing generators (SVG)
<code>H2.LIB</code> , <code>H2.libGet(id)</code> , <code>H2.libList(cat)</code>	Components of the library (copies)
<code>H2.basisFromComponent(basis, slot, comp)</code>	Design-basis values set by a component – merge them into the basis with <code>Object.assign</code>
<code>H2.assessPlant(sys)</code> , <code>H2.assessChain(result, blocks, settings)</code>	Impact assessment of a plant or of a Model-builder chain

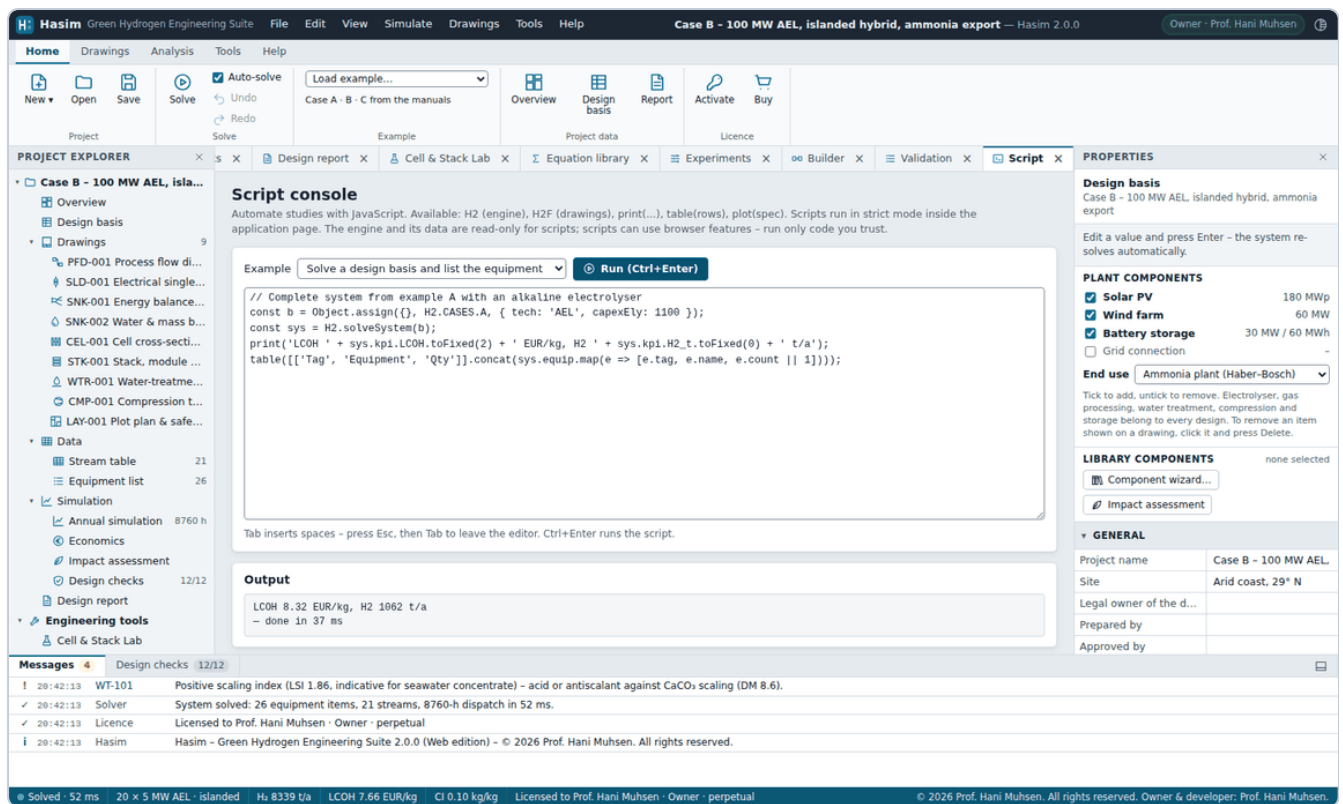


Figure 11.11 – Script console.

11.7 Validation

Runs the verification suite described in Chapter 12 and lists every check with manual value, Hasim value and deviation.

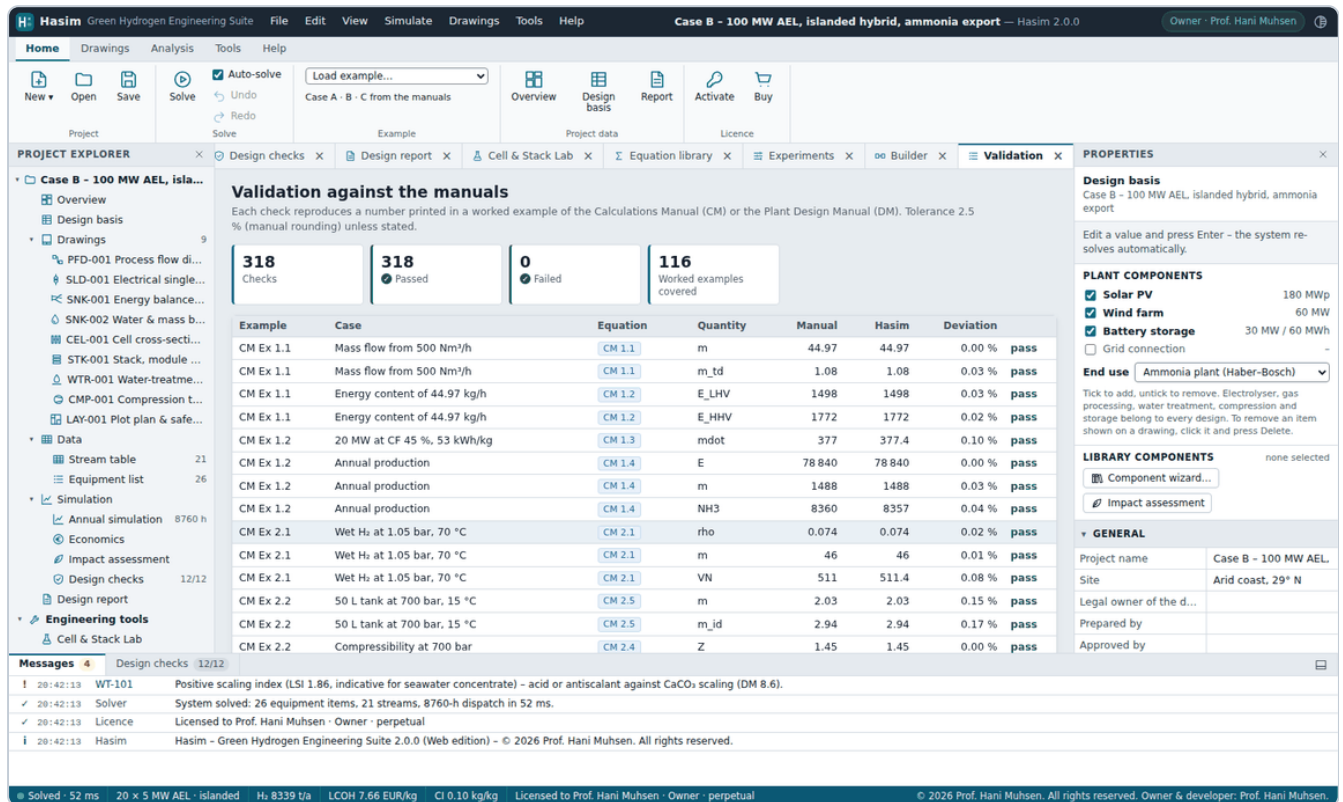


Figure 11.12 – Validation.

12 Verification and validation

Hasim is verified against the worked examples of both manuals: 318 numerical checks from 116 worked examples reproduce the printed results. All 318 checks pass; the largest deviation is 2.4 %, within the rounding of the printed values (tolerance 2.5 %).

- The annual **energy balance** closes to better than 10^{-6} GWh for all examples (reported as a design check).
- The **mass balance** of every stream node is conserved by construction; the water balance matches the DI-water demand of DM 6.1.
- Real-gas densities use the Abel–Noble equation, accurate to about 1 % up to 700 bar at ambient temperature.
- Drawings are checked for consistency with the solved data (equipment tags, stream numbers and values).
- A regression suite of the development checks the models introduced by the 2026 expert review – stack life and degradation, gas-processing losses, the RFNBO method, the compressor temperature limit, the major-accident summation, the SOEC energy balance – and the input guards of the solver.

13 Modelling assumptions and limitations

- Hasim provides **concept and pre-FEED level** engineering. Results must be reviewed by qualified engineers; applicable codes, standards and certified tools take precedence.
- The weather is a **synthetic typical year** scaled to the site's annual GHI and mean wind speed. Bankable energy yields require measured multi-year data.
- Costs are **indicative values** from the manuals and the component library; replace them with project quotations.
- The **impact assessment** is a screening assessment for comparing designs and components. Employment factors, embodied emissions and land use are typical published values (section 9.4) scaled with the size or cost of the equipment; they do not replace a life-cycle assessment to ISO 14040/14044, a socio-economic study or the environmental and safety permitting of the project. Seveso III tiers use the summation rule of Annex I for hydrogen and ammonia; other dangerous substances on the site and national implementations must be added.
- Cell model: LeRoy reversible voltage with Nernst term, symmetric Butler–Volmer kinetics, ohmic ASR and a limiting-current term; crossover by Fick permeation. SOEC: the heat for raising the steam and for operation below the thermoneutral voltage is supplied electrically or as external heat (section 8.1).
- Stack ageing accrues per operating hour at any load, which is conservative for plants that run long at part load; the replacement criterion is a 10 % voltage rise or the stated stack life (CM 6.6).
- Equipment sizing uses the simplified relations of the manuals (Souders–Brown separators, LMTD heat exchangers, Weibull wind, etc.). Vessel walls use ASME VIII-1 with the allowable stress of SA-516 Gr 70 (138 MPa) – conservative for the Cr-Mo steels of high-pressure tubes; pipelines use ASME B31.12 option A.
- Safety distances are consequence estimates for a reference leak (CM 19) and screening values of the NFPA 2 table; hazardous-area extents are screening values after IEC 60079-10-1. They do not replace a quantitative risk assessment, the hazardous-area classification or the separation-distance codes in force at the site.
- The RFNBO check covers the GHG intensity with hourly and monthly correlation. Additionality, geographic correlation and the rules for grid electricity counted as renewable (Delegated Regulation (EU) 2023/1184) must be verified with the PPA and the certification scheme.
- Scaling indices: the gypsum saturation uses the Davies activity model, valid to an ionic strength of about 0.5 mol/L; for seawater concentrates it is reported as “n/a”. The Langelier index is indicative above about 10 g/L, where the Stiff & Davis index applies.

14 Copyright, licence and support

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Support

Help ► User guide in the program · this manual and the Model-Building Examples · the support e-mail address (Help ► About shows it) · licence keys can be retrieved on the Hasim website · Hasim bought in the Microsoft Store belongs to your Microsoft account (account.microsoft.com).